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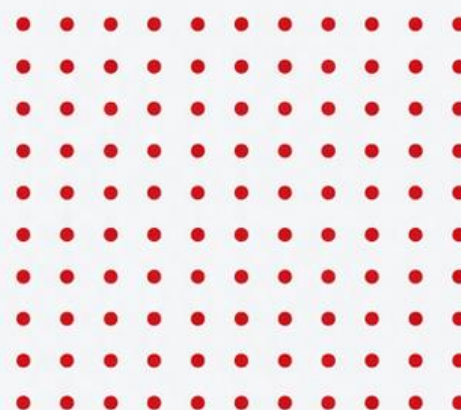
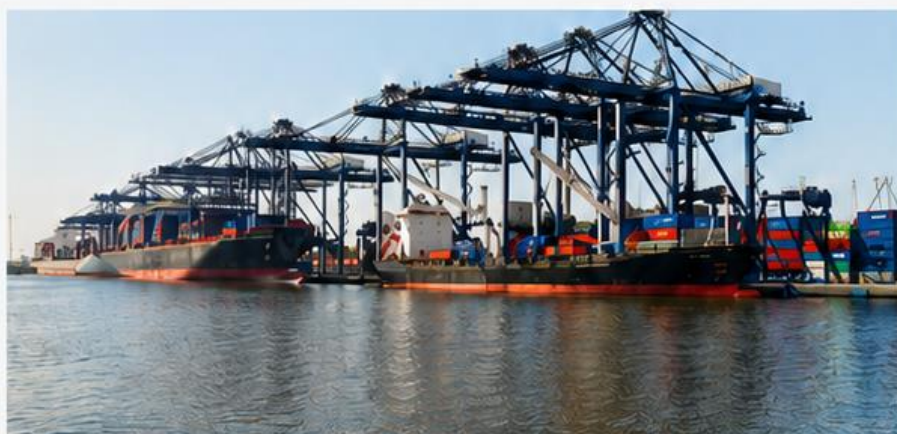


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LABOUR MARKET ASSESSMENT IN SINDH

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Acronyms

Acronym	Full Form
AQL	Acceptable Quality Level
BEOE	Bureau of Emigration and Overseas Employment
BLS	Basic Life Support
BOI	Board of Investment
BOQ	Bill of Quantities
BPO	Business Process Outsourcing
CBT&A	Competency-Based Training and Assessment
CIT	Certificate in IT
CoE	Centre of Excellence
CPR	Cardiopulmonary Resuscitation
CRM	Customer Relationship Management
CSSD	Central Sterile Services Department
DEFF	Design Effect
DIT	Diploma in IT
DTP	Desktop Publishing
EMR	Electronic Medical Record
EPZ	Export Processing Zone
EPZA	Export Processing Zones Authority
ETP	Effluent Treatment Plant
FC	Fulfilment Centre
FGD	Focused Group Discussion
GDP	Gross Domestic Product
GIS	Geographic Information System
GLP	Good Laboratory Practice
GMP	Good Manufacturing Practice
GPS	Global Positioning System
HACCP	Hazard Analysis and Critical Control Points
HAZOP	Hazard and Operability Study
HR	Human Resources
HSE	Health, Safety and Environment
ICT	Information and Communication Technology
ILO	International Labour Organization
ISCO-08	International Standard Classification of Occupations (2008)
LFPR	Labour Force Participation Rate
LFS	Labour Force Survey
LOTO	Lockout/Tagout
LSO	Local Support Organization
M-9	Motorway-9 (Karachi-Hyderabad)
MoCC&EC	Ministry of Climate Change and Environmental Coordination
MoFEPT	Ministry of Federal Education and Professional Training
MoHR	Ministry of Human Rights
MoIP	Ministry of Industries and Production
MoITT	Ministry of Information Technology and Telecommunication
MSME	Micro, Small and Medium Enterprise
NAC-TVS	National Accreditation Council for Technical and Vocational Streams
NAVTTTC	National Vocational and Technical Training Commission
NVQF	National Vocational Qualifications Framework
OHS	Occupational Health and Safety
O&M	Operations and Maintenance
PLC	Programmable Logic Controller
POS	Point of Sale
PPS	Probability Proportional to Size
PSIC	Pakistan Standard Industrial Classification
PSU	Primary Sampling Unit
PV	Photovoltaic
QA	Quality Assurance
QC	Quality Control
RF	Radio Frequency
RPL	Recognition of Prior Learning
RTG	Rubber-Tyred Gantry (crane)
SCADA	Supervisory Control and Data Acquisition
SEZ	Special Economic Zone
SMEDA	Small and Medium Enterprises Development Authority
SOP	Standard Operating Procedure

Acronym	Full Form
SSU	Secondary Sampling Unit
TEVTA	Technical Education and Vocational Training Authority
TVET	Technical and Vocational Education and Training
UC	Union Council
UN	United Nations
VFD	Variable Frequency Drive
WMS	Warehouse Management System

EXECUTIVE SUMMARY

This Labour Market Assessment has been prepared under the TVET Sector Support Programme to provide an employer-led evidence base for strengthening demand-driven TVET planning and delivery in Sindh. The assessment responds to the need for updated and practical labour market evidence in a province with a diversified economy, strong spatial concentration of employment, and persistent skills and inclusion gaps. It identifies priority sectors and occupations, quantifies demand–supply gaps at district and trade levels, defines green and environmental competencies for high-demand occupations, and proposes actionable measures to improve alignment between training provision and labour market needs over the next three years.

Approach and Coverage: The assessment combines a structured desk review with a province-wide Employer Skills Survey of 2,792 establishments, for which data were drawn from the Pakistan Bureau of Statistics Economic Census 2023. The survey used a stratified two-stage sampling design and collected information from business owners, managers, HR personnel, and operations leads directly responsible for hiring and workforce planning. The analysis covers sector-wise and district-wise demand, trade-wise shortages and surpluses, employer satisfaction with TVET graduates, workplace training practices, digital and green skill needs, and constraints affecting women and disadvantaged groups.

Labour Market Demand and Supply Gap Remains Large and Widens: Sindh faces a persistent and widening mismatch between demand for skilled labour and available supply. Employer-reported demand rises from 185,268 in 2018 to 331,652 in 2025, while supply increases only marginally from 65,412 to 72,746. The shortfall expands from 119,856 to 258,906 skilled workers, with unmet demand rising from 64.7% to 78.1%. The resulting demand–supply ratio deteriorates from about 3:1 to about 5:1. This is not a marginal issue but a structural scaling and alignment problem that requires expansion in high-demand trades and a reallocation of training capacity away from low-absorption pathways.

Demand is Concentrated in a Limited Number of Sectors: The findings show that sectoral demand is highest for Manufacturing with 95,675 positions, followed by Allied Health with 40,545 and Agribusiness with 39,037. Construction accounts for 25,001 positions, Energy 22,784, Information Technology 21,579, Water 20,970, and Hospitality and Tourism 20,649. Together, the top five sectors account for 67.3% of total reported demand. This pattern calls for Sindh’s skills system to prioritise industrial production, health-related services, agri-value chains, construction, and energy, while also responding to growing demand in digital, water, and service occupations.

Demand is Spatially Concentrated but not Confined to Karachi Alone: District-wise demand totals 331,652 positions and is heavily concentrated in Karachi. Karachi Central records the highest demand at 52,428, followed by Keamari 42,260, Malir 41,697, Korangi 32,607, and Karachi West 31,296. These five districts alone account for 60.4% of provincial demand, while total demand across Karachi districts increases to 68.4% when Karachi South and Karachi East are included. At the same time, demand from secondary hubs remains significant, particularly for Hyderabad 10,003, Khairpur 9,847, Larkana 9,323, and Sukkur 8,680. The planning implication is that Sindh needs both scale-focused delivery in Karachi and targeted, location-specific programming in non-Karachi districts.

Provincial Occupational Demand and Shortages are Increasingly Specific: The leading trades in the provincial labour market are Computer Operator 9,944, Stitching Machine Operator 9,056, CNC Machine Operator 8,468, Accountant 7,056, Medical Technician 6,960, Sales and Marketing 6,005, and Nurse 5,748. At the same time, the most severe shortages are concentrated in specific occupations rather than generic labels. The largest shortfall is in Software Technician, where demand of 27,410 against supply of 545 leaves a gap of 26,865. Other major shortages include Computer Operator, CNC Machine Operator, Accountant, Sales and Marketing, Construction Worker, Nurse, Quality Assurance, Sewing Machine Operator, Packaging Machine Operation, Plumbing and Pipe Fitter, and Industrial Stitching Machine Operator. By contrast, surpluses persist in D.I.T., Information Technology (C.I.T.), Fashion Design and Dress Making, Graphic Designer, and Web Designing and Development, indicating that part of the problem lies in the composition of training supply rather than in volume alone.

Sector-Specific Demand Patterns Sharpen the Planning Signal: Within sectors, the revised findings point to specific, high-demand occupations. Manufacturing demand is led by Stitching Machine Operator, CNC Machine Operator, Lathe Machine Operator, and Industrial/Automation

Software Developer. Allied Health demand is concentrated in Nurse, Medical Technician, X-Ray Technician, and Lab Technician. Energy demand is driven by Welder, Solar Installation Technician, Solar Technician, Electrician, and Electrical Technician. Construction demand is highest for Civil Technician, Plumber, Architecture Technician, Mason, and Heavy Machinery Operator. In Information Technology, Java Script, Web Designing, Computer Operator, Cyber Security, and PHP dominate. These patterns reinforce that sector strategy must be built around occupation families and task bundles, not broad sector labels alone.

Employers Value TVET, but Workplace Learning Systems Remain Weak: Employers are not dismissive of TVET, but they do not see it as sufficiently practical or consistent. Overall, 69% of employers report that they are satisfied or very satisfied with TVET graduates, 27% report a neutral view, and 4% are dissatisfied or very dissatisfied. During recruitment, employers rely mostly on personal interviews 44.6% and practical skills assessments 36.9%, which shows the importance of job-ready competencies. Yet only 20% of firms report having structured organisational training programmes, and only 32.4% have a dedicated training or learning and development department. Where employers do train, the dominant mode is in-house practical training 54.2%, while apprenticeship is reported by only 10.7%. This points to the need for stronger apprenticeship, internship, and co-delivered workplace training models.

Green, Digital, and Emerging Technology Skills are Cross-Cutting Requirements: Employers increasingly view green and digital skills as essential complements to mainstream occupations. The most cited green skill areas are environmental compliance at 21.8%, waste management and recycling at 21.2%, energy efficiency practices at 21.2%, and sustainable manufacturing processes at 20.3%. Employers prefer green skills to be integrated through practical green training 27.5% and embedding across relevant trades 25.7%, rather than through isolated classroom modules. Digital readiness is also highly important: 54.7% of employers rate digital and ICT skills as very important or extremely important, while 82.4% rate them at least moderately important. For emerging technologies, Artificial Intelligence 41%, 3D printing 30%, and robotics 22% are the most frequently cited priorities. This indicates that digital and green competencies should be mainstreamed across priority trades through modular, applied, and sector-linked training.

Inclusion Constraints Remain Limiting Factors: The overall skilled workforce demand profile remains male-dominated, with male demand at 252,780 or 76.2% and female demand at 78,872 or 23.8%. Female participation is relatively stronger in Allied Health with a 42.5% female share, Information Technology 35.2%, and Manufacturing 30.8%, but remains very low in Construction 7.2%. Employers reported that the main constraints to hiring and retaining women include workplace culture 29.9%, safety and security 25.2%, stereotyping 21.9%, and lack of flexible arrangements 20.5%. For disadvantaged groups, the largest barriers are accessibility constraints 32.4%, inadequate training opportunities 26.0%, lack of accommodations 23.4%, and discrimination 17.8%. These findings show that inclusive TVET planning must go beyond enrolment targets and address workplace conditions, access, safety, and placement support.

Some key findings of the study include:

- i. **Re-base Sindh's skills expansion on employer demand clusters, not on uniform province-wide targets:** The demand profile is spatially concentrated, with Karachi districts accounting for 226,898 of 331,652 projected skilled positions, or approximately 68.4% of total provincial demand. Karachi Central records the highest demand at 52,428, followed by Keamari 42,260, Malir 41,697, Korangi 32,607, Karachi West 31,296, Karachi South 15,261, and Karachi East 11,349. Operationally, this requires cluster-aligned delivery hubs and employer-linked training pipelines that can scale throughput in Karachi's industrial, port, services, and commercial belts, while maintaining targeted district-level interventions in secondary hubs such as Hyderabad 10,003, Khairpur 9,847, Larkana 9,323, and Sukkur 8,680 to prevent under-coverage outside Karachi.
- ii. **Shift from supply-led training plans to a market-led financing and intake model anchored in verified employer demand:** Training expansion should be guided by employer-reported demand, trade-wise shortages, placement potential, and district-specific labour market absorption rather than by legacy course portfolios or aspirational targets. Where labour demand is volatile or linked to changing technologies, training should be supported through employer-linked instruments such as apprenticeship co-financing, outcome-based payments tied to placement, and trainee vouchers for approved providers. This will help

ensure that training volume moves with real vacancies and verified demand rather than with supply-side assumptions.

- iii. **Standardise occupational classification and break down generic job labels into trainable trades:** The assessment shows that employers do not demand a single undifferentiated “machine operator” category. Demand is spread across specific roles such as CNC Machine Operator, Stitching Machine Operator, Lathe Machine Operator, Packaging Machine Operator, Textile Machine Operator, and other process-specific occupations. TVET planning should therefore move from broad job labels to a sector-linked trade taxonomy that distinguishes occupations by production process, equipment needs, task requirements, and certification standards. This is essential for making curricula, labs, and seat allocation responsive to actual labour market demand.
- iv. **Correct the analytical and programmatic gaps in services and retail by explicitly planning for customer-facing service roles:** Employer-side labour demand in Sindh is not only industrial. Wholesale, retail, repair, hospitality, customer service, malls, department stores, beauty and salon services, and digital commerce support should be treated as defined employment segments in both the sampling frame and the training portfolio. This should include standardised modules on customer service, POS and inventory management, basic digital commerce fulfilment, workplace communication, and job-readiness skills, with clear entry pathways for youth and disadvantaged groups.
- v. **Treat occupational stigma as a binding constraint and address it through a province-wide skills-to-careers approach:** Expanding training supply is not only a capacity issue; it is also linked to aspiration, social acceptance, and perceived career value of technical and vocational occupations. The practical package should include career guidance through schools and community channels, visible employer-backed wage and progression ladders, certified apprenticeship pathways, and recognition mechanisms that show how skills certification can lead to income progression. Without addressing stigma, high-demand trades may continue to face recruitment frictions even when training seats are available.
- vi. **Reinforce curriculum reform with assessment integrity, accreditation, and workplace standards enforcement:** Curriculum revision alone will not improve labour market outcomes if assessment systems remain weak and training quality varies across providers. Required reforms include independent third-party assessment and certification, regular audits of training centres against equipment and instructor standards, strengthened competency-based training and assessment, and employer-involved trade testing. These measures will improve the credibility of TVET certification, strengthen employer confidence, and reduce the risk of workers being qualified on paper but not job-ready in practice.

1. INTRODUCTION

This report presents the findings of the Labour Market Assessment for the TVET Interventions in Sindh conducted under the TVET Sector Support Programme (TVET SSP) in Pakistan, co funded by the European Union and the Federal Republic of Germany. The Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH is implementing the programme together with British Council in close cooperation with the National Vocational and Technical Training Commission (NAVTTTC) and other public and private sector organizations. The programme's objective is to support the development of Pakistan's TVET sector, focusing on training for men and women in professions with a high demand, especially in digital and green skills. The assessment has been designed to generate employer-led evidence on skilled workforce demand, labour supply, occupational skill gaps, and sector- and district-level labour market trends in Sindh. Its findings are intended to support demand-driven TVET planning, strengthen industry engagement, improve the relevance of training provision, and promote more inclusive access to skills and employment opportunities, particularly for women and disadvantaged groups.

1.1. Background

In Pakistan, the TVET sector is central to addressing skills shortages and improving employability, yet participation remains limited and inclusion gaps persist. UNESCO-UNEVOC country profile data for Pakistan reports enrolment in vocational programmes at secondary level at 3% and indicates a female share of TVET enrolment of 33.5%, underscoring both scale and gender inclusion constraints in the training pipeline (UNESCO-UNEVOC, 2024). Policy reforms have sought to reposition TVET as demand-driven and standardised. The National Skills Strategy 2009 to 2013 sets out a shift toward competency-based training (CBT) and stronger labour market linkages, providing a national reform anchor for provincial programme design and coordination (NAVTTTC, 2009).

Sindh is a priority province for demand-driven TVET because of its concentration of economic activity and its diverse sectoral base, with Karachi functioning as the primary manufacturing and trade hub. Asian Development Bank reporting characterizes Karachi as the gateway of Pakistan, accounting for 95% of foreign trade and contributing 30% of industrial production of the country, highlighting why employer demand is strongly clustered in Karachi and its surrounding industrial corridors (Asian Development Bank, n.d.). Sindh also holds a major share of national fisheries resources, with provincial investment facilitation material reporting 71% of Pakistan's marine fish resources located in Sindh, reinforcing the relevance of coastal value chains for skills planning (SEZMC, n.d.).

Within this context, the Sindh labour market assessment provides updated data and information on employer demand, occupational shortages and surpluses, and inclusion constraints across sectors and districts. This evidence is required to recalibrate TVET programmes, intake planning, and industry collaboration mechanisms toward occupations and sectors with verified demand stats, while embedding practical measures that improve access, retention, and progression for women and disadvantaged groups. The importance of addressing barriers faced by disadvantaged youth in skills systems is also emphasized in UNESCO-UNEVOC work on inclusive TVET pathways and the accessibility and acceptability constraints that shape participation and outcomes (Alla Mensah, 2021).

1.2. Purpose and Scope

The Labour Market Assessment for Sindh identifies priority sectors, occupation-specific skill requirements, demand and supply gaps, and district-level employment patterns. Furthermore, the assessment provides practical measures for strengthening TVET planning and delivery. It also presents a comprehensive labour market analysis to identify skill gaps in high-growth sectors, and translates evidence into actionable recommendations for TVET policymakers and implementers. It does so through the following means:

1. Assessment of key sectors and skill requirements by occupations, incorporating consistent and comparable skills data across the province.
2. Recommending highly demanded occupations for inclusion of green skills; also suggesting green and environmental skills for various occupations.

3. Skilled workforce demand and supply mapping for each division, district of Sindh, and identify trades where there is a shortage or surplus of skilled workforce. It also provides skill forecast and skill projections for identified growth sectors for 03 years.
4. Suggesting highly demanded occupations for disadvantaged groups (i.e. women, transgender, elderly, minorities, etc.) that should be introduced in TVET institutes.
5. Identifying employment trends in each district, including the types of jobs that are available, the industries that are growing, and the sectors that are experiencing a decline in employment.
6. Making recommendations for TVET policymakers and implementers based on crucial findings. These recommendations are practical and actionable and aim to address the challenges identified in the labour market assessment and skills mapping.

Table 1: Mapping of Objectives to Evaluation Questions, Indicators, and Data Sources

Objective	Evaluation Questions	Key Indicators and Evidence
1. Assess key sectors and occupation-wise skill requirements	Which sectors dominate employer-reported demand in Sindh? Which occupations recur across high-demand districts? How do employers assess candidates during recruitment?	Sector concentration: Manufacturing accounts for 95,675 positions (28.8%), Allied Health for 40,545 (12.2%), and Agribusiness for 39,037 (11.8%). Together, these sectors represent 52.8% of total demand. Second-tier sectors: Construction 25,001 (7.5%); Energy 22,784 (6.9%); Information Technology 21,579 (6.5%); Water 20,970 (6.3%); and Hospitality & Tourism 20,649 (6.2%). District occupation patterns: Karachi Central is led by industrial automation, medical technology, and digital occupations; Keamari by computer, stitching, CNC, accounting, and technician roles; and Malir by packaging, paper-cutting, electrical, and software-related occupations. Recruitment assessment: Interviews are used by 44.6% of employers and practical skills assessments by 36.9%. Graduate satisfaction: 69% of employers are satisfied or very satisfied with TVET graduates.
2. Recommend high-demand occupations for green skills and define relevant green competencies	Which high-demand occupations have strong green-skills relevance? Which green competencies do employers prioritise? How should green skills be incorporated into TVET programmes?	Priority competencies: Environmental compliance 21.8%; waste management and recycling 21.2%; energy-efficiency practices 21.2%; and sustainable manufacturing processes 20.3%. Preferred integration approaches: Practical training in green technologies 27.5%; integration of green concepts across relevant trades 25.7%; collaboration with environmental organisations 23.3%; and dedicated green courses or modules 22.6%. Programme implication: Green competencies should be embedded within high-demand mainstream trades rather than delivered only through stand-alone green programmes.
3. Map demand–supply shortages, surpluses, and the three-year labour-market outlook	Where are the largest occupational and district-level shortages and surpluses? Which trades require expansion or rationalisation? What is the short-term outlook for growth sectors?	Major occupational shortages: Software Technician 26,865; Computer Operator 14,290; CNC Machine Operator 8,468; Accountant 8,356; Sales & Marketing 7,405; Construction Worker 7,255; Nurse 7,230; Quality Assurance 5,889; Sewing Machine Operator 5,013; Packaging Machine Operation 4,457; Cane Inspector 4,242; Industrial Stitching Machine Operator 3,991; and Plumbing & Pipe Fitter 3,939. Key surpluses: DIT 3,420; Information Technology/CIT 2,432; Fashion Design & Dress Making 1,660; Graphic Designer 1,276; and Web Designing & Development 956. Geographic concentration: Karachi districts account for 226,898 positions, equivalent to approximately 68.4% of total demand. Demand–supply trend: Demand increased from 185,268 in 2018 to 331,652 in 2025, while supply increased only from 65,412 to 72,746. Unmet demand consequently rose from 64.7% to 78.1%, indicating sustained pressure in high-growth sectors.
4. Suggest suitable occupations and enabling measures for disadvantaged groups	Which occupations are feasible for disadvantaged groups? What constraints affect their recruitment and retention? Which enabling measures are required?	Reserved employment opportunities: A total of 8,150 reserved positions, including 5,918 for minorities (72.6%), 1,651 for transgender persons (20.3%), 448 for persons with disabilities (5.5%), and 133 for elderly persons (1.6%). Common occupations: Computer Operator 114; Machine Operator 102; Skilled Workers 90; Support Staff 90; and Sweeper 84. Barriers affecting women: Workplace culture and policies 29.9%; safety and security concerns 25.2%; occupational

Objective	Evaluation Questions	Key Indicators and Evidence
		segregation and gender stereotypes 21.9%; and lack of flexible work arrangements 20.5%. Barriers affecting disadvantaged groups: Accessibility constraints 32.4%; limited training opportunities 26.0%; lack of suitable workplace accommodations 23.4%; and social stigma or discrimination 17.8%.
5. Identify district-level employment trends and occupational patterns	Which districts account for the highest employment demand? Which occupations dominate high-demand districts? Where are the secondary employment hubs?	Highest-demand districts: Karachi Central 52,428; Keamari 42,260; Malir 41,697; Korangi 32,607; and Karachi West 31,296. Other Karachi districts: Karachi South 15,261 and Karachi East 11,349. Secondary employment hubs: Hyderabad 10,003; Khairpur 9,847; Larkana 9,323; Sukkur 8,680; Sanghar 8,272; and Shikarpur 8,153. Occupational differentiation: Karachi Central has strong demand for industrial automation, medical technology, and software occupations; Keamari for computer, stitching, CNC, accounting, and technician roles; and Malir for packaging, electrical, paper-cutting, and software-related occupations.
6. Provide actionable recommendations for TVET policymakers and implementers	Which courses should be introduced, revised, expanded, or reduced? Which measures would strengthen programme relevance, employment placement, industry collaboration, and inclusion?	Priorities for programme quality: Curriculum alignment with industry needs 32.3%; greater practical and hands-on training 24.6%; improved instructor quality 22.0%; and increased internships and industry exposure 20.4%. Priorities for system reform: Increased funding and infrastructure investment 31%; better coordination between industry, government, and training providers 26%; stronger quality assurance and accreditation 23%; and promotion of TVET as a viable career pathway 20%. Industry-TVET collaboration: Internships and apprenticeships 31%; industry participation in curriculum design and review 26%; formal advisory boards 22%; and industry contributions through equipment, instructors, or financing 21%. Institutional readiness: Only 32.4% of firms have a dedicated training department, while approximately 20% report having a formal employee training programme.

This assessment identifies priority growth sectors, occupation-specific skill requirements, and trade-specific shortages and surpluses in Sindh and connects these to actionable pathways for improving course relevance, seat allocation, and employer engagement. The findings confirm that demand is concentrated within a strategic set of sectors led by manufacturing, allied health, and agribusiness with secondary demand identified in construction, energy, information technology, water, hospitality and tourism, and ports and logistics. This concentration provides a robust basis for prioritizing programmes that directly respond to actual employer hiring needs in the context of Sindh province.

The scope of this study includes employer reported demand, skill needs, and hiring practices, complemented by training supply information where available and a targeted desk review for scoping the sector and its districts-wide dynamics. Within this scope, the assessment also examines employer hiring and competency checks to strengthen the link between training outcomes and recruitment practices.

Geographically, the assessment covers Sindh at provincial level with disaggregation by division and district where data allows, including separate profiling for Karachi districts. The district distribution shows demand is highly clustered in Karachi, with Karachi districts accounting for 68.4% of total demand. This has direct implications for TVET delivery planning, meaning the province requires a scale strategy for Karachi alongside targeted interventions in secondary hubs such as Hyderabad, Khairpur, Larkana, and Sukkur where meaningful demand also exists.

The analytical scope of the assessment includes trade-wise demand-supply mapping to identify shortages and surpluses and to guide programme adjustment in Sindh. Updated evidence shows severe shortages in key occupations such as Software Technician (26,865), Computer Operator (14,290), Construction Worker (7,255), and Nurse (7,230), alongside substantial deficits in specific machine-related and production trades including CNC Machine Operator (8,468), Sewing Machine Operator (5,013), Lathe Machine Operator (4,782), Packaging Machine Operation (4,457), and Industrial Stitching Machine Operator (3,991). By contrast, selected general programmes such as

Diploma in IT (DIT) (3,420) and Certificate in IT (CIT) (2,432) are in surplus supply, pointing to misallocation of training seats rather than only an overall shortage of training provision. This evidence is used to support short-term prioritisation of high-deficit trades and rationalisation of surplus programmes. The assessment also includes a three-year outlook for priority growth sectors, grounded in employer hiring expectations and sector trend evidence from the desk review, so that recommendations remain forward-looking and policy relevant.

The assessment integrates a strong inclusion and green skills lens because employer demand and training outcomes are shaped by workplace constraints and transition requirements. Employers report barriers to hiring and retaining women that are led by workplace culture and policies at 29.9% and safety and security at 25.2%, while barriers to hiring disadvantaged groups are led by accessibility constraints at 32.4% and inadequate training opportunities at 26.0%. These constraints are treated as part of the labour market reality that TVET interventions must respond to through course design, placement support, and employer engagement. On green skills, employers prioritise environmental compliance at 21.8%, waste management at 21.2%, and energy efficiency at 21.2%, and prefer integration through practical green training at 27.5% and embedding green competencies across relevant trades at 25.7%. Digital readiness is also incorporated as a research question, with most employers rating digital and ICT skills at least moderately important and 54.7% rating them very important or extremely important, which supports mainstreaming applied digital modules across priority trades.

The scope of this study is set to support decisions on which courses to introduce, revise, scale up, or scale down, priorities of districts for investment, and structure industry collaboration through internships, apprenticeships, and curriculum participation. The report outputs are useful for planning, including sector and district demand profiles, trade-wise gap mapping, priority lists for green skills inclusion, feasible occupations for disadvantaged groups, and an action-oriented set of recommendations linked to empirical evidence from the survey.

2. METHODOLOGICAL FRAMEWORK

This chapter outlines the methodological framework employed to conduct the Sindh Labour Market Assessment and to generate the empirical evidence-based results presented in Chapter 4. It presents the research design, data collection and analysis procedures, survey methodology, and the sampling approach, including sample size allocation across districts and sectors. Furthermore, the chapter documents the quality assurance measures implemented during fieldwork and analysis to ensure the consistency, reliability, and comparability of findings across the province.

2.1. Research Design

This assessment employed a cross-sectional, descriptive research design to establish an evidence base for skilled workforce demand, skill requirements, and TVET alignment in Sindh. The design is diagnostic in nature, specifically structured to identify sector and district priorities, occupation-specific skill needs, and actionable pathways for improving TVET planning and delivery. Importantly, the framework is not intended as a causal impact evaluation; instead, it utilizes structured employer responses to capture hiring demand, skill preferences, and recruitment constraints. These demand signals are then triangulated with available training supply data to identify critical shortages and surpluses.

The primary unit of analysis is the establishment, with respondents drawn from decision makers who have direct knowledge of workforce requirements and hiring practices, to capture demand by sector, district, occupation and trade categories, and where relevant by skill level. This structure enables comparable reporting across districts and supports aggregation to provincial level estimates. The design also incorporates an inclusion lens to examine constraints and opportunities for women and disadvantaged groups, and a green skills lens to identify high demand occupations with green relevance and the environmental competencies employers prioritise.

To ensure the research design directly supports the intended outputs, the survey instruments and analytical framework were aligned to the research priorities of this study to produce the core indicators reported in the findings chapter. These include sector-wise demand patterns, district-wise demand distribution, trade-wise demand and supply gaps, employer hiring and competency assessment practices, availability of workplace training, green skills priorities and integration preferences, and constraints affecting the hiring and retention of women and disadvantaged groups. The design thus enables TVET stakeholders to use the results for course prioritisation, seat allocation, curriculum alignment, and strengthening industry collaboration mechanisms such as internships, apprenticeships, and participation in standards and curriculum review.

2.2. Data Collection and Analysis

Data collection for the Sindh Labour Market Assessment combines a province wide Employer Skills Survey with a structured desk review, anchored to the official establishment frame from the Pakistan Bureau of Statistics Economic Census 2023. The sampling universe covers formal and semi formal establishments across all districts of Sindh. Use of the census frame enables stratification by district and by the study sector taxonomy and supports selection probabilities that reflect the distribution of active establishments. Within each district and sector stratum, primary sampling units were drawn from identifiable clusters and economic nodes to ensure coverage of major employment centres and value-chain locations. These nodes included ports and trade catchments in Karachi South and Keamari, the Port Qasim area, the Korangi and Landhi industrial belts, Nooriabad in Jamshoro, and coastal fisheries nodes in Thatta Sujawal and Badin. This approach preserved the economic weight of Karachi while applying allocation floors for non-Karachi districts so that coastal and agrarian belts remained visible in the estimates.

Primary data were collected through a structured Employer Skills Survey administered to respondents with direct responsibility for hiring and workforce planning, including owners' managers and human resource and operations leads. The instruments captured establishment characteristics, current employment and demand by occupation and trade, recent hiring and separations, vacancies and time to hire, hard to fill roles, competency requirements including green and digital skills, workplace training practices, preferences for certification, and employer expectations over a three-year period. Relevance of questions was ensured across manufacturing allied health, agribusiness,

construction, energy, information technology, water, hospitality and tourism, ports and logistics, and other services. Where an enterprise operated multiple plants or distinct units, interviews were conducted at the appropriate unit level to avoid masking within firm variation in staffing and skill needs.

Survey tools were programmed for tablet-based data capture with built in validations to improve data quality. These included skip logic range checks and mandatory fields for key variables. GPS and timestamp tagging supported field monitoring and audit trails. Prior to full rollout, a pilot was conducted in selected urban and coastal districts to test question comprehension interview duration and routing logic and to refine the instrument and field protocols based on observed issues.

Quality assurance protocols were applied throughout fieldwork and data management. Field supervisors conducted spot checks and back checks (follow-up verification) on a random 10% of completed interviews and verified location data against known clusters and estate catchments. Daily monitoring focused on identifying outliers in interview duration item nonresponse and patterned answers. The data management team ran consistency checks to reconcile total headcount with occupational and trade breakdowns and to flag improbable ratios such as vacancies exceeding total employment in micro establishments. These procedures were designed to strengthen internal consistency and comparability across districts and sectors.

Data management and statistical analysis were conducted using IBM SPSS Statistics and MS Excel, with each software used for distinct analytical purposes. After field data cleaning, validation, and coding, the dataset was imported into SPSS for statistical processing. SPSS was used to generate frequency distributions, descriptive statistics, cross-tabulations, and percentage distributions across key variables, including establishment size, sector, district, occupation, gender, training availability, recruitment practices, employer satisfaction, and skill demand categories. For Section C of the questionnaire, which included several multiple-response questions, SPSS multiple response sets were created to analyse questions where respondents could select more than one option, such as training types, skills assessment methods, green skills priorities, challenges in hiring women and disadvantaged groups, workforce development constraints, and suggestions for improving TVET programmes.

Cross-tabulations were conducted to examine patterns across districts, sectors, enterprise size, and selected respondent categories. Where required, weighted analysis was applied to reflect the sampling design and to support province-level estimates while retaining district- and sector-level disaggregation. SPSS was also used for consistency checks, including validation of coded responses, identification of missing or outlier values, verification of multiple-response coding, and review of internal consistency across employment, demand, and occupation-related variables. MS Excel was used primarily for data visualization, formatting of tables and charts, and presentation of final outputs. This combined approach ensured that the analysis was not limited to simple charting, but was based on structured statistical processing, multiple-response analysis, and systematic disaggregation relevant to the objectives of the Labour Market Assessment.

2.3. Literature Review

This literature review draws on official statistics, TVET policy documents, labour market diagnostics, and international guidance to identify what must be measured at establishment level to understand occupation-wise demand, competency requirements, recruitment constraints, workplace training practices, green and digital skill needs, and inclusion barriers. This approach is consistent with labour market assessment practice, where employer surveys are used to translate economic and sector trends into measurable training and workforce planning requirements (International Labour Organization [ILO], 2018; ILO, 2024).

A first design implication from the literature is that Pakistan's enterprise structure is dominated by micro and small establishments, where hiring decisions are often made by owners or managers rather than formal human resource departments. The Pakistan Economic Census 2023 provides the most relevant establishment frame for understanding this structure and for informing sampling, stratification, and sector coverage (Pakistan Bureau of Statistics [PBS], 2024). This has direct implications for questionnaire design in Sindh: the instrument must be concise, operationally clear, and suitable for respondents who may not maintain formal HR records. It must therefore capture practical indicators such as current employment, occupation-wise demand, vacancies, recent hiring,

hard-to-fill roles, training practices, and skill requirements in language that employers can answer reliably.

Sindh's labour market cannot be understood through provincial aggregates alone. The province includes a dominant metropolitan and port-industrial economy in Karachi, a secondary industrial and services corridor around Hyderabad and Jamshoro, coastal fisheries and cold-chain activity, agrarian-processing belts, and market-town service economies. Labour market evidence and economic profiling therefore support a district- and cluster-sensitive approach to sampling and analysis (PBS, 2022; Finance Division, 2024). This justifies the use of district stratification, sector quotas, and targeted coverage of industrial belts, ports and logistics clusters, coastal districts, and agrarian districts so that Karachi's economic weight does not mask demand patterns in the rest of Sindh.

Skills mismatch is a central theme in the labour market and TVET literature. ILO guidance defines skills mismatch as a divergence between the skills required by employers and the skills available among workers, and recommends the use of observable indicators such as vacancies, time to fill positions, hard-to-fill occupations, recruitment channels, and employer-reported competency gaps (ILO, 2018; ILO, 2024). For this assessment, this means that demand should not be measured only by asking employers for broad future needs. Instead, the questionnaire must capture concrete demand signals, including occupation-wise vacancies, expected annual demand, reasons for recruitment difficulty, and the specific skills or competencies that employers find lacking.

Sindh-focused skills diagnostics reinforce the need to move beyond broad job titles and to assess competency bundles. Earlier research on skills disconnect in Sindh has highlighted gaps between training provision and employer expectations, particularly in practical exposure, workplace readiness, communication, problem-solving, and job-specific technical skills (British Council Pakistan, 2015; Pakistan Institute of Development Economics [PIDE], 2016). This suggests that a trade-based list alone is insufficient. For example, an employer's demand for a "machine operator" may involve several distinct competency areas, including equipment handling, safety routines, production discipline, basic maintenance, quality checking, and record keeping. The questionnaire therefore needs to collect information on both occupations and the task-level competencies required to perform them effectively.

Pakistan's TVET reform architecture provides the policy basis for translating employer demand into training recommendations. The National Skills Strategy and subsequent competency-based training reforms emphasize demand-driven training, competency-based training and assessment, and stronger linkages between industry and training providers (National Vocational and Technical Training Commission [NAVTTTC], 2009; NAVTTTC, 2020). The National Vocational Qualifications Framework further supports the use of occupational standards, competency units, assessment criteria, and recognition of prior learning as mechanisms for aligning training with labour market needs (NAVTTTC, 2020; NAVTTTC, 2024). For the present assessment, this means employer feedback must be captured in a form that can inform curriculum revision, certification, assessment standards, and institute-level seat planning.

Quality assurance and accreditation are also important in the literature because employer confidence in TVET graduates depends on whether certification reliably signals workplace competence. The accreditation and quality assurance framework for TVET institutions emphasizes standards related to training delivery, assessment, institutional capacity, and programme quality (NAVTTTC/NAC-TVS, 2022). This supports the inclusion of survey questions on employer satisfaction with TVET graduates, preferred assessment methods during recruitment, willingness to participate in curriculum review, and perceptions of practical training quality. These indicators help determine whether the issue is only insufficient training supply or also weak credibility and relevance of existing programmes.

Firm-level training behaviour is another important evidence stream. International labour market literature shows that firms respond to skills shortages through on-the-job training, apprenticeships, informal mentoring, short courses, and internal upgrading, especially where external training supply is weak or mismatched (ILO, 2018; ILO, 2024). In Pakistan's TVET reform context, employer engagement and workplace learning are repeatedly emphasized as mechanisms for improving the employability of graduates and reducing the gap between training and job performance (GIZ, 2023; NAVTTTC, 2020). The questionnaire therefore includes items on whether employers offer training, what types of training they provide, whether they have dedicated learning and development

functions, and what support is needed to expand apprenticeships, internships, and employer-led training.

Green skills literature provides a further rationale for the assessment. ILO guidance on green skills and greening TVET emphasizes that environmental competencies are often embedded within existing occupations rather than appearing only as separate “green jobs” (ILO, 2019; ILO, 2022). For example, energy efficiency practices may be required in electrical and maintenance trades, waste management in manufacturing and municipal services, safe chemical handling in industrial production, and resource efficiency in construction and utilities. This is particularly relevant for Sindh’s industrial, construction, energy, water, fisheries, and agri-processing sectors. The questionnaire therefore measures both the green skill areas employers consider relevant and the ways in which they prefer such competencies to be introduced into TVET programmes.

Digital skills are also increasingly treated in the literature as cross-cutting workforce requirements rather than only ICT-sector skills. Pakistan’s digital policy direction highlights the role of digital capabilities in productivity, services, innovation, and workforce transformation (Ministry of Information Technology and Telecommunication, 2018). In employer-facing skills assessments, this requires measuring not only demand for IT occupations but also the importance of digital competencies across sectors, including workplace software use, digital communication, data handling, inventory systems, customer relationship management, and emerging technologies. The questionnaire therefore includes digital/ICT skills, emerging technologies, and future digital skill demand as specific areas of inquiry.

The inclusion literature shows that labour market participation is shaped not only by skills supply but also by workplace conditions, mobility, social norms, and employer practices. Gender diagnostics for Pakistan highlight persistent barriers to women’s labour force participation, including mobility constraints, workplace safety, care responsibilities, and discriminatory norms (UN Women, 2024). Legal and policy frameworks, including the Protection Against Harassment of Women at the Workplace Act, provide an institutional basis for assessing whether workplaces are enabling and safe for women (Government of Pakistan, 2010). For this assessment, it is therefore important to measure employer-reported constraints in hiring and retaining women, including workplace culture, flexible arrangements, occupational segregation, and safety concerns.

The same logic applies to disadvantaged groups, including persons with disabilities, transgender persons, minorities, and elderly workers. Inclusion-oriented TVET literature stresses that access to training and employment depends on reasonable accommodation, accessible infrastructure, non-discrimination, and practical pathways into occupations with realistic placement potential (Alla Mensah, 2021; Ministry of Human Rights, 2021; United Nations, 2023). The questionnaire therefore captures employer quotas or reserved seats, occupations considered feasible for disadvantaged groups, and barriers such as accessibility, lack of supportive policies, inadequate training opportunities, and social stigma. This allows the assessment to recommend not only courses but also enabling measures required for inclusive employment.

Overall, the literature supports a questionnaire and analytical framework that is establishment-aware, sector-sensitive, competency-based, and inclusion-focused. It confirms the need to collect employer demand at the level of occupations, trades, competencies, recruitment constraints, training practices, and workplace readiness. It also supports the integration of green and digital skills as cross-cutting competency areas across priority trades. Finally, the literature demonstrates that TVET recommendations should be based not only on the volume of labour demand but also on the credibility of certification, the quality of practical training, the strength of employer engagement, and the feasibility of inclusion for women and disadvantaged groups.

Table 2: Evidence Themes Translated into Questionnaire Constructs

Evidence Theme from Literature	Why It Matters for Sindh Labour Market Assessment	Questionnaire Constructs to Capture
Establishment structure dominated by micro and small firms, as reflected in PBS establishment data	Many employers, especially small establishments, do not have formal HR systems, written job descriptions, or structured workforce planning records. The questionnaire therefore needs to be simple, practical, and answerable by owners, managers, and operations staff.	Enterprise size, ownership, location, respondent role, current employment, full-time and part-time workers, occupation-wise headcount, and basic hiring information

Evidence Theme from Literature	Why It Matters for Sindh Labour Market Assessment	Questionnaire Constructs to Capture
District and cluster-based labour market structure	Labour demand differs across Karachi's port-industrial economy, Hyderabad–Jamshoro corridor, North Sindh logistics belt, coastal fisheries/cold-chain areas, and agrarian market-town districts. This requires district-wise and sector-wise data rather than only provincial aggregates.	District, industrial zone, sector/economic activity, sector-specific occupations, district-wise demand, and cluster-relevant skill needs
Sector-specific production functions and compliance requirements	Skills demand is shaped by the nature of work in each sector, such as machine operation in manufacturing, documentation in logistics, hygiene in allied health, safety in construction, and cold-chain handling in fisheries.	Sector module selection, occupation/trade lists, task-based competency prompts, compliance-related skills, safety requirements, and quality-related skills
Skills mismatch measured through observable labour market signals	Reliable skills-gap analysis requires concrete evidence of demand and shortage rather than aspirational responses. Observable indicators help identify where training supply is insufficient or misaligned.	Occupation-wise demand, annual workforce requirement, vacancies, hard-to-fill occupations, recruitment difficulties, demand-supply gaps, and shortage/surplus trades
Competency-based training and qualification frameworks	Pakistan's TVET reform agenda emphasizes competency-based training, assessment, occupational standards, certification, and recognition of prior learning. Employer data must therefore be linked to competencies, not only job titles.	Required competencies, preferred qualifications/certification, practical assessment methods, recognition of prior learning, and employer interest in curriculum/standards review
Employer recruitment and assessment practices	Employer confidence in TVET graduates depends on whether graduates can demonstrate job-ready competence during recruitment and probation.	Methods used to assess job candidates, including interviews, practical skills assessments, written tests, group activities, and other screening methods
Firm-based training as a response to skills gaps	Many firms manage skill shortages internally through on-the-job training, informal mentoring, apprenticeships, and short practical courses. Understanding these practices is essential for designing employer-led training models.	Availability of training programmes, types of training offered, in-house practical training, apprenticeships, online courses, dedicated training/L&D department, and constraints to workplace training
Green skills as cross-cutting tasks within mainstream occupations	Green skills are often embedded in existing trades rather than limited to separate green jobs. This is relevant for construction, energy, manufacturing, water, logistics, fisheries, and agri-processing.	Energy efficiency practices, waste management and recycling, environmental compliance, sustainable manufacturing, practical green technologies, and preferred methods for integrating green skills into TVET
Digital and ICT skills as cross-sector workforce requirements	Digital skills are increasingly required across sectors, including manufacturing, retail, logistics, health, services, and ICT. They should be measured as both sector-specific and cross-cutting competencies.	Importance of digital/ICT skills, emerging technology needs, AI, robotics, 3D printing, digital marketing, CRM, computer operations, future digital skills, and suggestions for digital skills development
Inclusion barriers as workplace and access constraints	Women, persons with disabilities, transgender persons, minorities, and elderly workers face constraints related to workplace culture, safety, transport, accessibility, discrimination, and limited training pathways.	Quotas/reserved seats, occupations suitable for disadvantaged groups, barriers to hiring and retention, workplace accessibility, reasonable accommodation, flexible work arrangements, safety/security concerns, and inclusion support needs
Industry–TVET collaboration as a mechanism for relevance	Stronger employer involvement is needed to align curricula, practical training, assessment, internships, apprenticeships, and competency standards with actual labour market demand.	Employer willingness to participate in curriculum/standards review, preferred collaboration models, internships, apprenticeships, industry advisory boards, equipment/instructor support, and feedback on TVET quality

The evidence themes presented in Table 2 show how the literature review was translated into research questions and measurable variables for the Sindh Labour Market Assessment. These constructs ensured that the survey did not only collect general employer opinions, but also captured structured information on workforce demand, skills mismatch, competency requirements, workplace training, green and digital skills, inclusion barriers, and industry–TVET collaboration. Table 3 further

identifies the key source categories that informed these design choices and explains how each evidence stream supported the final survey structure and analytical framework.

Table 3: Key Sources Used and the Design Choices They Support

Source Category	Illustrative Sources Cited in This Section	Design Choices Supported
Official establishment and labour market statistics	Pakistan Bureau of Statistics Economic Census 2023; Pakistan Labour Force Survey 2020–21; Pakistan Population and Housing Census 2023	Use of establishment-based sampling frame, district and sector stratification, enterprise-size categories, labour market indicators, and province/district-level disaggregation
National economic and macro-labour context	Pakistan Economic Survey 2023–24; Finance Division, Government of Pakistan	Framing of economic sectors, employment context, macroeconomic trends, and broader labour market conditions influencing skills demand
Pakistan TVET policy and reform framework	NAVTTTC National Skills Strategy 2009–2013; National Vocational Qualifications Framework; CBT&A implementation guidance; NVQF Manual Version III	Competency-based questionnaire design, link between employer demand and occupational standards, certification, assessment, recognition of prior learning, and curriculum relevance
TVET accreditation and quality assurance sources	NAVTTTC/NAC-TVS accreditation and quality assurance framework	Inclusion of questions on employer satisfaction, quality of TVET graduates, assessment methods, trainer quality, institutional credibility, and programme relevance
Sindh-focused skills diagnostics	British Council Pakistan, <i>The Skill Disconnect in Sindh</i> ; PIDE skill disconnect evidence	Use of competency bundles, employer-centred gap framing, practical skills focus, workplace readiness indicators, and sector/cluster-sensitive interpretation
International skills mismatch and labour market measurement guidance	ILO guidance on skills mismatch measurement; ILO skills anticipation and matching resources	Use of demand-side indicators such as vacancies, hard-to-fill occupations, recruitment channels, reasons for hiring difficulty, and demand-supply gap analysis
Green skills and just transition guidance	ILO, <i>Skills for a Greener Future</i> ; ILO, <i>Greening TVET and Skills Development</i> practical guide	Integration of green skills as task-based competencies within mainstream trades; inclusion of energy efficiency, waste management, environmental compliance, and green technology training
Digital policy and technology transformation sources	Digital Pakistan Policy, Ministry of Information Technology and Telecommunication	Inclusion of ICT skills, digital workplace skills, emerging technologies, AI, robotics, 3D printing, and future digital skill requirements
Gender and inclusion evidence	UN Women National Gender Assessment Report; Protection Against Harassment of Women at the Workplace Act; Ministry of Human Rights policy sources; UNESCO-UNEVOC disadvantaged youth evidence; United Nations reporting on transgender rights	Inclusion of workplace safety, flexible arrangements, accessibility, reasonable accommodation, anti-discrimination, employer openness, and feasible occupations for women and disadvantaged groups
Programme and development partner sources	GIZ TVET Reform Support Programme; TVET Sector Support Programme materials; ILO Decent Work Country Programme for Pakistan 2023–2027	Alignment with demand-driven TVET reform, employer engagement, inclusion, decent work, apprenticeship pathways, and system-level coordination between government, industry, and training providers

2.4. Survey Method

The Sindh Labour Market Assessment was conducted through a structured Employer Skills Survey administered to key decision-makers, including owners, senior managers, HR personnel, and operations leads. The survey was designed to generate occupation-wise demand signals, including current employment, vacancies, hiring difficulty, and near-term hiring intentions. It also captured competency requirements across technical, digital, and green skill areas, as well as workplace training practices such as in-house training, apprenticeships, and certification preferences. This structure follows standard employer survey logic that links workforce planning, hard-to-fill roles, and competency gaps with actionable training design and qualification alignment (International Labour Organization, 2020).

Data were collected using tablet-based tools with embedded validation controls, including skip logic, range checks, mandatory fields, and GPS and timestamp tagging to support an auditable data trail. Supervisory protocols included daily dashboard review, spot checks, back-checks on a defined subsample, and consistency scripts to reconcile occupational breakdowns with total headcount and to flag improbable patterns. These controls are aligned with established survey quality assurance practices for establishment-based surveys (Lohr, 2010; Kish, 1965).

The sampling universe was anchored to the official establishment frame drawn from the Pakistan Bureau of Statistics Economic Census 2023, which provided the baseline for establishment counts and distribution across geography and industry. The frame was supplemented through field listing in sampled clusters to capture eligible active establishments in fast-changing industrial belts, commercial nodes, and sector-specific locations.

2.5. Sampling Technique

The study uses a two-stage stratified random sampling design to balance statistical precision with field practicality across Sindh's diverse districts and economic zones. Districts are treated as the primary strata to ensure complete geographic coverage and to prevent Karachi's establishment concentration from dominating provincial estimates. Within districts, sector quotas have been applied so that priority industries remain analytically visible, and in large districts with multiple industrial belts, district by sector may be treated as explicit strata to keep data representative across belts.

At stage one, primary sampling units were selected using probability proportional to size, where the size measure is the number of establishments in the EA on the frame. Primary sampling units are area clusters that match context, including enumeration blocks or municipal blocks in dense urban settings and village clusters or comparable administrative units in peri urban and rural settings. At stage two a rapid listing of eligible establishments was conducted inside each selected primary sampling unit, tagging each establishment by sector, and selecting a fixed number of establishments using simple random sampling or systematic sampling from the ordered list. This approach is standard for establishment surveys because it reduces travel cost while retaining defensible inference when weights are correctly specified (Cochran, 1977; Kish, 1965; Lohr, 2010).

Survey weights are computed as the inverse of the selection probabilities at each stage and adjusted for nonresponse within district and sector cells. Where primary sampling units are selected with probability proportional to establishment counts and a fixed number of establishments is selected per primary sampling unit, base weights are typically near constant within the strata, simplifying analysis and improving stability. Post stratification adjustments may be applied to align weighted totals with trusted control totals from the frame, particularly at the district by sector level.

To maintain the planned level of precision, the survey design incorporated an adjustment for expected non-response and ineligible cases. Based on field access arrangements, official coordination, and planned quality assurance protocols, an expected response rate of 98.7% was applied in the final sample size calculation. During implementation, field teams followed structured access, replacement, and verification procedures to minimise non-response and ensure that completed interviews remained aligned with the approved district and sector allocations.

2.6. Sample Size

2.6.1. Sample Size Determination

The sample size for the Employer Skills Survey was determined using the standard single-proportion sample size formula. The calculation was based on a 95% confidence level, a planning proportion of 0.50, and a margin of error of approximately $\pm 3\%$. The use of $p = 0.50$ was appropriate because it provides the maximum variance and therefore produces a conservative sample size when the true population proportion is unknown.

The base sample size was calculated as follows:

$$n_0 = \frac{Z^2 \cdot p(1 - p)}{e^2}$$

Where:

- $Z = 1.96$ (95% confidence level)
- $p = 0.50$ (assumed population proportion)
- $e = 0.03$ (margin of error)

$$n_0 = \frac{1.96^2 \cdot 0.50 \cdot (1 - 0.50)}{0.03^2} \approx 1,067$$

Design Effect Adjustment: To account for clustering and stratification across districts and economic sectors, the base sample size was adjusted using a design effect (DEFF)¹ of 2.54:

$$n_1 = n_0 \times \text{DEFF} = 1,067 \times 2.54 = 2,710.18$$

Non-Response Adjustment: Factoring in an anticipated response rate of 98.7% to cover non-response and ineligible cases:

$$\text{Adjusted Sample} = \frac{n_1}{\text{Response Rate}} = \frac{2,710.18}{0.987} = 2,746$$

Operational Sample Allocation: To ensure minimum representation for smaller districts, satisfy sector-specific allocation needs, and account for rounding, the final operational sample size was established at 2,792 completed establishment interviews. This sample size ensures comprehensive geographic and sectoral coverage across Sindh while maintaining the desired statistical precision for the Labour Market Assessment.

2.6.2. Allocation across Strata, Clusters, and Operational Choices

For this assessment, districts served as the primary strata, represented in accordance with the Economic Census 2023 establishment frame. The final sample of 2,792 completed establishment interviews was distributed across districts using a proportional allocation approach based on the number of establishments and the relative economic weight of each district. To safeguard the reliability of district-level interpretation, smaller non-Karachi districts were assigned minimum allocation floors so that all districts remained visible in the analysis and the findings were not dominated by Karachi's establishment concentration.

The district-level sample allocation followed the principle below:

District sample allocation = District share of establishments × Total sample size

Where required, proportional allocations were adjusted through minimum district floors and rounding so that the final total remained 2,792 completed interviews. This ensured that high-establishment districts received appropriate sample weight, while smaller districts retained sufficient representation for meaningful analysis.

Within each district, Primary Sampling Units (PSUs) were defined as area-based clusters. In dense urban locations, these included enumeration blocks, municipal wards, industrial estates, commercial centres, and other identifiable economic clusters. In peri-urban and rural locations, PSUs included village clusters, union council areas, market towns, and sector-specific economic nodes. PSUs were selected using probability proportional to size (PPS), with the number of establishments used as the size measure.

After PSU selection, field teams conducted a listing of eligible establishments within each selected cluster. Establishments were then classified by sector and selected using simple random or systematic sampling from the ordered list. This two-stage approach balanced statistical representativeness with field practicality, reduced travel time, and ensured coverage of major employment centres, industrial belts, coastal value-chain locations, agrarian market towns, and service-sector clusters.

Final survey weights were computed using the stage-wise selection probabilities and adjusted for non-response within district and sector cells where required. Post-stratification adjustments were applied, where necessary, to align the sample with district-level establishment totals from the

¹ Design Effect (DEFF): The ratio of variance under a complex survey design (clustering/stratification) to the variance of a simple random sample of equal size. A DEFF >1 indicates that a larger sample size is necessary to achieve the target precision.

Economic Census 2023. This allocation approach ensured that the survey results could support both province-level interpretation and district- and sector-level analysis relevant to TVET planning.

2.6.3. Sectoral Distribution of Sample

The sectoral distribution indicates that Agri, Livestock (327), Wholesale, Retail & Repair (293), and Allied Health Services (211) constitute the primary pillars of the sample, closely followed by Construction (214). This concentration reflects the core drivers of employment within Sindh's economy. Key established sectors also maintain a significant presence, notably Textile & Garments (168), Hospitality & Tourism (168), and Chemicals & Pharmaceuticals (171).

Growth-oriented and infrastructure-related segments, such as Water supply, sewerage, and waste management (165), Energy (148), and Maritime, Ports & Logistics (128), represent an increasing focus on essential services and industrial modernization. Meanwhile, specialized or emerging niche sectors including Fisheries (62), Sports Goods & Equipment (27), and Mining and Quarrying (12) represent critical location-specific economic activities that contribute to the diversity of the province's labour market.

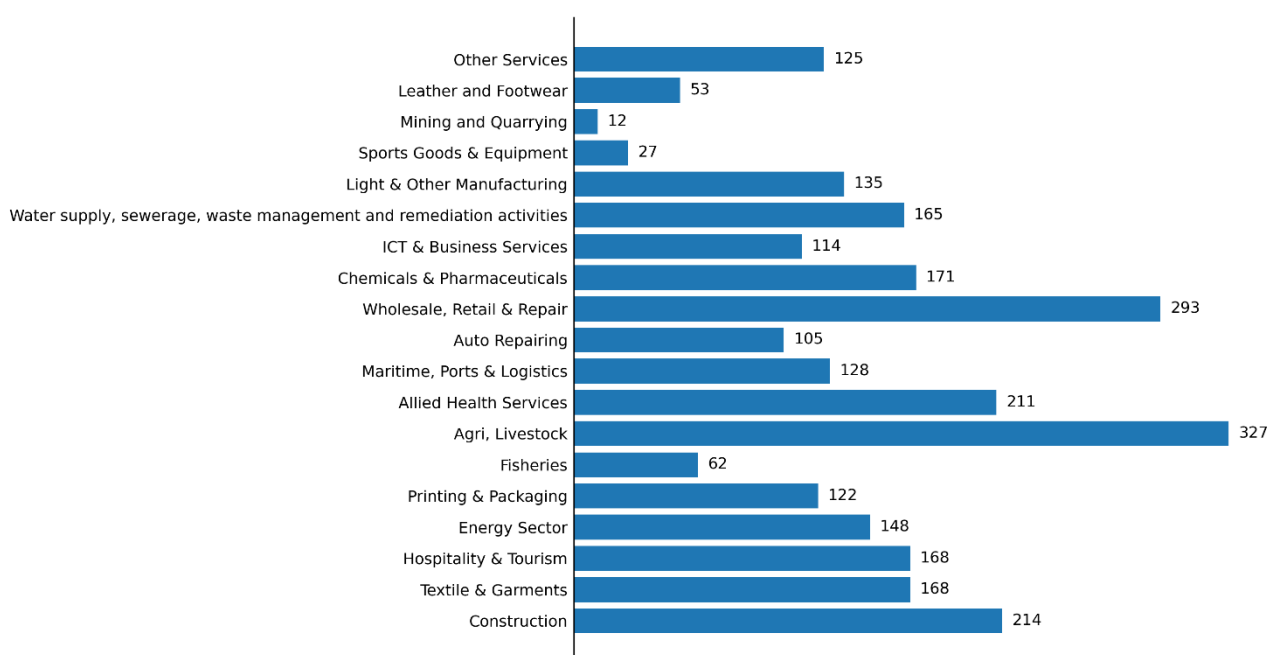


Figure 1: Total Sample Size at Sector Level

2.6.4. Sample Distribution by District

The district-wise distribution reflects a comprehensive provincial coverage, with a strategic focus on the primary economic engines of Sindh. The sample is heavily concentrated in the Karachi metropolitan area, which accounts for approximately 53% of the total sample (1,443 completions). This urban cluster is led by Korangi (223), Karachi South (217), and Karachi East (204), representing the highest density of industrial and commercial activity.

Hyderabad (149) serves as the secondary urban hub, followed by significant representation from Northern Sindh's key cities, including Sukkur (82), Khairpur (74), and Larkana (71). Furthermore, the assessment ensures geographic diversity by including interior and coastal districts such as Tharparkar (71), Shaheed Benazirabad (68), and Sanghar (67). This broad representation across all 30 administrative units ensures that the findings capture diverse workforce types and economic zones, from specialized mining in the south-east to traditional agriculture in the central plains.

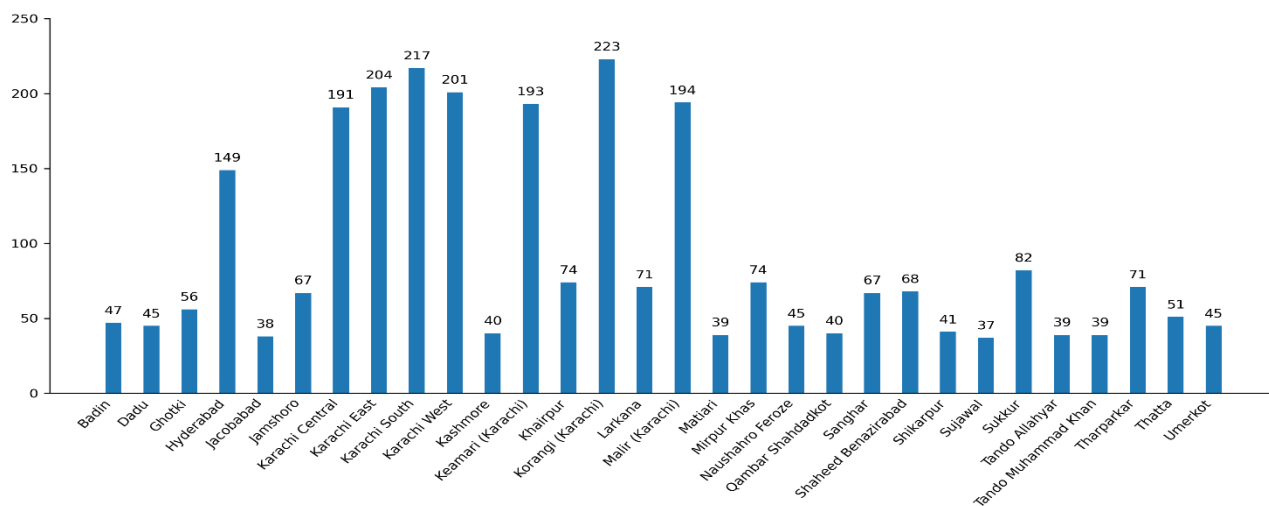


Figure 2: Total Sample Size at District Level

3. SECONDARY DATA ANALYSIS FOR DISTRICT PROFILING

3.1. Regional Overview

This chapter establishes the empirical baseline for district profiling in Sindh and translates that baseline into survey design implications. In addition to descriptive reporting, the analysis provides evidence that informs (i) district stratification, (ii) identification of sector clusters requiring explicit sampling visibility beyond proportional allocation, and (iii) specification of competency bundles captured through sector-specific modules in the employer questionnaire.

Sindh's labour market is shaped by a strong metropolitan centre and multiple secondary economic corridors, each with distinct establishment structures, occupational compositions, and constraints on labour supply participation. District profiling therefore has two analytical tasks. First, it maps where establishments and jobs are predominantly found, to prevent provincial estimates from being dominated by Karachi and to ensure smaller but policy-relevant clusters remain statistically visible. Second, it documents sectoral and occupational patterns that determine what the instrument must measure, including recruitment channels, vacancy duration, skills mismatches, training practices, and compliance-relevant competencies (for example, safety procedures in ports and logistics, or basic quality documentation in food handling and processing). Pakistan's official statistical system, including the Labour Force Survey and the latest census architecture, provides the most defensible basis for these baseline indicators (Pakistan Bureau of Statistics, 2022 & 2023).

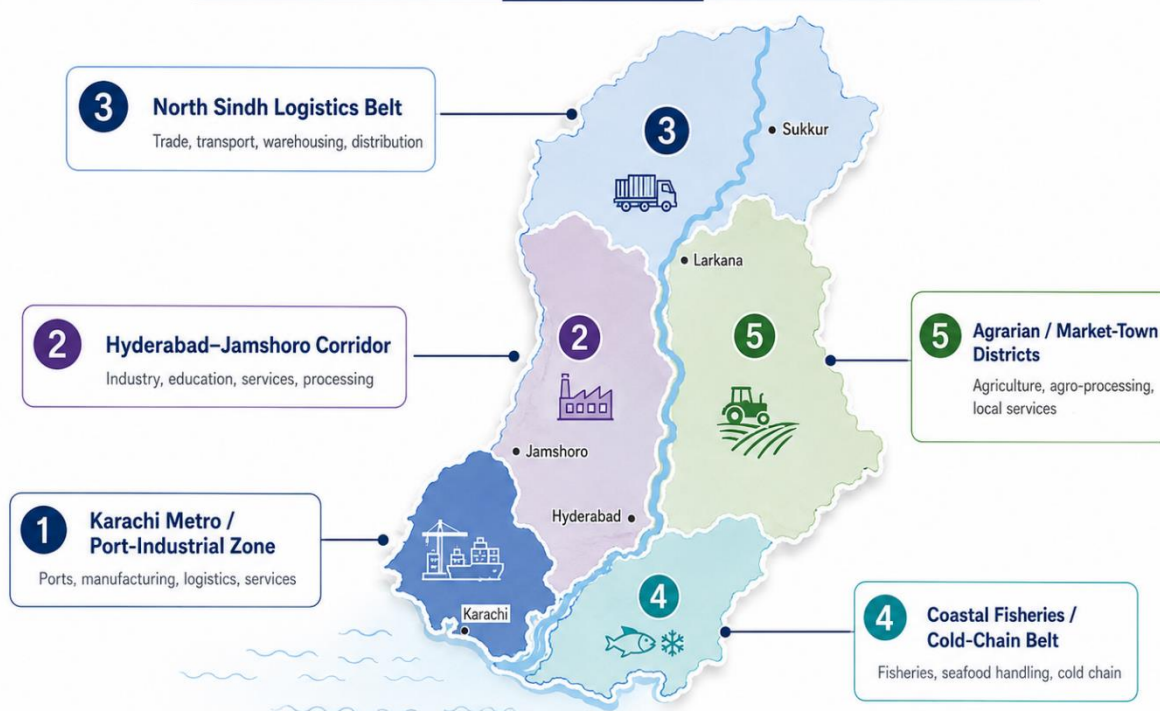
The secondary-data analysis links Sindh's spatial economy, including the Karachi metropolitan core, the Hyderabad-Jamshoro corridor, the North Sindh logistics belt, the coastal fisheries belt, and agrarian-processing districts, to labour-market structures and observed skills-demand patterns, which are examined alongside findings from the primary data.

3.1.1. Geographical and Socio-Economic Context of Sindh

Sindh occupies a strategic position as Pakistan's principal maritime gateway and as a major agrarian and processing province anchored around the Indus corridor. The province's economic geography combines a dense coastal-metropolitan economy led by Karachi, an inland industrial and services corridor centred on Hyderabad and Jamshoro, a northern logistics and distribution belt around Sukkur, coastal fisheries and cold-chain districts, and agrarian and market-town districts along the Indus and eastern belt. This geography matters because labour markets in Sindh are not uniform: the nature of establishments, production systems, and hiring practices varies sharply by corridor, and these variations directly affect how employers articulate skills demand and how workers access employment.

To illustrate this spatial-economic structure more clearly, Figure 3 presents Sindh's five conceptual labour market zones used for analytical framing in this assessment.

Sindh's Five Labour Market Zones



Note: Conceptual labour market zones for analytical framing; not administrative boundaries.

Figure 3: Sindh's Five Labour Market Zones²

The Karachi metropolitan region is shaped by port-linked logistics, wholesale and retail trading ecosystems, industrial estates, and a large services base. Port operations, warehousing, freight forwarding, and customs-related services form a distinct employment and skills cluster tied to Karachi Port and Port Qasim catchments. These activities operate within a legal and regulatory environment that historically enabled export-oriented and trade-intensive operations through dedicated regimes, including the EPZ framework and investment facilitation instruments (Export Processing Zones Authority, 1980; Government of Pakistan, 2012).

Outside Karachi, the Hyderabad-Jamshoro axis represents a secondary industrial and services corridor connected to Karachi markets through major transport routes and industrial estates. The corridor typically combines light manufacturing, packaging, construction-related services, and urban services expansion. From a survey standpoint, this corridor is important because its employer structure differs from Karachi's, with a greater share of mid-sized firms and mixed manufacturing-service establishments, and because it often functions as a recruitment alternative for semi-skilled labour that would otherwise seek employment in Karachi. District profiling therefore treats this belt as a separate economic space requiring explicit representation in the sample design, rather than allowing the Karachi concentration to define "urban Sindh" by default.

North Sindh districts anchored around Sukkur serve as a connectivity and distribution belt between upper Sindh and other provinces. Logistics intermediation, wholesale trade, warehousing, construction, and public service employment tend to be more visible in these districts as compared with high-end export-linked manufacturing. These characteristics translate into a skills demand profile centred on drivers and fleet-related skills, inventory handling, market-based retail competencies, and multi-tasking roles in smaller establishments. District profiling must therefore distinguish between port-linked logistics in Karachi catchments and inter-district logistics and trade in the Sukkur belt, because the required skill bundles and training pathways are not identical.

² Note: These zones are conceptual labour market zones used for analytical framing and do not represent formal administrative boundaries.

The coastal belt districts, including Thatta, Sujawal, and Badin, combine fisheries, cold-chain nodes, and coastal agriculture, with spillovers into processing, transport, and construction. Labour demand here is typically organized around seasonal and value-chain-specific tasks, including handling, grading, icing, cold-room operations, transport scheduling, and basic equipment upkeep. For sampling and instrument design, the implication is straightforward: coastal districts must remain visible through allocation floors and targeted PSUs, and the questionnaire must capture practical competencies in cold-chain handling, basic quality and hygiene routines, and safety practices relevant to wet and cold environments.

Finally, agrarian districts along the Indus and in the eastern belt combine crop and livestock production with first-stage processing and market-town services. These districts are usually characterized by mixed livelihood patterns where households may combine farm work, casual non-farm employment, and microenterprise activity. For the employer survey, the critical linkage is that agrarian and processing establishments often have informal hiring practices and limited structured training, so the questionnaire must be designed to measure informal skills acquisition, learning-by-doing, and the extent to which employers rely on prior experience versus certified training.

3.1.2. Population Demographics and Settlement Patterns

Sindh's settlement pattern is dominated by a major metropolitan core and a network of secondary cities and market towns embedded within agrarian districts. This spatial structure is central to labour-market interpretation because urban districts exhibit higher establishment density, more diversified occupations, and more formal employment relationships, while rural and peri-urban districts typically display greater reliance on informal work arrangements and seasonal cycles. The latest population and housing census architecture provides the baseline for district-level population distributions and for distinguishing dense urban blocks from rural settlement clusters in field operations (Pakistan Bureau of Statistics, 2023).

From a labour market assessment perspective, three demographic features are operationally decisive:

- First, urban concentration implies that labour demand and training supply are disproportionately located in Karachi and a small number of secondary cities. If sampling relies only on proportional allocation, Karachi's establishment mass can mechanically dominate estimates of skills demand, masking constraints and mismatches in coastal and agrarian districts. District profiling therefore justifies the use of minimum allocation floors and cluster targeting in non-Karachi districts to preserve analytical visibility.
- Second, participation gaps by gender are a defining feature of Sindh's labour market baseline. The desk-review context indicators already point to a large difference between male and female refined participation rates. This is not only a diagnostic statistic; it determines field protocols and subgroup quotas because the assessment must capture employer-side constraints and enabling practices related to women's employment, including shift structures, workplace facilities, transport arrangements, and harassment prevention mechanisms. These issues cannot be inferred reliably from establishment characteristics alone and must be measured directly in the employer instrument, with district profiling guiding where women's employment is more feasible and where barriers are likely to be binding.
- Third, settlement heterogeneity within districts matters for cluster design. Large districts contain multiple labour-market "micro-spaces" such as industrial estate catchments, market towns, peri-urban settlement belts, and rural village networks. For survey execution, this justifies treating area-based PSUs as context-matched clusters, rather than forcing a single PSU template across Sindh. District profiling therefore identifies where enumeration blocks and municipal blocks are the correct unit, and where village clusters or union-council aligned units are operationally more appropriate.

3.1.3. Economic Activities and Major Sectors

Sindh's economy combines services, industry, and agriculture in ways that are spatially concentrated and institutionally shaped by its gateway function. Within Sindh, however, the relevant unit of analysis for the labour market assessment is not only "sector shares" at the provincial level, but the district-specific composition of establishments and the cluster logic that determines what employers demand.

For survey instrument design, the sector baseline must accomplish four things.

1. **Confirm which sectors are dominant employers by corridor and district:** Karachi-linked districts require explicit modules for logistics and ports-adjacent services, wholesale and retail ecosystems, industrial manufacturing segments, and services occupations. Non-Karachi districts require stronger visibility for agriculture-linked processing, construction trades, and district-level trade and services.
2. **Translate sector structure into measurable skill bundles:** Sector profiling must identify the competency categories that employers can realistically report, such as equipment operation roles in logistics, preventive maintenance roles in manufacturing and construction, hygiene and quality routines in hospitality and food-linked activities, and retail operations and customer-facing skills in trade.
3. **Align the employer survey with establishment realities:** Many establishments in Sindh, particularly outside Karachi, may not have formal HR functions or written job descriptions. The questionnaire must therefore use task-based and competency-based prompts rather than relying on formal job-title structures. District profiling informs where this approach is essential.
4. **Identify where regulation and compliance affect hiring and training:** In port-adjacent operations, industrial estates, and export-oriented segments, employers often face compliance requirements that shape the training they provide and the skills they demand. These include safety practices and documentation routines that should be measured as part of “skills requirements,” not treated as peripheral governance issues (Export Processing Zones Authority, 1980; Government of Pakistan, 2012).

To operationalize this baseline across subsequent sections of the chapter, district profiling draws on Labour Force Survey data and census frameworks to describe employment structures, combined with desk-review evidence on industrial estates, cluster locations, and economic corridors. This approach identifies districts requiring explicit stratification or quota layering. The resulting analytical mapping of Sindh’s labour market is directly applied to sampling design and to the specification of sector modules in the questionnaire.

Table 4: Sindh-Specific Socio-Economic and Labour Market Indicators³

Indicator	Sindh-Specific Value / Reference
Crude labour-force participation rate, total	30.90%
Refined labour-force participation rate, total	43.80%
Refined labour-force participation rate, male	70.20%
Refined labour-force participation rate, female	15.20%
Unemployment rate, total	3.90%
Key legal and institutional frameworks relevant to industrial and skills clusters	EPZ framework (1980); SEZ legal regime (2012); Sindh TEVTA Act, 2009

3.2. Sectoral Skill Gap Analysis

Sindh’s labour demand is structured around a metropolitan core anchored by Karachi’s port, industrial, and services ecosystem, secondary corridors around Hyderabad-Jamshoro–Nooriabad, a northern logistics and market belt anchored by Sukkur, and coastal and agrarian value chains along Thatta-Sujawal-Badin and interior districts. This geography necessitates that “skills gaps” should not be treated as generic shortages. They are best interpreted as sector-specific capability gaps that manifest in hiring frictions, quality losses, downtime, compliance exposure, and safety risks, with measurable consequences for firm performance and workforce inclusion (Finance Division, 2024; PBS, 2022).

³ Source note: Baseline labour market indicators draw on PBS labour statistics, while the institutional and cluster framing draws on the EPZ framework, SEZ legal regime, and the Sindh TEVTA Act. National wage, national sector employment share, national youth unemployment, and national GDP growth indicators have been removed from this table to retain a Sindh-specific focus.

3.2.1. Analytical Approach used in this Assessment

The assessment frames skill gaps across four linked layers:

1. **Task and technical competence:** ability to execute core job tasks to required standards, including tool use, machine operation, and basic diagnostics.
2. **Process and quality discipline:** adherence to standard operating procedures, inspection routines, workflow control, and defect prevention.
3. **Compliance and safety capability:** occupational health and safety routines, regulated practices, documentation, and incident prevention, particularly where special regimes or buyer compliance apply (for example EPZ and SEZ-linked operations).
4. **Digital and green add-ons:** incremental competencies embedded in existing occupations, such as basic digital systems use (ticketing, WMS, EMR, POS) and energy-efficiency, solar-ready, and safe handling practices, consistent with how labour markets typically absorb environmental transition requirements (ILO, 2019; NAVTTC, 2020).

This structure is deliberately aligned with competency and qualification language so that findings can be translated into modular training packages and certification pathways under national frameworks rather than remaining descriptive (NAVTTC, 2020; NAVTTC/NAC-TVS, 2022).

3.2.2. Sector Synthesis: Demand Signals and Competency Implications

Construction: The assessment indicates that construction-related demand is concentrated in Karachi, Hyderabad, and Sukkur, particularly around building works, civil works, plumbing, electrical support, masonry, and site supervision. The key skill gaps are visible in workmanship consistency, site safety routines, task sequencing, and the application of energy-efficient and solar-ready installation practices that increasingly intersect with building systems (NAVTTC, 2018a; MoHR, 2021).

Textiles and Garments: The findings show that textile and garment-related demand is concentrated in Karachi's industrial belts and secondary stitching clusters. The main competency gaps are visible in line discipline, defect control, stitching quality, machine handling, production efficiency, and preventive maintenance. The sector also requires stronger compliance awareness in areas such as safe chemical handling, effluent-related routines, and workplace safety (NAVTTC/NAC-TVS, 2022).

Hospitality and Tourism: The assessment indicates that hospitality and tourism demand is linked to Karachi's hospitality economy as well as selected coastal and heritage locations. Skill gaps are visible in customer service, food safety routines, front-office operations, housekeeping standards, service workflows, and the use of basic digital systems for booking, billing, and customer handling (Ministry of Education, 2018).

Renewable and Alternative Energy: The findings show growing technician-level demand in solar installation, commissioning, electrical work, diagnostics, and operations and maintenance. The key gaps are visible in safe work practices, system troubleshooting, documentation, equipment handling, and consistent application of technical standards across varied field sites. These gaps reflect the wider green transition and the need to embed renewable energy competencies within electrical and maintenance-related training pathways (Ministry of Climate Change, 2021; NAVTTC, 2020).

Printing and Packaging: The assessment indicates that printing and packaging demand is closely tied to Karachi's industrial base and the Hyderabad–Nooriabad corridor. The key competency gaps are visible in machine setup, press and finishing operations, workflow discipline, packaging quality, inspection routines, and documentation readiness, particularly where firms supply regulated or quality-sensitive buyers (NAVTTC/NAC-TVS, 2022).

Agriculture, Livestock, and Fisheries Value Chains: The findings show that coastal and agrarian districts require practical competencies linked to post-harvest handling, cold-chain operations, grading, hygiene, livestock support, fisheries handling, and basic equipment maintenance. Gaps are visible in spoilage reduction, cold-room handling, quality routines, safe handling practices, and maintenance of small processing and storage equipment (Ministry of Climate Change, 2021).

Allied Health Services: The assessment indicates strong demand for mid-level health and diagnostic support roles, particularly nurses, medical technicians, laboratory technicians, X-ray technicians, and support staff. The main competency gaps are visible in infection prevention, sample

handling, sterilization routines, patient support, basic documentation, and use of digital record systems in support functions (NAVTTTC/NAC-TVS, 2022).

Maritime, Ports, and Logistics: The findings show that Karachi South, Keamari, and Port Qasim-linked catchments generate demand for equipment operators, warehouse staff, documentation roles, terminal support workers, and compliance-oriented functions. The key gaps are visible in dangerous goods handling, yard and terminal safety, warehouse systems discipline, inventory handling, documentation accuracy, and logistics workflow management (BOI, 2016; Government of Pakistan, 1980).

Wholesale, Retail, and Repair: The assessment indicates that wholesale, retail, and repair remain widespread and labour-intensive across Sindh, with strong relevance for youth employment. Gaps are visible in customer service, inventory discipline, POS and stock management, basic digital transactions, workplace communication, and evolving repair competencies linked to new equipment and vehicle technologies (PBS, 2022).

Chemicals and Pharmaceuticals: The findings show that chemicals and pharmaceuticals are concentrated in Karachi's industrial belts and require strong process discipline and compliance-oriented skills. Competency gaps are visible in documentation culture, GMP-adjacent routines, QA/QC support, utilities operations and maintenance, safe chemical handling, and process safety awareness (BOI, 2016; NAVTTTC/NAC-TVS, 2022).

ICT and Business Services: The assessment indicates that Karachi and Hyderabad's service ecosystems generate demand for IT support, operations support, BPO, customer service, CRM, and digital workflow roles. The key gaps are visible in ticketing discipline, troubleshooting, customer handling, cybersecurity hygiene, workplace software use, and applied digital problem-solving, consistent with the Digital Pakistan policy direction (MoITT, 2018).

Public Administration and Education: The findings show that public administration and education-related establishments create demand for institutional support roles, including lab support, AV and ICT support, records management, office support, and facility maintenance. These roles are important for the functioning of training institutions and public service delivery systems (NAVTTTC, 2020).

Light and Other Manufacturing and Sports Goods: The assessment indicates that Nooriabad and Karachi-based estates sustain demand in fabrication, machining support, plastics, assembly, finishing, and export-linked production tasks. Gaps are visible in machine discipline, measurement and metrology basics, preventive maintenance, quality checking, and workplace safety routines (NAVTTTC/NAC-TVS, 2022).

3.3. Labour Market Analysis for Selected Districts

This section provides a concise district-level labour-market snapshot to justify (i) which districts are treated as high-visibility strata and (ii) which sector modules must be prioritized in the employer survey. It summarizes the structural logic from secondary evidence; the survey outputs remain the primary source of district-level estimates (PBS, 2022; Finance Division, 2024).

3.3.1. District Grouping Logic for Sindh

Sindh's labour market is best interpreted as a set of economic zones rather than as uniform district averages. For profiling and instrument planning, districts are grouped into five functional zones aligned with the sector taxonomy and cluster geography already established in the desk review:

1. Karachi metropolitan and port-industrial catchments,
2. Hyderabad Jamshoro Nooriabad corridor,
3. North Sindh services and logistics belt,
4. Coastal belt (fisheries, cold chain, and distributed energy services), and
5. Agrarian and market-town districts with processing and large informal services bases (EPZA, 1980; BOI, 2016; NAVTTTC, 2020; Finance Division, 2024).

This zoning is used only to guide sampling visibility and questionnaire modularization; it does not assume district outcomes.

3.3.2. District Profiling Focus Areas

To align it with primary survey findings, district profiling is confined to four analytical elements that are measured directly through the survey and validated empirically:

- **Sector presence and cluster location:** Identification of sectors with material presence in each zone that warrant minimum sample visibility (BOI, 2016; EPZA, 1980).
- **Occupation families covered by modules:** Broad occupation families captured through survey modules, rather than numeric demand projections (NAVTTTC, 2020).
- **Skills and compliance context:** Presence of standards and compliance regimes shaping competency-related questions (e.g., safety, documentation, and quality systems), without estimating compliance rates (NAVTTTC/NAC-TVS, 2022).
- **Inclusion feasibility:** Assessment of sectors serving as plausible entry points for women, youth, persons with disabilities, and transgender persons, used to justify subgroup quota design rather than to report participation levels (MoHR, 2021; Ministry of Education, 2018).

Table 5: Selected District Zones and What they Imply for Survey Module Coverage

Zone for profiling	Indicative districts	Primary reason for inclusion in profiling	Survey implications
Karachi metro and port-industrial catchments	Karachi South, Keamari, Korangi, Karachi East, Malir, Bin Qasim	Ports, logistics, SEZ/EPZ-linked industrial belts, dense services and manufacturing clusters (EPZA, 1980; BOI, 2016)	Treat as explicit high-visibility strata with sector quotas; ensure coverage of ports-logistics, chemicals-pharma, packaging, wholesale-retail, and ICT modules
Hyderabad Jamshoro corridor	Hyderabad, Jamshoro, Nooriabad catchment	Industrial corridor linked to Karachi markets and estates; mixed manufacturing and services (Finance Division, 2024)	Ensure adequate visibility for light manufacturing, packaging, and renewables module items
North Sindh belt	Sukkur, Khairpur, Larkana, Ghotki	Corridor logistics, wholesale and services hubs, and public services spillovers (Finance Division, 2024)	Ensure coverage for wholesale-retail-repair and inland logistics questions
Coastal belt	Thatta, Sujawal, Badin	Fisheries value chain, cold chain, coastal construction, and distributed energy services (Ministry of Climate Change, 2021)	Ensure fisheries and cold-chain competency items are included where sampled; include renewables tasks where relevant
Agrarian and processing districts	Sanghar, Mirpur Khas, Naushahro Feroze, Tharparkar	Agriculture and processing with large market-town services base (PBS, 2022)	Ensure agriculture-livestock-fisheries and retail-service modules are fielded where sampled

Table 6: Occupational Coverage Map for Questionnaire Modules (District-Neutral)

Sector module	Core occupation families to be captured	Competency themes to be asked uniformly across districts
Maritime, Ports and Logistics	Equipment operators, warehouse staff, documentation roles	Safety practices, documentation workflows, inventory and warehouse routines
Chemicals and Pharmaceuticals	Production support, QA/QC support, utilities and maintenance	SOP discipline, basic quality-system awareness, utilities O and M tasks
Printing and Packaging	Pre-press, press, finishing, QA support	Workflow discipline, machine handling routines, basic QA and safety
Renewable and Alternative Energy	PV installation and service roles, electrical maintenance	Installation steps, commissioning and O and M routines, electrical safety
Wholesale, Retail and Repair	Sales, stock control, basic repair roles	POS and inventory routines, customer service, basic diagnostics where relevant

Sector module	Core occupation families to be captured	Competency themes to be asked uniformly across districts
Agriculture, Livestock and Fisheries	Farm and processing support, grading and handling roles	Post-harvest handling, cold chain basics where present, hygiene and safety
Construction	Trade roles and supervised helpers	Task proficiency, site safety routines, work organization and supervision
ICT and Business Services	Support and operations roles	Ticketing routines, basic troubleshooting, digital productivity practices
Allied Health Services	Technician aides and records roles	Hygiene protocols, documentation and records routines, service workflows

3.4. Gender and Inclusion Considerations

Gender and inclusion have been embedded as design conditions within the Sindh Labour Market Assessment, reflected in labour demand measurement, skills-gap interpretation, and the feasibility of training and employment pathways. If employer demand is documented without an inclusion lens, the assessment risks producing a skills roadmap that is technically correct for the median firm but operationally irrelevant for groups that face structural barriers to entry, retention, and progression. The desk review therefore treats inclusion as a cross-cutting axis to be measured consistently across sectors and districts, and to be carried through into sampling, questionnaire modules, and analysis.

Sindh’s inclusion challenge stems from a dual labour market: formal metropolitan Karachi and informal, mobility-constrained non-metropolitan districts. In this context, women’s economic participation is constrained by a number of demand-side and supply-side factors that employers routinely cite: limited safe transport, concerns about shift timing, workplace facilities, and risk of harassment, plus limitations arising from unpaid care responsibilities that reduce availability for full-time or late-hour work. These constraints are consistently highlighted in national gender diagnostics and sectoral evidence on commuting and workplace safety, making it essential that the assessment measures not only “whether women are hired” but also the enabling conditions that determine whether hiring is sustained. (UN Women, 2024; International Labour Organization, n.d.).

For employers, gender inclusion is also a compliance and governance issue. Firms that operate in larger value chains, regulated industrial estates, export-linked activities, and public-facing services face reputational and contractual expectations around workplace conduct and grievance handling. Pakistan’s legal framework on workplace harassment provides an anchoring reference point for what employers are expected to have in place, including complaint mechanisms and organizational responsibility, and it is directly relevant to survey modules that measure institutional readiness for inclusive hiring. (Government of Pakistan, 2010).

Youth inclusion is best treated as a transition problem rather than a headcount problem. Employers in Sindh often recruit young workers into entry roles in retail, logistics, construction support, hospitality, and basic production lines, but the progression path is frequently weak: limited structured training, informal supervision, and high churn when wages fail to compensate for transport costs and irregular hours. For the assessment, the key is to measure the “conversion rate” from entry roles to skilled roles through employer-provided training, apprenticeship style arrangements, and recognition of prior learning. This keeps the section aligned with what the survey can credibly measure, and avoids speculative claims that would later conflict with field findings. (Finance Division, 2024).

Persons with disabilities require a practical measurement approach anchored in job design and workplace accommodation rather than broad statements about participation. In employer settings, inclusion depends on whether tasks can be reconfigured, whether physical access exists, whether assistive arrangements are feasible, and whether supervisors have basic capacity to manage accommodations without productivity loss. The assessment should therefore treat disability inclusion as an employer capability indicator: (i) accessibility of premises and facilities, (ii) reasonable accommodation practices, (iii) role suitability and task modification, and (iv) non-discrimination in recruitment and retention policies. This approach avoids over-claiming on prevalence and keeps the analysis consistent with establishment-level data.

Transgender inclusion is structurally important in Sindh because discrimination can exclude an entire segment of potential workers from mainstream hiring channels even when skills exist. Evidence from UN system reporting emphasizes that stigma and violence risks translate into labour-market exclusion, with predictable consequences for occupational sorting into insecure and informal work. For this assessment, the operational requirement is to measure employer openness and safeguards, not to assume participation rates. Survey modules should therefore include (i) equal opportunity policies that explicitly cover gender identity, (ii) workplace conduct and grievance mechanisms, and (iii) any prior experience hiring or accommodating transgender workers in customer-facing and back-office roles. (United Nations, 2023).

3.5. TVET and Skills Development Landscape

Sindh's skills system sits at the intersection of a national standards regime and a provincial delivery architecture. At the national level, the core direction of reform is the move from time-bound, content-based training toward competency-based training and assessment, with formal qualification levels, defined occupational standards, and pathways that include Recognition of Prior Learning (RPL) for workers trained informally (National Vocational and Technical Training Commission [NAVTTTC], 2020). In Sindh, this national logic is operationalized through the province's TVET governance and institute network, which is mandated to plan, regulate, and coordinate technical and vocational provision in line with provincial priorities while interfacing with national certification and quality assurance requirements (Government of Sindh, 2010).

3.5.1. TVET Landscape in Sindh: Institutions, Delivery Channels, and Reform Direction

For district profiling and the employer survey, the relevant point is not simply “how many institutes exist” but how the system behaves as a supply chain for skills: how qualifications are defined, how training quality is assured, and how employers interact with providers. The NVQF provides the national backbone for this supply chain by defining qualifications as learning outcomes linked to industry-defined competencies; it also formalizes progression routes, equivalences, and credit concepts, and explicitly positions RPL as a mechanism to certify informally acquired skills in an economy where informal training is widespread (NAVTTTC, 2020).

Sindh's TVET provision is therefore best understood as a mixed ecosystem: public institutes operating under provincial oversight, private providers delivering market-facing short courses, and firm-based training (including informal apprenticeships and structured on-the-job learning). The strategic challenge, consistently highlighted in Pakistan's TVET reform discourse, is to make these channels legible to employers through credible assessment and certification, and to align course content with task profiles that dominate Sindh's cluster economy (for example, port and logistics compliance workflows in Karachi's port districts; GMP-adjacent documentation and utilities roles in pharma and chemicals belts; solar installation and operations and maintenance roles in the Hyderabad-Jamshoro corridor and coastal districts) (NAVTTTC, 2020).

Support from the development partners has been oriented toward precisely this institutional modernization: strengthening competency standards, improving governance and delivery quality, and strengthening public-private coordination so that training responds to employer demand rather than supply-side inertia (GIZ, 2023). This is material for the survey instrument because it justifies modules on employer engagement with training providers, perceived credibility of certificates, and the extent to which firms rely on in-house training versus external TVET institutes.

3.5.2. Challenges in the TVET Sector

The binding constraint for Sindh is not “absence of training,” but the credibility and relevance of training relative to employer task demands. Evidence from Sindh-focused diagnostic work has emphasized the “disconnect” between what training institutions produce and what firms can immediately use, with mismatches in curriculum relevance, practical exposure, and signalling value of qualifications in hiring (British Council Pakistan, 2015). This disconnect tends to be amplified in fast-changing sectors (ICT and business services; renewable energy services; modern warehousing and fulfilment) and in compliance-sensitive segments (ports and logistics safety regimes; pharma

documentation and clean practices; chemical handling and environmental controls), where employers require both technical skills and procedural discipline.

From a system design standpoint, the NVQF reform logic directly responds to these issues by tying qualifications to occupational standards, requiring competency-based assessment, and embedding monitoring and evaluation as a quality assurance function during transition from traditional training models (NAVTTTC, 2020). In practice, however, implementation challenges remain predictable and should be treated explicitly in the report narrative because they shape what the survey must measure:

1. **Weak demand signalling and fragmented employer engagement:** Many firms do not systematically communicate skill specifications to providers, while providers often lack structured mechanisms (sector skills councils, active advisory boards, routine tracer studies) to update programmes. This is why global skills anticipation guidance treats employer-facing instruments (establishment surveys, sector consultations, vacancy analytics, and structured qualitative panels) as essential complements to administrative training data (International Labour Organization [ILO], 2024).
2. **Variable credibility of credentials in hiring:** Where employers do not trust certification as proof of job-ready competence, they substitute probation, informal screening, referrals, or in-house testing. The NVQF's emphasis on competency-based assessment and standardized outcomes is meant to reduce this uncertainty, but the transition period typically produces uneven adoption across regions and provider types (NAVTTTC, 2020).
3. **Practical training constraints and equipment gaps:** Task-intensive sectors (industrial electrical, HVAC, welding, packaging lines, laboratory practices) require hands-on exposure and safe workshop infrastructure. Where training is theory-heavy, employers face higher induction costs and longer time-to-productivity, reinforcing preferences for informal apprenticeships or firm-based training (British Council Pakistan, 2015).
4. **Inclusion barriers as a labour supply constraint:** Participation constraints for women and marginalized groups (mobility, harassment risk, facilities, shift patterns, accessibility) reduce the effective pipeline for sectors that could otherwise absorb them. These constraints are not peripheral; they are labour market frictions that employers also experience as "scarcity", especially in services and health-adjacent roles (Government of Sindh, 2010).

These constraints justify a survey design that measures not only "vacancies and skill gaps" but also the institutional reasons firms do or do not engage with formal TVET, and the specific points at which training fails to translate into employability.

3.5.3. Rationale for Centres of Excellence for Skills Development: Linking Cluster Economies to Credible Skills Pipelines

Within Sindh's spatially concentrated economy, the most defensible TVET strengthening strategy is not uniform expansion, but cluster-linked specialization: centres of excellence (CoEs) that deliver high-credibility training, assessment, and industry-facing services for priority sectors and compliance regimes. Conceptually, CoEs are justified when (i) demand is largely clustered geographically (industrial estates, port catchments, corridor economies), (ii) training requires specialized equipment or regulated practice, and (iii) employer trust depends on standardized assessment and workplace-aligned curricula. The NVQF supports such specialization because it is built around occupational standards, competency units, and structured progression pathways that can be delivered in modular form and assessed consistently (NAVTTTC, 2020).

For Sindh, CoE logic aligns naturally with the province's cluster map already established in the desk review. CoEs should be framed in the report as quality anchors rather than "new institutes": they provide benchmark training delivery, assessor capacity, employer coordination, and demonstration effects that can diffuse into surrounding provider networks. Development partner programming has also emphasized system strengthening and modernization of TVET delivery and governance, which is consistent with the CoE approach when it is embedded in a broader reform programme rather than treated as a stand-alone infrastructure intervention (GIZ, 2023).

4. FINDINGS AND ANALYSIS OF SURVEY DATA

4.1. Size-Wise Distribution of Establishment

Small establishments account for the largest share of the study sample at 62.9%, making them the dominant enterprise size category by a wide margin. Medium establishments represent 13.3%, followed by micro establishments at 12.7% and large establishments at 11.1%. The distribution shows that the sample is strongly concentrated among small establishments, while micro, medium, and large establishments have broadly comparable shares.

This pattern implies that the overall findings largely reflect the hiring practices, workforce demand, and skills constraints of small firms. However, interpretation should also consider differences across medium and large establishments, where formal HR systems, structured training arrangements, and occupational specialization are generally more developed.

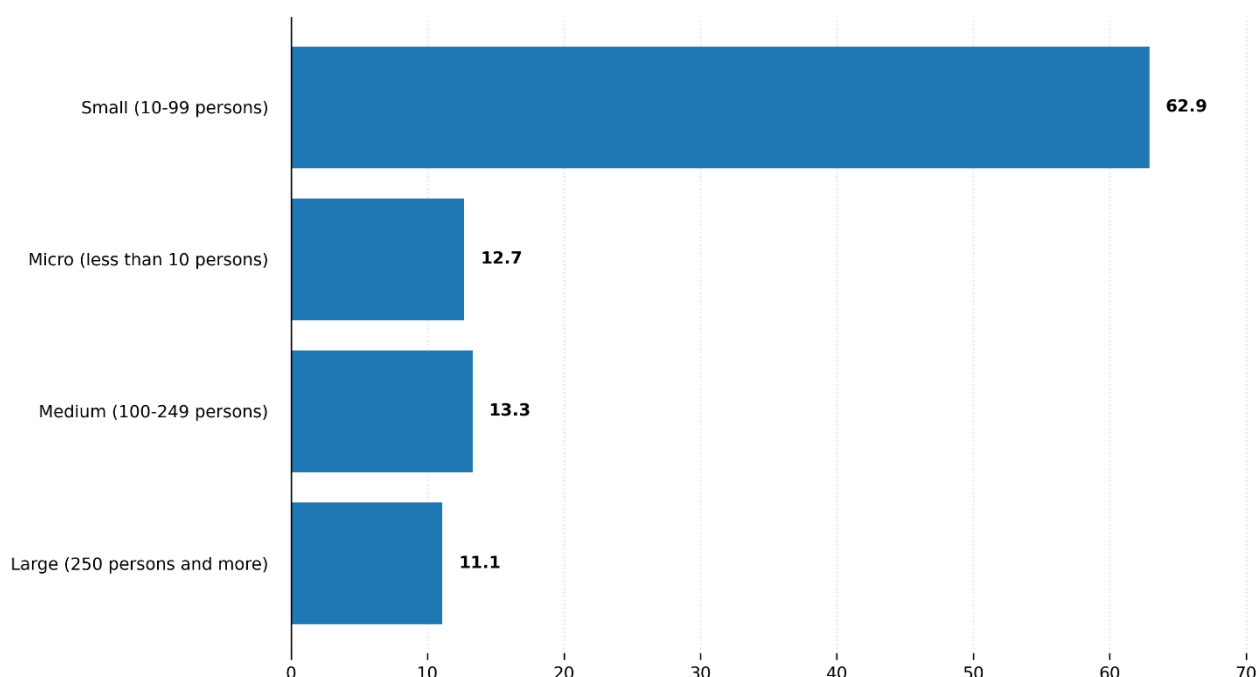


Figure 4: Distribution of Establishments Based on their Size (%)

4.2. Sector-Wise Distribution of Establishments

Manufacturing is the largest sector at 19.7%, followed by Agribusiness at 14.3% and Wholesale, Retail and Repair at 10.3%. A second tier includes Allied Health Services at 8.2% and Construction at 7.8%. Mid-sized shares are reported for Other Services at 6.0%, Hospitality and Tourism at 5.9%, Water Supply, Sewerage and Waste Management at 5.4%, and Energy at 5.1%. Printing and Packaging accounts for 4.6%, while Information Technology and Maritime, Ports and Logistics each account for 4.2%. Automobile Repairing represents 4.0%, while Mining and Quarrying remains marginal at 0.4%. At the establishment level, the sample is intentionally concentrated in manufacturing-linked and market-facing sectors, so interpretation should prioritise these sectors while treating Mining and Quarrying as too small for reliable standalone inference.

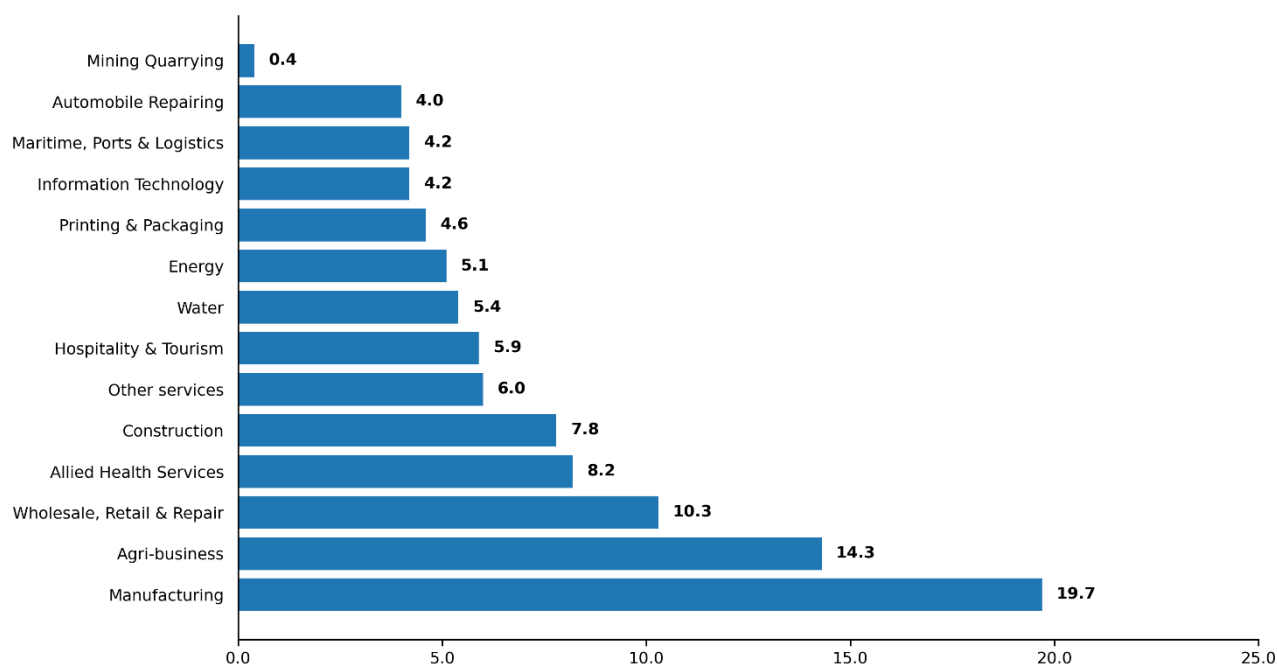


Figure 5: Sector-wise Distribution of Establishments

4.3. Annual Skilled Workforce Demand and Supply

Figure 6 shows a growing mismatch between skilled labour demand and supply in Sindh. Demand increased from 185,268 positions in 2018 to 331,652 in 2025, while supply grew only from 65,412 to 72,746. Consequently, the unmet demand increased from 119,856 (64.7%) to 258,906 (78.1%). The demand–supply ratio worsened from approximately 3:1 to 5:1, indicating that workforce development has not kept pace with labour market expansion. The findings highlight a structural skills gap and emphasize the need for stronger alignment between TVET programmes, industry demand, and emerging sectoral employment opportunities.

To close this gap, interventions should be sequenced around four operational priorities:

- modernize curricula, equipment, and assessment standards to match the dominant sector/trade demand patterns;
- scale short-cycle, competency-based certification and upskilling pathways that expand supply rapidly without compromising quality;
- institutionalize employer-led delivery through apprenticeships, internships, and placement-linked training agreements so that training converts to employability; and
- strengthen provincial labour market intelligence and annual demand forecasting to prevent persistent under-supply in high-absorption technical occupations and over-supply in low-absorption trades.

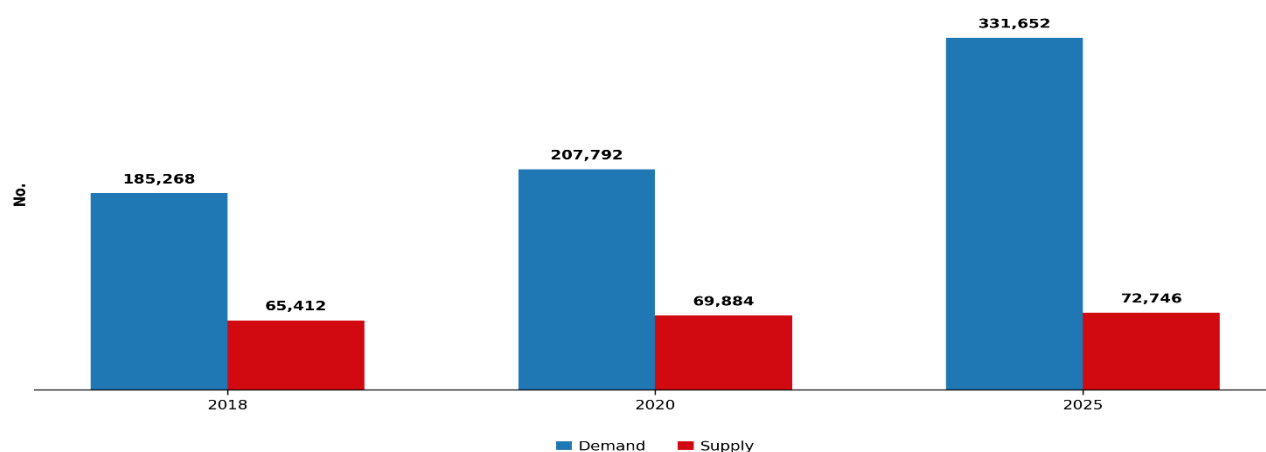


Figure 6: Annual Skilled Workforce Demand and Supply

4.4. Sector-Wise Skilled Workforce Demand

Figure 7 breaks down the annual skilled workforce demand of 331,652 reported in Section 4.3 by sector. The demand profile is highly concentrated in a limited number of sectors. Manufacturing dominates with 95,675 positions, representing 28.8% of total demand, followed by Allied Health with 40,545 positions or 12.2%, and Agribusiness with 39,037 positions or 11.8%. Collectively, these three sectors account for 175,257 positions, or 52.8% of total demand, indicating that Sindh's skills requirements are anchored in industrial production, health-related services, and agri-value chain activity. Adding Construction with 25,001 positions or 7.5% and Energy with 22,784 positions or 6.9% raises the top-five sectors' share to 67.3%, reinforcing the case for prioritising training pipelines in these high-absorption sectors.

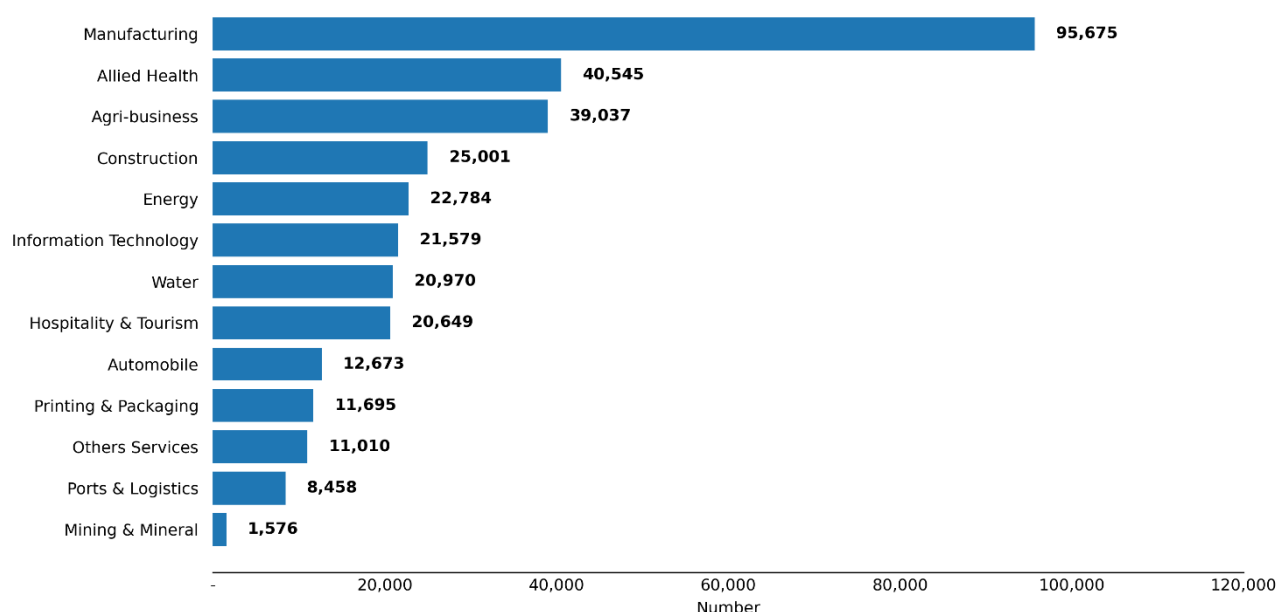


Figure 7: Sector-wise Skilled Workforce Demand in Sindh

A second tier of demand is visible in Information Technology (21,579; 6.5%), Water (20,970; 6.3%), and Hospitality & Tourism (20,649; 6.2%), each representing meaningful, policy-relevant employment volumes that will require targeted occupational pathways. By contrast, relatively smaller demand levels are reported for Automobile (12,673; 3.8%), Printing & Packaging (11,695; 3.5%), Others Services (11,010; 3.3%), Ports & Logistics (8,458; 2.6%), and Mining & Mineral (1,576; 0.5%). This distribution implies that Sindh's TVET response must be designed around scale in the dominant sectors and precision programming in the mid-tier sectors.

4.5. Gender-wise Skilled Workforce Demand at Sectoral Level

Table 7 presents the sector level distribution of skilled workforce demand by gender. Out of the total annual demand of 331,652, male demand is 252,780 which is 76.2% and female demand is 78,872 which is 23.8%. This confirms that the overall demand profile is strongly male dominated, with female demand representing under one quarter of total skilled workforce requirements.

Sector level patterns show substantial variation in female participation. Allied Health has the highest female share, with 17,247 females out of 40,545 total demand which is 42.5%. Information Technology follows with 7,599 females out of 21,579 which is 35.2%, while Manufacturing records 29,501 females out of 95,675 which is 30.8%. In absolute terms, female demand is mainly situated in a small number of sectors. Manufacturing accounts for 29,501 female positions, Allied Health for 17,247, and Information Technology for 7,599. Together these three sectors comprise 68.9% of total female demand. By contrast, female demand remains minimal in Construction at 1,807 out of 25,001 which is 7.2% and Automobile at 255 out of 12,673 which is 2.0%. Mining and Mineral has zero female demand against a total demand of 1,576. This distribution indicates that Sindh gender inclusive skills planning should be designed around scaling female entry pathways where demand already exists at volume, while also addressing binding constraints that keep female participation

negligible in sectors with sizeable overall demand such as Construction, Energy, and Transport linked activities.

Table 7: Gender-wise Skilled Workforce Demand at Sectoral Level

Sector	Male	Female	Total
Manufacturing	66,174	29,501	95,675
Allied Health	23,298	17,247	40,545
Agri-business	33,187	5,850	39,037
Construction	23,194	1,807	25,001
Energy	19,136	3,648	22,784
Information Technology	13,980	7,599	21,579
Water	16,418	4,552	20,970
Hospitality & Tourism	17,034	3,615	20,649
Automobile	12,418	255	12,673
Printing & Packaging	9,628	2,067	11,695
Other Services	9,195	1,815	11,010
Ports & Logistics	7,542	916	8,458
Mining & Mineral	1,576	-	1,576
Grand Total	252,780	78,872	331,652

4.6. Trade-Wise Skilled Workforce Demand

Figure 8 presents the distribution of employer-reported skilled workforce demand across leading occupations in Sindh. The highest demand is observed for Computer Operators (9,944), followed by Stitching Machine Operators (9,056) and CNC Machine Operators (8,468). Other major occupations include Accountants (7,056), Medical Technicians (6,960), Sales and Marketing professionals (6,005), and Nurses (5,748). The demand pattern demonstrates that Sindh's labour market requires a diverse skills portfolio spanning industrial production, healthcare, administration, and commercial services.

Additional high-demand occupations include Electricians (5,207), Civil Technicians (5,049), Mobile App Developers (4,806), Lathe Machine Operators (4,782), Plumbers (4,532), and Packaging Machine Operators (4,456). These trends indicate two key priorities for TVET planning. First, training capacity should be expanded in industrial, construction, and technical trades through updated curricula, modern equipment, and competency-based approaches. Second, digital, healthcare, administrative, and business-oriented programmes should be strengthened to respond to emerging market needs. Greater industry engagement through apprenticeships, internships, and placement partnerships will be essential to improve graduate employability and labour market outcomes.

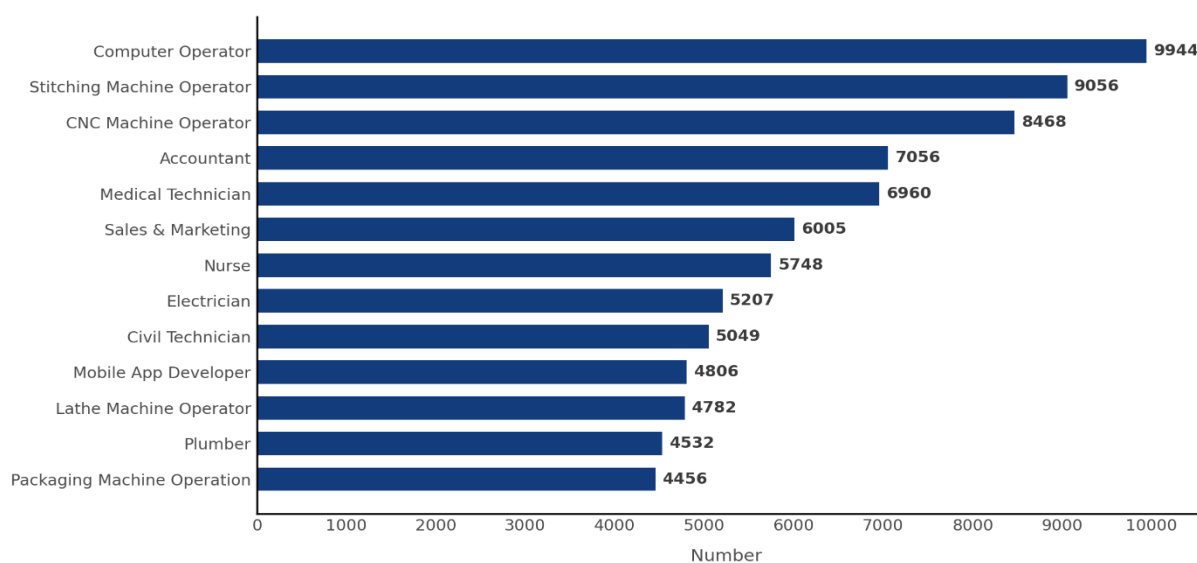


Figure 8: Demand for Leading Trades in Provincial Labour Market

4.7. Gender-Wise Workforce Demand at Occupation Level

Table 8 presents skilled workforce demand by qualification level and gender in Sindh. Total demand is 331,652, of which male demand is 252,780 which is 76.2% and female demand is 78,872 which is 23.8%. Demand is concentrated at Level 4, with total demand of 117,059 which is 35.3% of overall demand. Level 1 demand is 52,587 which is 15.9%, Level 2 demand is 50,766 which is 15.3%, and Level 3 demand is 46,573 which is 14.0%. Level 5 demand is 21,466 which is 6.5% and Level 6 to Level 8 demand is 43,201 which is 13.0%. This distribution indicates that the provincial labour market is primarily pulling for intermediate level skills, with Level 4 forming the main demand anchor.

Gender composition varies sharply by level. Female participation is highest at Level 6 to Level 8, where female demand is 18,892 out of 43,201 which is 43.7%. At Levels 1 and 2, female demand is 14,679 out of 52,587 which is 27.9% and 13,360 out of 50,766 which is 26.3%. Level 3 female demand is 10,340 out of 46,573 which is 22.2% and Level 5 female demand is 5,353 out of 21,466 which is 24.9%. The weakest female participation is at Level 4, where female demand is 16,248 out of 117,059 which is 13.9%. This pattern implies two operational priorities for Sindh. First, scale Level 4 pathways because they account for the largest share of demand, while explicitly redesigning entry, workplace readiness, and placement mechanisms to reduce the gender gap at this level. Second, protect and expand the Level 6 to Level 8 pipeline where female demand is comparatively strong, ensuring that training supply, certification, and employer linkage are aligned with the occupations driving this higher-level demand.

Table 8: Gender-Wise Workforce Demand across Different Occupation Levels

Level	Male	Female	Total
Level 1	37,908	14,679	52,587
Level 2	37,406	13,360	50,766
Level 3	36,233	10,340	46,573
Level 4	100,811	16,248	117,059
Level 5	16,113	5,353	21,466
Level 6-8	24,309	18,892	43,201
Total	252,780	78,872	331,652

4.8. Sector-Wise Leading Trades

The data highlights the importance of different sectors of Sindh's economy, each of which relies on a skilled workforce to sustain growth.

4.8.1. Manufacturing Sector

Figure 9 presents the leading trade-wise skilled workforce demand reported for the manufacturing sector in Sindh. Demand is strongly concentrated in a relatively small group of production-oriented and machine-linked occupations. Stitching Machine Operator emerges as the largest single trade with demand for 9,016 workers, followed by CNC Machine Operator with 8,412 and Lathe Machine Operator with 4,782. Demand for Industrial/Automation Software Developer is also notable at 4,308, indicating that manufacturing labour requirements are no longer confined to conventional shop-floor trades alone, but increasingly include automation-related digital roles.

Among the next tier of occupations, Quality Assurance accounts for 2,750, Textile Machine Operator for 2,708, and Sales & Marketing for 2,596. Further demand is observed for Dyeing Machine Operator (2,096), Production Helper (2,064), Computer Operator (1,970), and Software Developer (1,960). Cutting Machine Operator, with demand of 1,584, also remains an important trade within the sector. This pattern suggests that manufacturing demand in Sindh is distributed across three broad layers: core machine-operation trades, production-support occupations, and a growing segment of digital and managerial support functions linked to modern industrial processes.

The leading trades shown in Figure 9 account for 44,246 workers in total, underscoring that manufacturing workforce demand in Sindh is heavily skewed toward machine operation, production support, and process-oriented technical roles. The prominence of stitching, CNC, and lathe machine operators reflects the continued importance of shop-floor technical skills in Sindh's industrial base,

while the sizeable requirement for industrial and general software developers points to an emerging shift toward technologically enabled and partially automated production systems. Overall, the figure supports a skills development response that prioritises advanced machine operation, quality assurance, textile and process-related trades, and automation-linked digital competencies aligned with changing industrial demand.

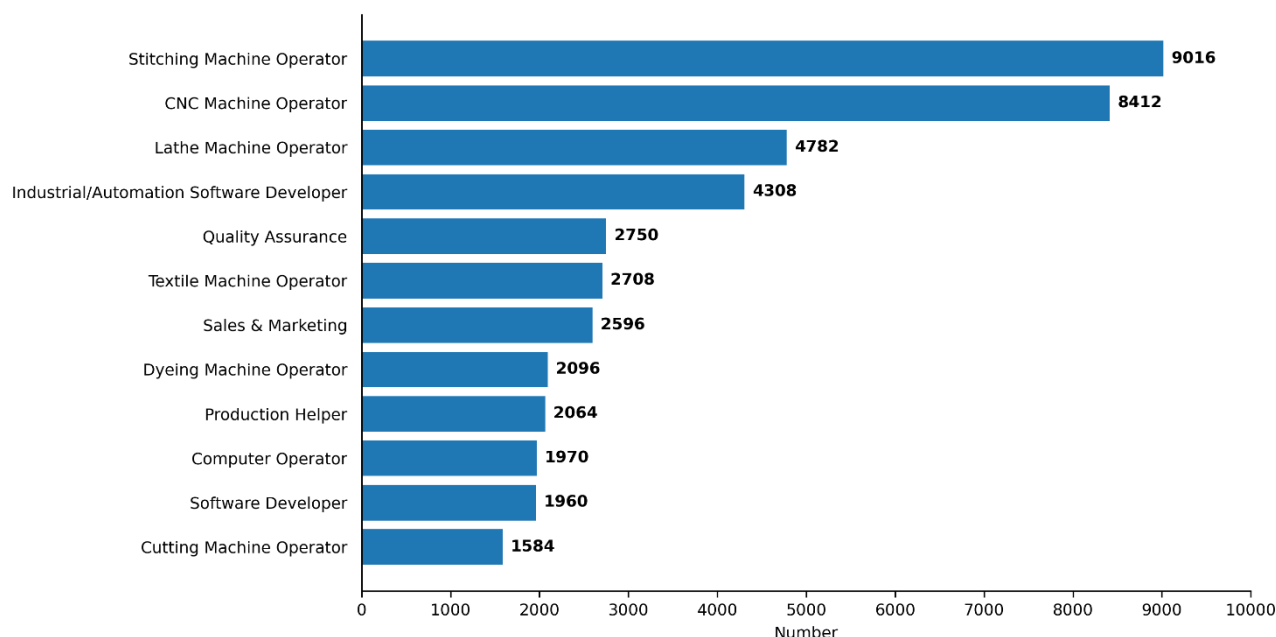


Figure 9: Workforce Distribution across Different Positions in the Manufacturing Sector

4.8.2. Allied Health Sector

Figure 10 presents the leading trade wise skilled workforce demand reported for the Allied Health sector in Sindh. Demand within the leading trades shown is concentrated in a few occupations. Nurse has the highest demand at 7,200, followed closely by Medical Technician at 7,176. X-Ray Technician is 3,868, Laser Machine Operator is 3,066, and Lab Technician is 2,376. Computer Operator accounts for 1,486. Smaller demand volumes are reported for Doctor 840, Accountant 642, Children Attendant/Aya 600, Paramedic 540, and BS Nursing 498.

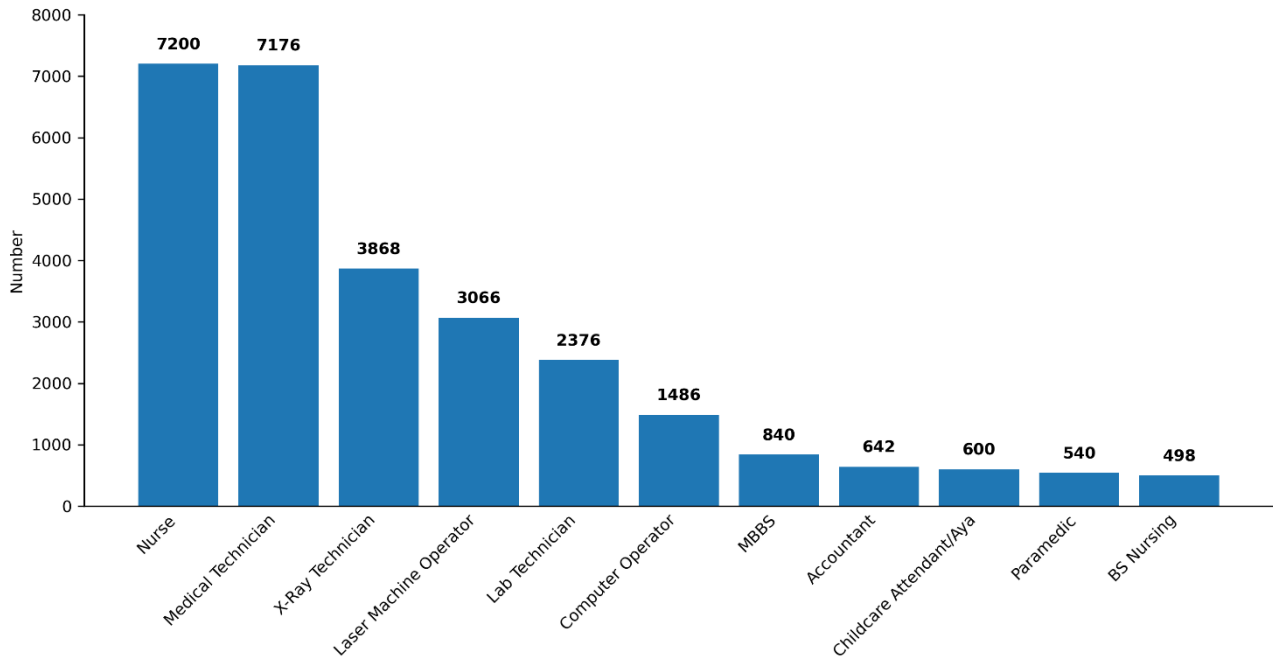


Figure 10: Workforce Distribution across Different Positions in Allied Health Sector

The leading trades demand shown in Figure 10 sum to 28,292 which is 69.8% of total Allied Health sector demand of 40,545. Within the leading trades shown, Nurse and Medical Technician together account for 14,376 which is 50.8%. As a share of total Allied Health demand, Nurse represents 17.8% and Medical Technician represents 17.7%. This pattern indicates that the Allied Health labour market is primarily pulling for mid-level clinical and diagnostic skills rather than higher end specialist roles. For Sindh, this supports scaling competency-based training and certification pathways for nursing, medical technician, radiology, and laboratory streams, backed by structured clinical placements and employer linked assessment, because these trades represent the bulk of measurable demand in the sector.

4.8.3. Agri-business Sector

Figure 11 presents the leading trade wise skilled workforce demand reported for the Agri-business sector in Sindh. Demand within the leading trades shown is concentrated in mechanization and inspection roles. Agri Machine Operator is the largest trade with demand of 4,440, followed by Cane Inspector with 4,242. Packaging Machine Operation is 3,136. Accountant is 1888 and Trained Computer Operator accounts for 1,162. Additional reported trades include Fish Drying Machine Operator 924, Packing Machine Operator 917, Sales and Marketing 839, Livestock Technician 694, and Veterinary Technician 552.

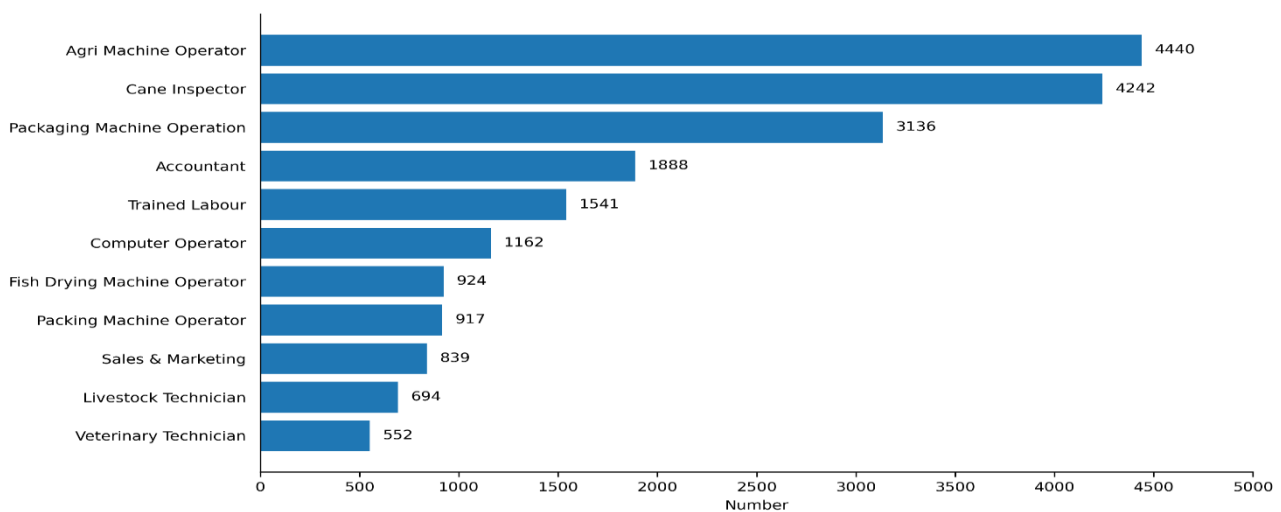


Figure 11: Workforce Distribution across Different Positions in Agribusiness

The dominant trades presented in figure 11 sum to 20,335 which is 52.1% of total Agri business demand of 39037. Within these leading trades, Agri Machine Operator and Cane Inspector together comprise 8,682 positions (42.7 percent), with Agri Machine Operator accounting for 11.4 percent and Cane Inspector for 10.9 percent of total agribusiness demand, highlighting the strong linkage of skills demand to mechanized field operations and post-harvest handling and processing. The prominence of machine operation, inspection, and packaging-related roles indicates that agribusiness skills demand in Sindh is closely tied to mechanized field operations and post-harvest handling and processing. This pattern supports the scaling of competency-based training aligned with agricultural machinery operation, cane and quality inspection, and packaging functions, supported by employer-validated equipment standards and structured on-site practical exposure.

4.8.4. Energy Sector

Figure 12 presents the leading trade-wise skilled workforce demand reported for the Energy sector in Sindh. Demand in the leading trades is concentrated in a relatively small group of technical and field-based occupations. Welder is the largest trade with demand of 2,704, followed by Solar Installation Technician with 2,320. Helper and Solar Technician each record demand of 2,064, while Electrician shows demand of 2,025. The remaining leading trades include Electrical Technician (1,194), Driver (992), Fitter (704), and Accountant (637).

The leading trades identified in Figure 12 account for 14,704 positions, representing 64.5% of the total Energy sector demand (22,784). Within these leading trades, Welders emerge as the most prominent trade, comprising 18.4% of the leading trades and 11.9% of the total sectoral demand. Furthermore, solar-related roles demonstrate high significance; Solar Installation Technicians and Solar Technicians collectively account for 4,384 positions, representing 19.2% of the overall demand within the Energy sector.

This pattern indicates that the Energy sector demand profile in Sindh is driven primarily by field-level technical skills related to welding, electrical work, solar installation, and operational support. For Sindh, this supports scaling up competency-based training in welding and electrical trades, while also establishing standardized solar training pathways with safety compliance, equipment handling, and practical on-site exposure, since these occupations form a major share of measured demand in the sector.



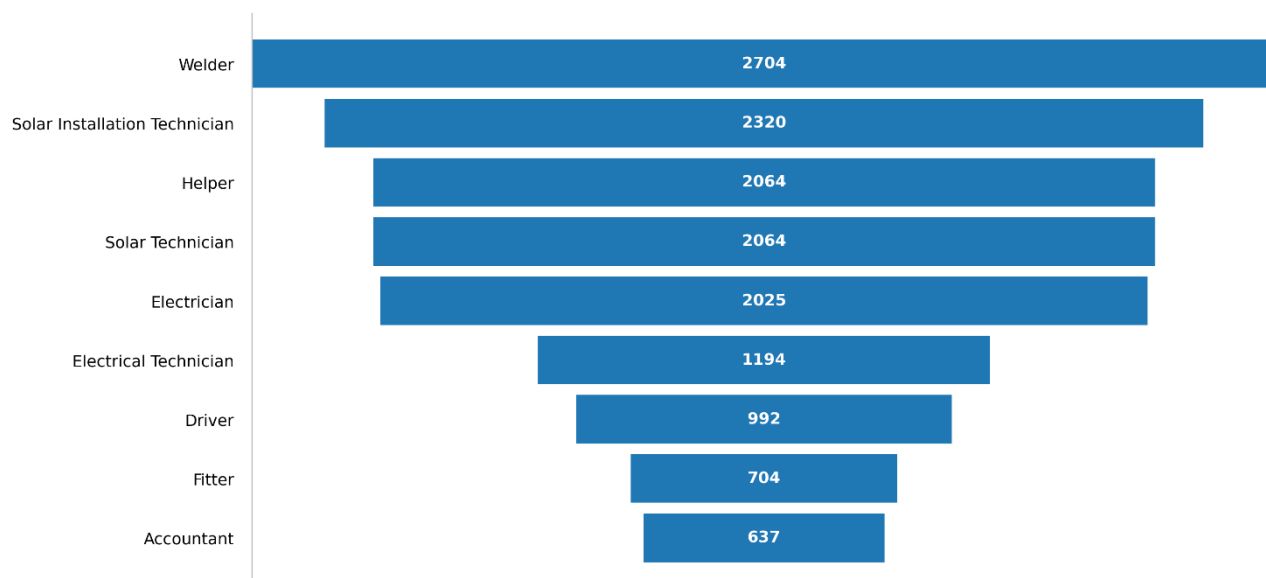


Figure 12: Workforce Distribution across Different Positions in Energy Sector

4.8.5. Construction Sector

Figure 13 presents the leading trade-wise skilled workforce demand reported by employers in Sindh's Construction sector. The demand profile is largely concentrated in technical, site-based, and equipment-related occupations, highlighting the sector's need for skilled workers capable of supporting planning, execution, and mechanized construction activities. Civil Technician represents the highest demand with 4,346 positions, followed by Plumber (2,714) and Architecture Technician (2,333). Other major occupations include Heavy Machinery Operator (1,636), Mason (1,365), and Shuttering Carpenter (1,130). Additional demand exists for Building Electrician (666), Computer Operator (558), Construction Worker (500), Mixture Machine Operator (447), and Concrete Machine Operator (411).

Together, these leading occupations account for 16,106 positions, representing approximately 64.4% of the total Construction sector demand of 25,001 positions. Civil Technicians alone represent about 17.4% of overall construction demand, while Plumbers contribute approximately 10.9%. Combined, these two trades account for nearly 28.2% of sector workforce requirements.

The findings demonstrate the importance of expanding demand-driven technical training pathways in Sindh's construction sector. Priority should be given to civil technology, plumbing, masonry, carpentry, electrical work, and heavy machinery operations through competency-based curricula, improved practical facilities, industry-led apprenticeships, and workplace assessments to enhance employability and meet sector needs.

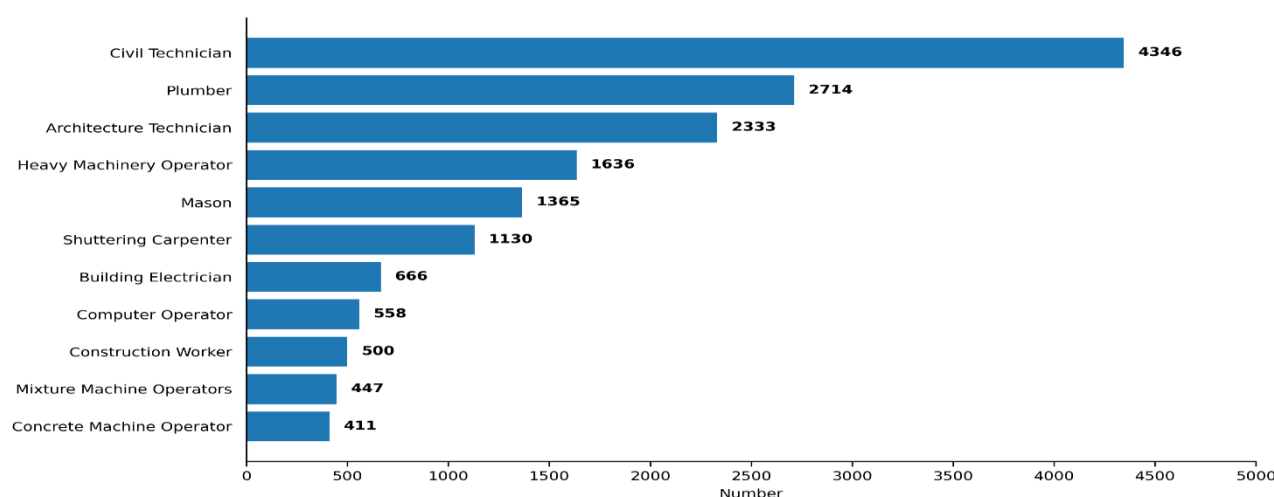


Figure 13: Workforce Distribution across Different Positions in Construction Sector

4.8.6. Water Sector

Figure 14 presents the leading trade wise skilled workforce demand reported for the Water sector in Sindh. Demand in the leading trades shown is concentrated in operations and maintenance roles. Suction Machine Operator is the largest trade with demand of 4,518. Water Pump Operator follows with 2,277 and Plumber with 1,701. The next group includes Sewerage Technician 1,007, Sweeper 999, and Electrician 819. Smaller demand volumes are reported for HTV Driver 666, Data Entry Operator 665, Fire and Safety Management 658, Sanitation Technician 648, and Environmental Engineer Water 442.

The leading trades shown in Figure 14 sum to 14,400, which is 68.7% of total Water sector demand of 20,970. Suction Machine Operator alone accounts for 21.6% of total Water demand, while Water Pump Operator accounts for 10.9%. Together, these two trades account for 32.4% of total Water demand.

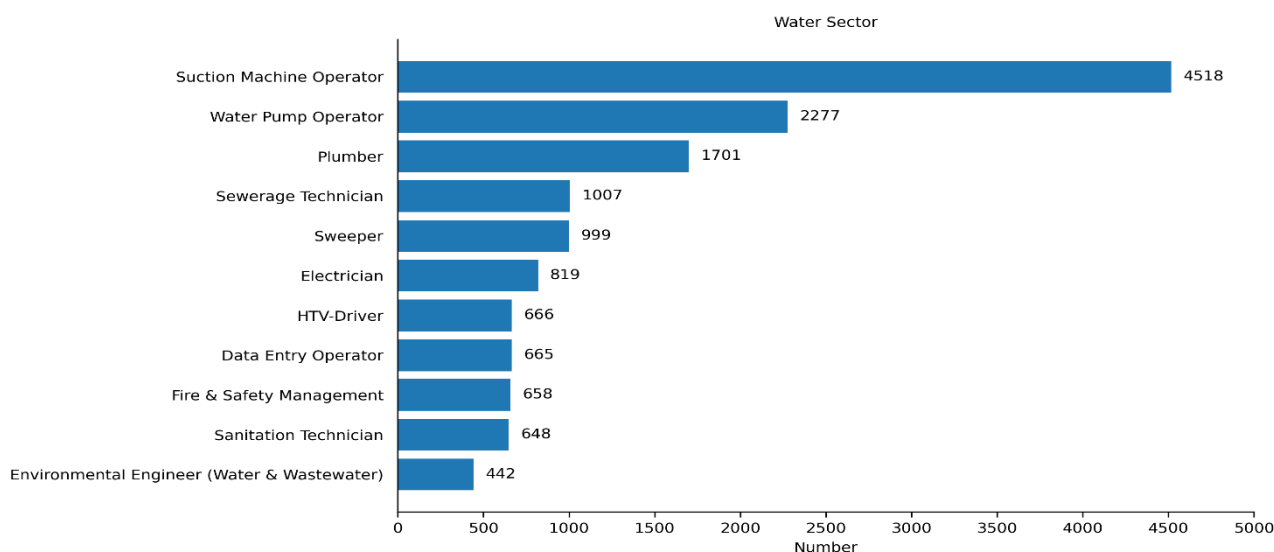


Figure 14: Workforce Distribution across Different Positions in Water Sector

4.8.7. Information Technology

Figure 15 presents the leading trade-wise skilled workforce demand in Sindh's Information Technology sector. The demand profile is concentrated in the IT domains such as web-development, digital services, and emerging technology occupations. Java Script is the highest-demand trade with 3,360 workers, followed by Web Designing with 2,916 and Computer Operator with 1,590. These three occupations alone account for a large share of the demand reflected in the figure, indicating that the IT labour market in Sindh is strongly oriented toward front-end development, web-related services, and operational digital support roles.

A second tier of demand is visible in more specialized and technology-intensive occupations. Cyber Security records demand of 1,392, while PHP accounts for 984 and Artificial Intelligence for 762. Additional demand is observed for IT Technician (722), Call Centre Operator (720), and Block Chain Development (672). Smaller but still notable demand appears in Digital Marketing (540), Fiber Optic Technician (480), CRM (Customer Relationship Management) (447), and Game Developer (372). This distribution shows that Sindh's IT sector is not limited to conventional software roles; it also increasingly requires skills in digital infrastructure, customer systems, security, and new technology applications.

The leading trades shown in Figure 15 account for 14,957 positions, underscoring the breadth of employer demand across both foundational and specialized digital occupations. The prominence of Java Script, Web Designing, and Computer Operator suggests continued market demand for practical, job-ready digital skills, while the sizeable requirements in Cyber Security, Artificial Intelligence, and Block Chain Development point to a gradual shift toward more advanced and innovation-driven areas. For Sindh, this profile supports a skills development strategy that simultaneously expands high-volume training in web development and digital operations, while also

building specialized pathways in cybersecurity, AI, digital infrastructure, and related emerging technologies.

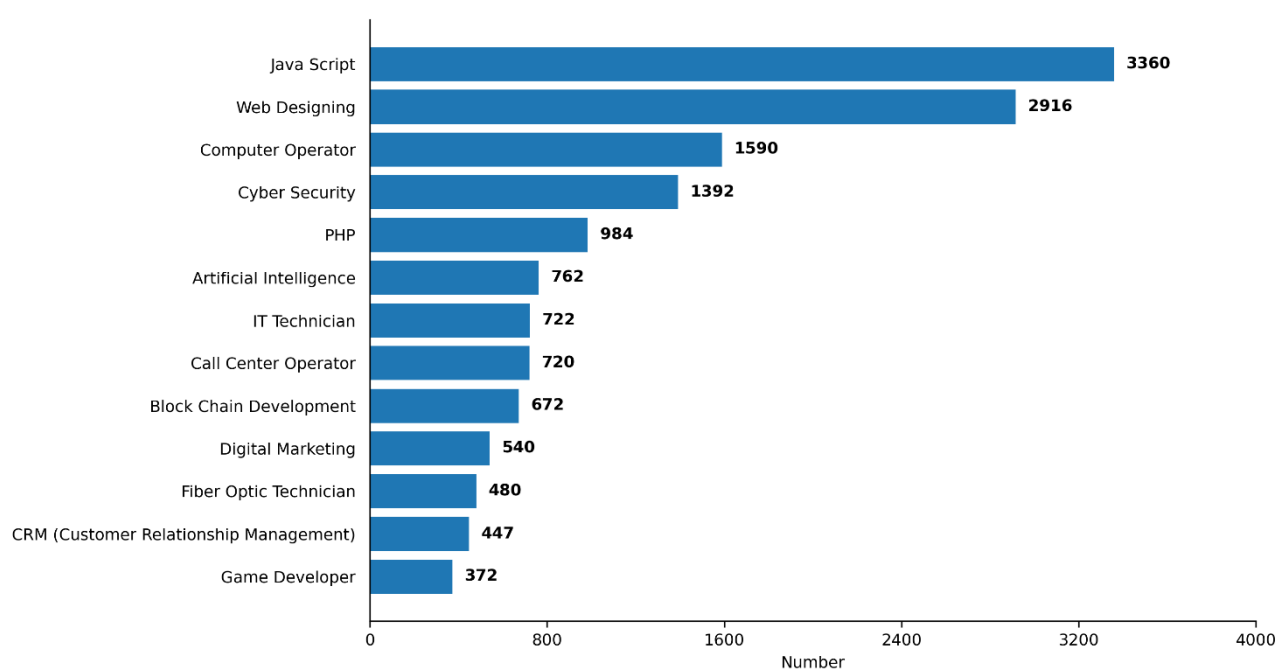


Figure 15: Demand of Information Technology Sector

4.9. District-wise Demand in Provincial Labour Market

Table 9 presents district wise skilled workforce demand in Sindh, totalling 331,652. Demand is highly concentrated in Karachi districts. Karachi Central reports the highest demand at 52,428 which is 15.8% of total provincial demand. Keamari and Malir follow with 42,260 and 41,697 which is 12.7% and 12.6%. Korangi and Karachi West report 32,607 and 31,296 which is 9.8% and 9.4%. Together,

these five districts account for 60.4% of total demand. When Karachi South and Karachi East are included, total demand across Karachi districts rises to 68.4%.

Outside Karachi, demand is more dispersed and appears to be anchored in a small set of regional hubs. Hyderabad and Khairpur report 10,003 and 9,847 demand which is 3.0% each. Larkana reports 9,323 which is 2.8%. Sukkur reports 8,680 which is 2.6%. Sanghar and Shikarpur report 8,272 and 8,153 which is 2.5% each. The remaining districts each contribute a relatively small share of total demand, with the lowest reported in Thatta at 1,139 which is 0.3%.

The distribution implies that TVET planning and placement systems must be designed around two realities. First, Karachi dominates provincial demand and requires a scale focused pipeline of training delivery, industry linkages, and placement services. Second, Sindh also has secondary demand hubs where targeted sector specific programmes can generate strong employment outcomes without requiring migration to Karachi. This reinforces the importance of district level programme prioritization, strengthening employer engagement and placement mechanisms in the highest demand districts, and aligning institute level capacity with where demand is concentrated.

Table 9: District-wise Demand in Sindh

District	Demand of Skilled Workforce	District	Demand of Skilled Workforce
Karachi Central	52,428	Sanghar	8,272
Keamari	42,260	Shikarpur	8,153
Malir	41,697	Tando Muhammad Khan	6,623
Korangi	32,607	Umerkot	5,891
Karachi West	31,296	Tando Allahyar	5,107
Karachi South	15,261	Ghotki	5,007
Karachi East	11,349	Mirpur Khas	4,952
Hyderabad	10,003	Jacobabad	3,841
Khairpur	9,847	Matari	2,600
Larkana	9,323	Jamshoro	2,380
Sukkur	8,680	Shaheed Benazirabad (Nawabshah)	1,966
Tharparkar	1,851	Sujawal	1,773
Kashmore	1,742	Dadu	1,563
Badin	1,393	Qambar Shahdadkot	1,330
Naushahro Feroze	1,318	Thatta	1,139

4.10. Occupation in Top 5 Leading Districts

Below is the summary of study aspects on leading occupations in prominent districts of Sindh, offering policymakers insights into the dynamics of industrial needs and the support required from public institutions.

4.10.1. Karachi Central

Figure 16 presents the occupation-wise skilled workforce demand reported for Karachi Central. Unlike the earlier pattern, the updated figure shows a more diversified demand structure spread across digital, technical, medical, and machine-related occupations. Industrial/Automation Software Developer is the largest occupation with demand of 3,832, followed by Medical Technician with 3,252 and Mobile App Developer with 2,842. Together, these three occupations account for 9,926 positions, representing about 41.8 percent of the total demand shown in the figure.

The next tier of occupations' demand includes Laser Machine Operator with 2,712, Suction Machine Operator with 2,637, and Lathe Machine Operator with 2,624. These are followed by Welder with 1,920, Full-Stack Developer with 1,680, Printing Machine Operator with 1,169, and Front-End Developer with 1,080. Altogether, the occupations shown in the figure sum to 23,748, indicating that labour demand in Karachi Central is distributed across a combination of industrial automation, health-related technical roles, software development, and machine-based production work.

Karachi Central's labour market is anchored in a hybrid demand structure: on one side, there is strong demand for digital and software-oriented occupations such as industrial automation software developers, mobile app developers, full-stack developers, and front-end developers; on the other, there is substantial demand for industrial and technical roles such as laser machine operators,

suction machine operators, lathe machine operators, welders, and printing machine operators. The sizeable demand for medical technicians further indicates the importance of technical health-service occupations in the district's employment structure. For Sindh, this supports a district-specific response that expands training pipelines in industrial automation, software and app development, medical technical skills, and precision machine operations, supported by employer-linked practical training and placement pathways.

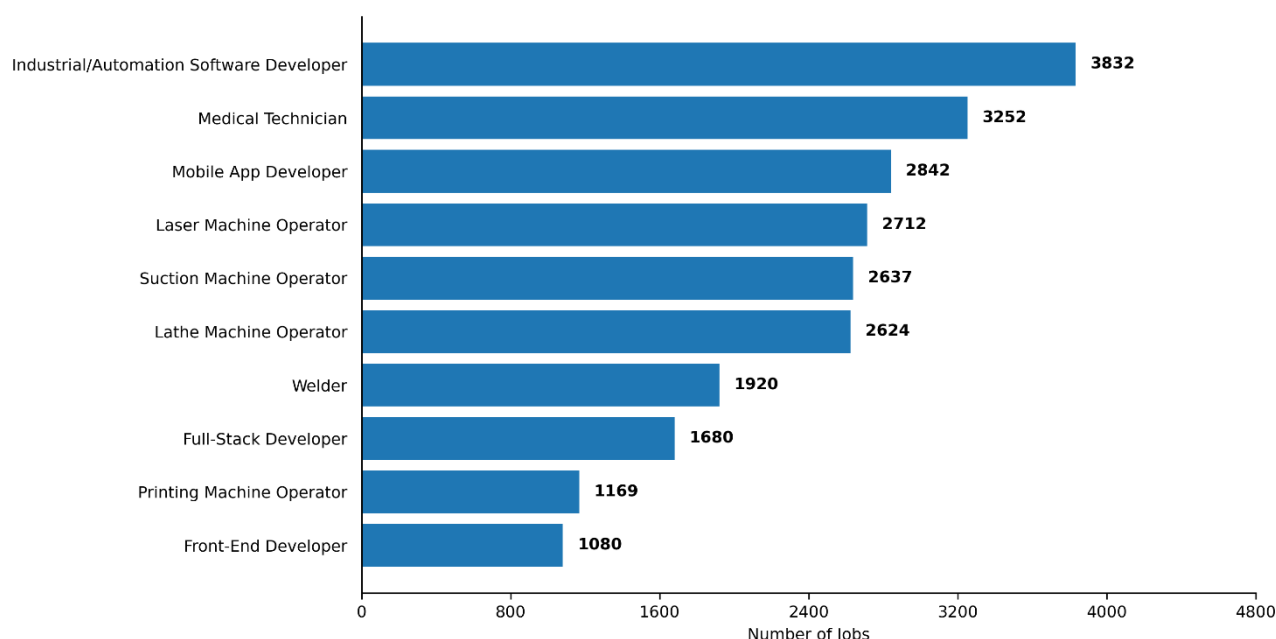


Figure 16: Leading Occupations in Karachi Central

4.10.2. Keamari

Figure 17 presents the occupation-wise skilled workforce demand reported for Keamari district. The demand profile reflects a diverse labour market structure, combining industrial, technical, healthcare, and service-related occupations. Sewing Machine Operator records the highest demand with 4,680 positions, accounting for 20.3% of the total demand shown in the figure. Nurse is the second-highest occupation with 2,364 positions (10.2%), followed by Fitter with 2,250 positions (9.8%). Together, these three occupations represent 9,294 positions, equivalent to approximately 40.3% of the total demand presented.

A second group of occupations demonstrates substantial workforce requirements. Sales and Marketing accounts for 1,881 positions (8.2%), followed by Welder (1,820; 7.9%) and Quality Assurance (1,716; 7.4%). The figure initially included two separate categories of Machine Operator, with reported demands of 1,544 and 1,669 positions. As the available data does not specify machine types, these have been consolidated into a single Machine Operator category with a combined demand of 3,213 positions, representing 13.9% of the total demand. Other important occupations include Crane Operator (1,616; 7.0%), Plumber (1,370; 5.9%), Packaging Machine Operator (1,120; 4.9%), and Stitching Machine Operator (1,040; 4.5%).

The findings indicate that Keamari's labour demand is strongly driven by manufacturing and technical trades, complemented by healthcare and commercial occupations. The district requires targeted TVET responses focusing on machine operation, sewing and stitching, fitting, welding, quality assurance, packaging, plumbing, and crane operations, while strengthening nursing and healthcare pathways through industry-linked training, practical exposure, and placement mechanisms.

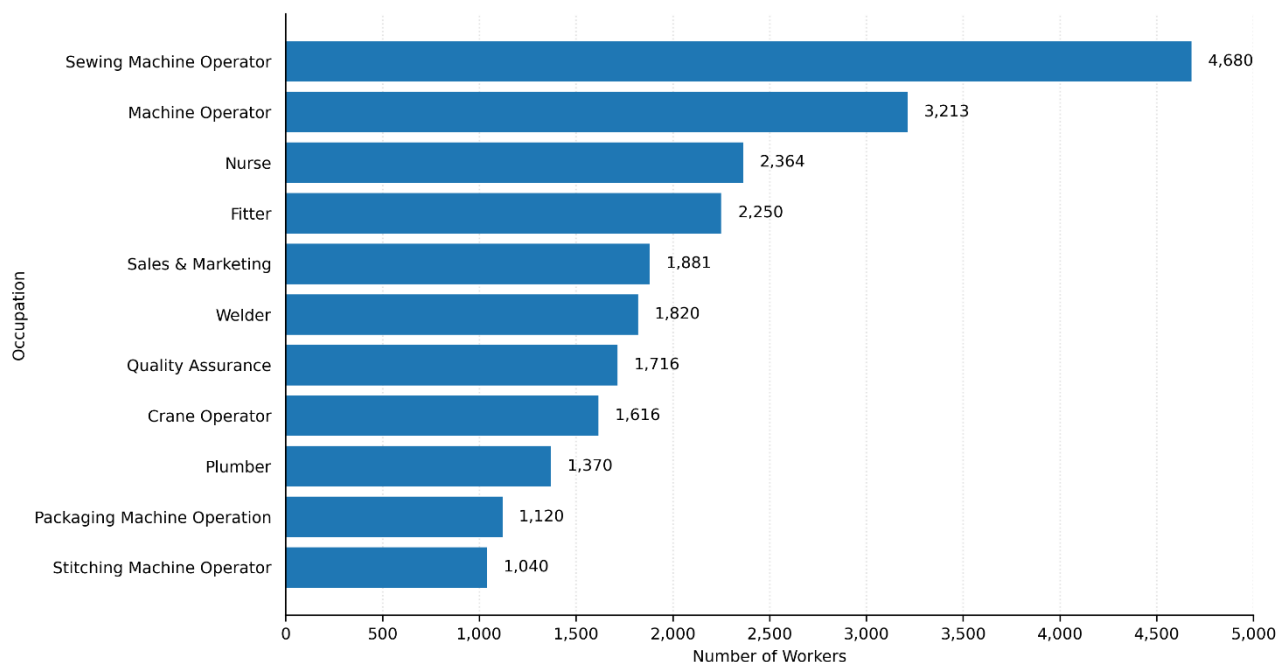


Figure 17: Occupation-wise Skilled Workforce Demand in Keamari

4.10.3. Malir

Figure 18 presents the occupation-wise skilled workforce demand reported for Malir. Demand is concentrated in a mix of production-related, machine-based, electrical, and digital occupations. Packaging Machine Operation is the highest-demand occupation with 3,142 workers, followed by Micro Paper Cutting Machine Operator with 2,212. Electrician and Software Developer also show strong demand at 2,047 and 2,032, respectively. Together, these four occupations account for 9,433 positions, which is about 58.5 percent of the total demand shown in the figure, indicating that Malir's labour market is anchored in packaging and processing operations, supported by technical and digital roles.

A second cluster of occupations' demand includes Graphic Designer with 1,108, Accountant with 1,096, Loading Machine Operator with 1,012, and Textile Machine Operator with 984. The remaining occupations comprise Fish Drying Machine Operator (910), Fabric Cutting Machine Operator (904), and Civil Technician (670). Altogether, the occupations shown in Figure 18 sum to 16,117 workers. This profile suggests that workforce demand in Malir is driven primarily by industrial packaging, machine operation, and electrical trades, while also reflecting a meaningful requirement for software, design, and administrative support skills.

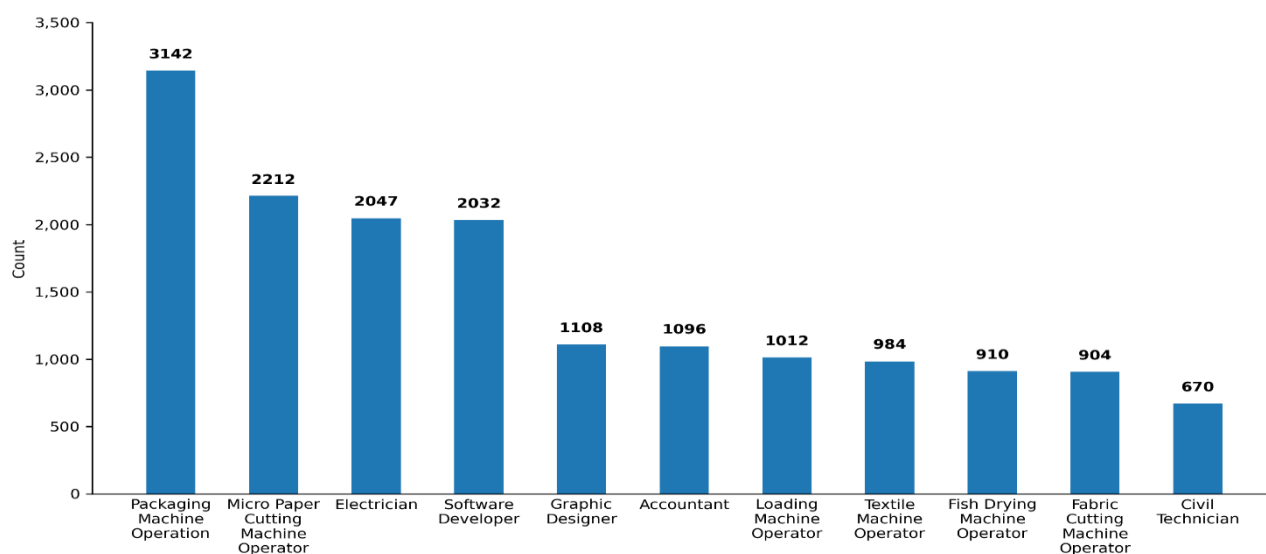


Figure 18: Leading Occupations in Malir

4.10.4. Korangi

Figure 19 presents the occupation wise skilled workforce demand reported for Korangi. Demand is concentrated in production and quality related occupations. Production Supervisor is the highest demand occupation with 2,373 which is 20.3% of the total demand shown in the figure. Stitching Machine Operator follows with demand of 1,776 which is 15.2% and Quality Assurance demand is 1486 which is 12.7%. Together these three occupations account for a demand of 5,635 which is 48.3% of the total.

A second cluster of occupations includes Helper with demand of 1,133 which is 9.7%, Sales and Marketing with 1035 which is 8.9%, and Nurse with 978 which is 8.4%. Technical occupations reported include Solar Installation Technician with demand of 796 which is 6.8%, Machine Operator with 788 which is 6.8%, Cutting Machine Operator with 672 which is 5.8%, and Knitting Machine Operator with 624 which is 5.4%. The demand profile indicates that Korangi is primarily pulling for supervision, stitching, and quality control functions alongside supporting shop floor and technical roles. For Sindh, this supports a district specific response that prioritizes production supervision pathways, stitching and knitting related training, and quality assurance certification, complemented by employer linked practical exposure for machine operation and installation trades.

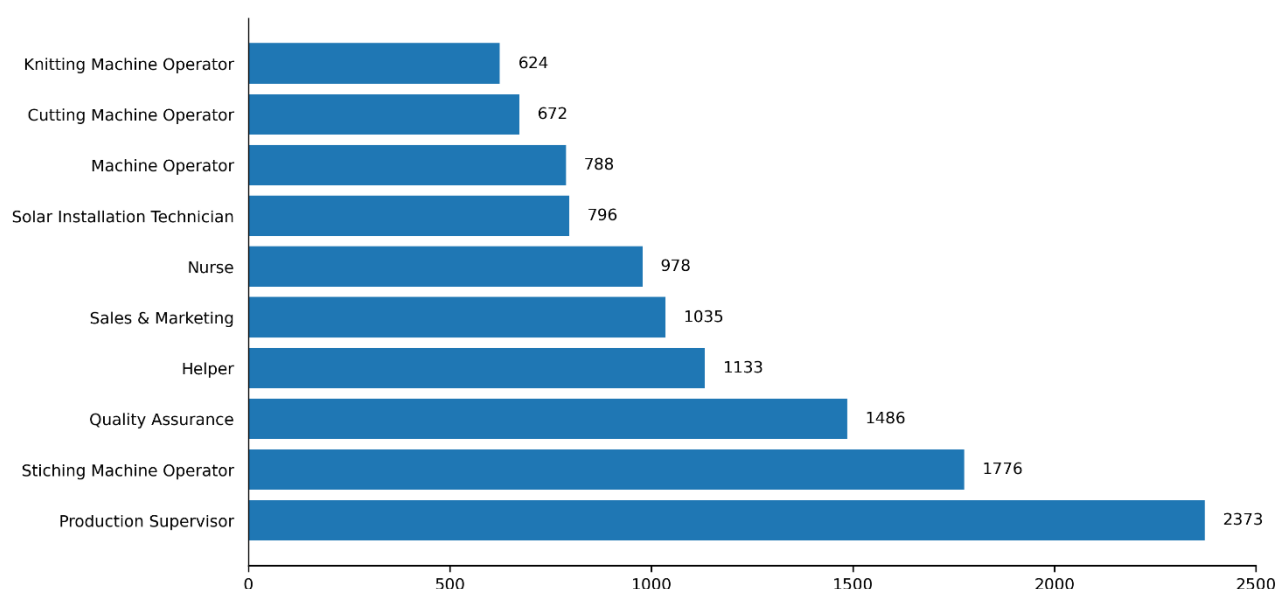


Figure 19: Leading Occupations in Korangi

4.10.5. Karachi South

Figure 20 presents the occupation wise skilled workforce demand reported for Karachi South. Demand is distributed across technology, services, and technician roles. IT Technician trade has the highest demand occupation with 588 which is 13.6% of the total demand shown in the figure. Architecture Technician is 516 which is 11.9% and Textile Design Technician demand is 492 which is 11.4%. CRM Customer Relationship Management is 476 which is 11.0% and Artificial Intelligence is 465 which is 10.7%. The top five occupations together account for 2,537 which is 58.6% of the total demand shown.

The remaining occupations in highest demand are: Concrete Machine Operator with 420 which is 9.7%, Event Manager with 372 which is 8.6%, Sales and Marketing with 348 which is 8.0%, Electrical Technician with 328 which is 7.6%, and Textile Machine Operator with 324 which is 7.5%. This profile indicates that Karachi South demand is anchored in IT support and emerging digital functions alongside a mixed set of technician and operator roles. For Sindh, this supports a district specific response that prioritizes IT technician pipelines, CRM aligned customer services training, and applied digital skills pathways, while maintaining targeted technical training for electrical and concrete machine operations and textile related roles linked to employer placement.

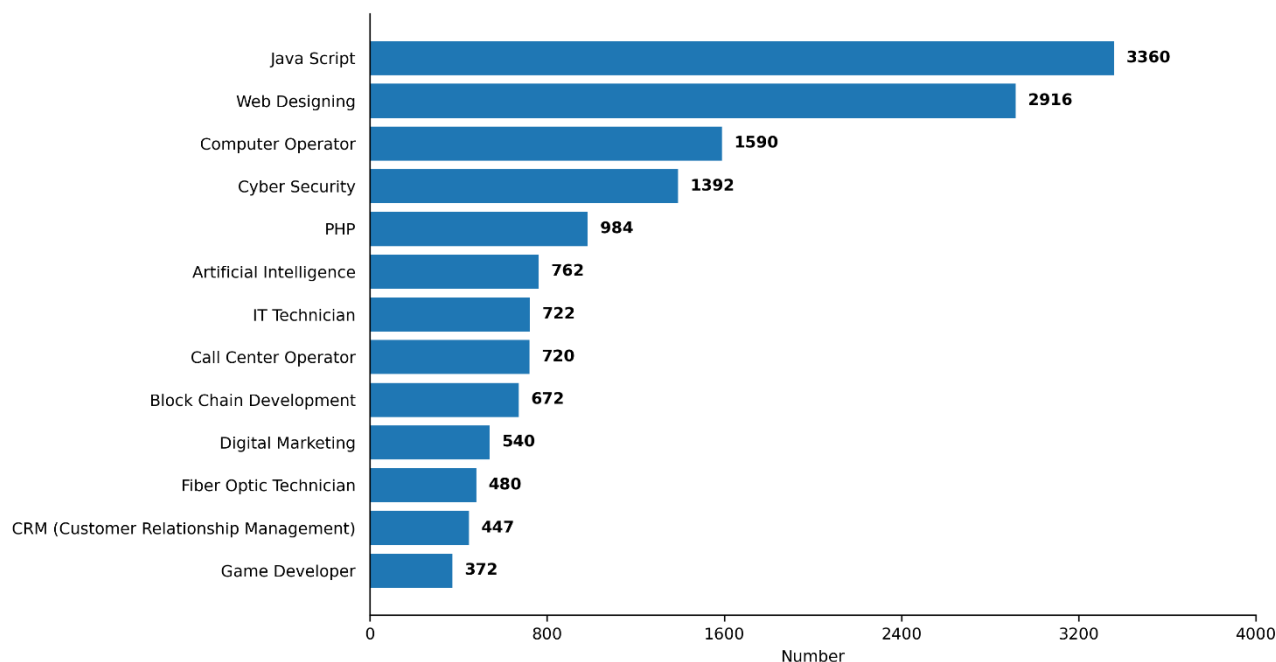


Figure 20: Leading Occupations in Karachi South

4.11. Analysis of Skilled Workforce Demand & Supply Gaps

Table 11 in the appendix provides a comprehensive overview of trade-wise skills supply, labour market demand, and resulting gaps in Sindh. The table highlights a pronounced misalignment between training output and labour market requirements, with gaps ranging from acute shortages in high-demand production and service roles to notable surpluses in a limited number of general IT and design-oriented programmes.

4.11.1. Acute Shortages in Manufacturing, Production, and Machine-Operation Trades

The most pronounced gaps are concentrated in machine-intensive and production-oriented trades that are central to Sindh's industrial structure. CNC Machine Operator shows a shortage of 8,468, while Lathe Machine Operator records a gap of 5,961. Textile Machine Operator shows a deficit of 2,708, while other significant shortages are visible in trades such as Sewing Machine Operator 5,013, Industrial Stitching Machine Operator 3,991, Packaging Machine Operation 4,457, Dyeing Machine Operator 2,584, Cutting Machine Operator 2,093, Micro Paper Cutting Machine Operator 2,513, Printing Machine Operator 1,321, and Fabric Cutting Machine Operator 968. Cane Inspector also shows a sizeable shortfall of 4,242. These figures indicate that Sindh's manufacturing and processing sectors continue to demand job-ready shop-floor workers, yet training supply in these trades remains limited or altogether absent.

4.11.2. Large Deficits in Software, ICT Support, and Specialised Digital Skills

The data also show that digital labour demand in Sindh is increasingly occupation-specific rather than general. Software Technician records the single largest shortfall in the table, with demand of 27,410 against supply of only 545, leaving a gap of 26,865. Computer Operator demand faces major deficits, including a shortfall of 14,290 in one entry and 1,290 in another. In addition, Java Script shows a gap of 3,260, Web Designing 2,671, Cyber Security 1,392, PHP 984, CRM 2,859, Digital Marketing and SEO 552, Artificial Intelligence 897 in one entry, and Data Analyst 676. This suggests that employers are not merely looking for broad digital literacy; rather, they require applied and role-specific competencies linked to software support, web development, digital customer management, emerging technologies, and data functions.

4.11.3. Shortages in Business, Administrative, and Quality-Related Occupations

A notable feature of the updated data is the scale of deficits in commercial, managerial, and quality-related roles. Accountant exhibits a shortfall of 8,356, Sales and Marketing 7,405, Quality Assurance 5,889, Manager 3,287, and Supply Chain and Logistics 1,244. Additional deficits are visible in trades such as Product Manager, Storekeeper, Business Management, and Logistic and Supply Technician. This pattern broadens the labour market picture beyond narrowly technical skills and shows that Sindh's firms also face shortages in business operations, supervisory functions, inventory management, and quality systems. These occupations are especially important for industrial competitiveness, distribution networks, and modern service-sector growth.

4.11.4. Persistent Gaps in Construction and Allied Technical Trades

Construction and infrastructure-related trades remain significantly under-supplied. Construction Worker shows a gap of 7,255, while Electrician records a shortage of 2,384 and Civil Technician 1,829. Plumbing and Pipe Fitter faces a deficit of 3,939, while Welder shows a deficit of 981. Other notable shortages include Mason 1,395, Building Electrician 1,336, Crane Operator 1,666, Architectural Technician 1,100, Maintenance Technician 1,059, Tiles Fixer 552, and Heavy Machinery Operator 567. These shortages reflect continued labour demand arising from urban construction, industrial installation, infrastructure maintenance, and utility expansion. The evidence suggests that current training provision in field-based technical trades is not keeping pace with actual market requirements.

4.11.5. Critical Unmet Demand in Healthcare and Allied Health Occupations

The healthcare sector also exhibits major workforce shortages. Nurse records a deficit of 7,230, while Lab Technician shows a shortfall of 3,158. Medical Technician and Pharmacy Technician record gaps of 1,494 and 1,479, respectively, while Doctor shows a deficit of 840. X-Ray Technician and Paramedic also remain under-supplied, with gaps of 584 and 540. These figures point to structural weaknesses in Sindh's health-related training pipeline, especially in middle-level technical and allied health roles that are essential for effective service delivery across both public and private providers.

4.11.6. Emerging Deficits in Energy, Utility, and Environmental Service Occupations

The updated table also points to growing demand in energy, municipal, and environmentally linked trades. Solar PV System Technician records a shortfall of 2,259, while the Solar Technician field faces a gap of 2,718. Water Treatment Plant Operator records a deficit of 852, Environmental Engineer in Water and Wastewater 642, and Sanitation Technician 725. These gaps are important because they reflect growing labour demand in renewable energy, water management, sanitation, and environmental service delivery. Training supply in these trades has not yet expanded in line with emerging sectoral needs.

4.11.7. Under-Covered Demand in Hospitality, Service, and Routine Semi-Skilled Roles

A wide range of semi-skilled and routine service occupations also show large gaps, despite often receiving limited attention in formal skills planning. Helper has a shortfall of 3,941, Trained Labour 2,744, Driver 2,723, Chef 3,597, Cook 1,209, Waiter 1,312, Cleaner 750, Call Centre Operator 910, Sweeper 1,017, Auxiliary Staff 724, and Childcare Attendant / Aya 642. These results suggest that a substantial portion of labour demand in Sindh is concentrated in practical entry-level service occupations for which formal training supply is negligible. This creates a strong case for short-duration, competency-based, and workplace-linked training models that can respond rapidly to employer requirements.

4.11.8. Limited Surpluses in General IT, Design, and Selected Technician Programmes

Although most occupations demonstrate significant shortages, a limited number of trades show a surplus of trained workers compared to current market demand. DIT records the highest surplus with 3,420 workers, followed by CIT with 2,432. Other notable surplus trades include Fashion Design and Dress Making (1,660), Graphic Designer (1,276), and Web Designing and Development (956). Smaller surpluses are observed in Mechanical Technician (672), Electrical Technician (574), and Machinist (566). These patterns indicate that certain training streams are producing graduates beyond the absorption capacity of the labour market, particularly where programmes are broader in scope, less specialized, or insufficiently connected with employer requirements. The findings highlight the need for periodic review of enrolment targets, curriculum relevance, and training pathways to ensure resources are redirected towards occupations with demonstrated demand. In contrast, specialized and application-oriented roles continue to face persistent shortages, requiring expanded training capacity and stronger industry alignment.

4.11.9. Implications for Workforce Planning in Sindh

Overall, Sindh's skills ecosystem remains structurally misaligned with labour market requirements. The most severe shortages are concentrated in production trades, machine operation, specialised digital occupations, commercial and quality-related roles, healthcare, construction, and routine service work. Meanwhile, surpluses remain concentrated in a few broad-based IT and design programmes. This suggests that the province's training system is still oriented toward general supply creation rather than demand-responsive workforce planning.

From a policy perspective, this has three implications:

- i. Training capacity needs to be expanded in the highest-deficit trades, especially software and applied digital roles, industrial machine operation, accounting and sales support, construction trades, healthcare technicians, and utility-linked occupations.
- ii. Greater emphasis should be placed on apprenticeship models, workplace-based training, and modular short-cycle certification in occupations where demand is high but formal supply is weak.
- iii. The overall training portfolio should be rebalanced by gradually rationalising persistent surplus programmes and redirecting seats, equipment, and instructional resources toward trades with demonstrated labour market demand.

4.12. Employment Opportunities for Disadvantaged Groups

Figure 21 presents the distribution of seats allocated for disadvantaged groups in Sindh. The largest share is allocated to minorities, with 5,918 seats, representing 72.6% of the total seats reported in the figure. Transgender persons account for 1,651 seats (20.3%), while persons with disabilities are allocated 448 seats (5.5%). The allocation for elderly persons is comparatively limited, with 133 seats (1.6%). Overall, the total number of seats allocated across all disadvantaged groups is 8,150.

The distribution indicates a significant concentration of opportunities among minority groups, which account for nearly three-quarters of all reported seats. In comparison, persons with disabilities and elderly groups collectively represent only 7.1% of the available seats, highlighting an imbalance in current inclusion measures. While minority-focused pathways remain important, broader accessibility mechanisms are required to ensure equitable participation across all disadvantaged groups.

For Sindh's skills development response, two key priorities emerge. First, minority-focused training and employment pathways should be strengthened through employer partnerships, career support, and placement mechanisms. Second, targeted measures for persons with disabilities and elderly participants should be enhanced through accessible training environments, adapted learning approaches, and supportive services to ensure that allocated opportunities translate into meaningful participation and employment outcomes.

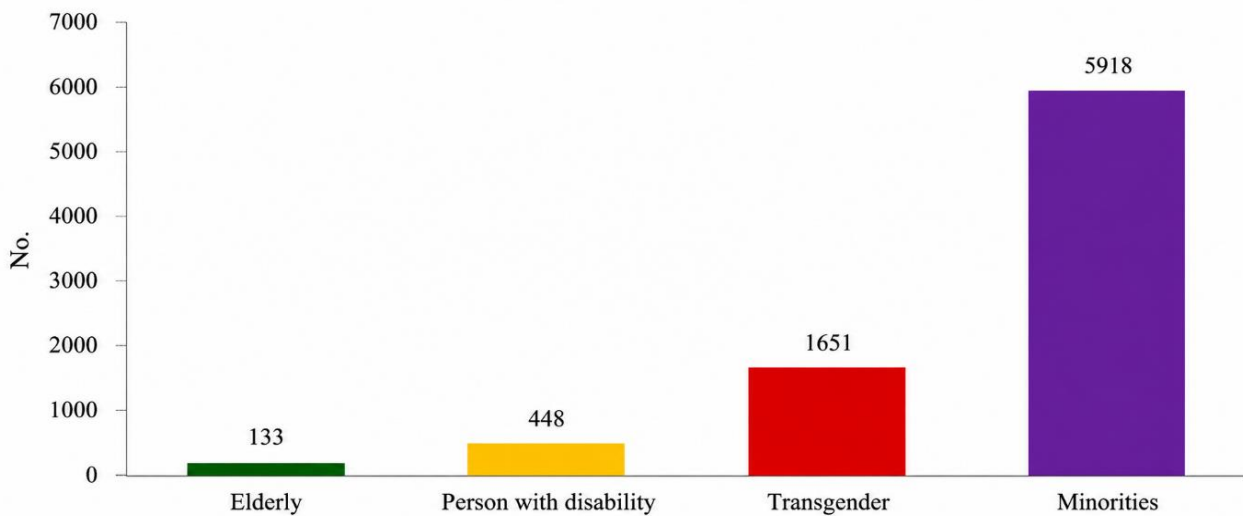


Figure 21: Distribution of Quota or Reserved Seats for Disadvantaged Groups

4.13. Skills for Disadvantaged Groups

The analysis of skills and occupations for disadvantaged groups reveals a significant concentration of employment in low- to semi-skilled roles, with limited representation in specialized or high-skill professions. The most common occupations include Computer Operators (114), Machine Operators (102), Skilled Workers (90), Support Staff (90), and Sweepers (84), as shown in Table 10.

Sanitation-related roles, including Cleaners (72), Janitorial Staff (72), and Sanitary Workers (24), constitute a notable portion of employment, reflecting the continued reliance on disadvantaged groups in essential but lower-status occupations. Support and auxiliary roles, such as Support Staff (90), Security Guards (54), Administrative Staff (24), Receptionists (24), Record-keepers (24), and Office Attendants (12), further indicate that a significant share of employment remains concentrated in support functions with limited career progression. Similarly, food and domestic service roles are represented by Cooks (18), Dishwashers (24), Housekeepers (6), and Commercial Cooking Staff

(6), reinforcing the concentration of disadvantaged groups in low-paying and labour-intensive occupations.

Table 10: Occupation-Wise Seats for Disadvantaged Groups

Occupation	Number	Occupation	Number
Computer Operator	114	AutoCAD	6
Machine Operator	102	Childcare Attendant / Aya	6
Skilled Workers	90	Commercial Cooking	6
Support Staff	90	Construction Worker	6
Sweeper	84	Cutting Saila – Cutter	6
Accountant	72	Dialysis Technician	6
Cleaner	72	Doctor	6
Janitorial Staff	72	Dress Making	6
Security Guard	54	Floor Manager	6
Software Developer	36	GIS & Remote Sensing	6
Supervisor	30	Housekeeper	6
Administrative Staff	24	Hygiene Staff	6
Dishwasher	24	Juice Bar Attendant	6
Engineer	24	Low-skilled Worker	6
Labourer	24	Medical Assistant	6
Receptionist	24	Medical Technician	6
Record-keeper	24	Mining	6
Sanitary Worker	24	Mirror Work	6
Cook	18	Motor Operator	6
Dispenser	18	Needle Work	6
Lab Assistant	18	Office Boy	6
Packager	18	Peon	6
Clerk	12	Pharmacist	6
Driver	12	Printing Worker	6
Generator Operator	12	Safety Officer	6
Handicraft	12	Stitching / Tailoring / Sewing	18
Lab Technician	12	Telephone Operator	6
Mechanic	12	Ward Boy	6
Nurse	12	Watchman	6
Office Attendant	12	Water Plant In-charge	6
Plumber	12	Welding	6
Technician	12	Total	1,374

A relatively small number of individuals are employed in skilled or semi-professional roles, such as Accountants (72), Software Developers (36), Supervisors (30), Engineers (24), Lab Assistants (18), and Nurses (12). This highlights the challenges disadvantaged groups face in accessing technical and higher-education-based occupations. The presence of Record-keepers (24), Lab Technicians (12), Pharmacists (6), Medical Assistants (6), and Medical Technicians (6) suggests that some individuals have transitioned into specialized fields; however, their overall representation remains limited. Highly specialized trades such as GIS and Remote Sensing (6), AutoCAD (6), and Dialysis Technician (6) remain almost negligible, indicating limited access to advanced technical education and certification pathways.

The list also includes skilled and semi-skilled trades with small representation, such as Stitching / Tailoring / Sewing (18), Handicraft (12), Dress Making (6), Construction Worker (6), Plumber (12), Welding (6), and Mechanic (12). Despite demand for these trades in local markets, their limited presence among disadvantaged groups suggests a gap in vocational training outreach, career counselling, and programme alignment with broader labour market and social integration needs. Other niche occupations, including Mining (6), Safety Officer (6), Mirror Work (6), and Printing Worker (6), demonstrate scattered and inconsistent participation across employment sectors.

Overall, the employment distribution indicates that disadvantaged groups are largely concentrated in low-skill, low-wage, and support-sector occupations, with minimal representation in technical, managerial, and leadership roles. This pattern underscores broader systemic challenges, including limited access to quality education and vocational training, social discrimination, inadequate career

counselling, weak placement support, and limited workplace accessibility. Addressing these disparities requires targeted interventions, including inclusive TVET programmes, market-driven skills development, accessible training facilities, employer sensitization, and stronger workplace accommodation measures.

4.14. Employers' Satisfaction with TVET Graduates

Figure 22 presents employers' satisfaction levels with TVET graduates in Sindh. Overall perception is positive. 59% of employers report they are satisfied and 10% report they are very satisfied. This means 69% of employers express a positive satisfaction rating. A further 27% report a neutral position. Only 3% report dissatisfied and 1% report very dissatisfied, indicating a combined negative rating of 4%.

The distribution suggests that TVET graduates are broadly meeting employer expectations, but the large neutral share signals inconsistency in workplace readiness and skills match across trades and institutions. For Sindh, this supports improvements in training quality and employer relevance through stronger industry engagement in curriculum updates, competency-based assessment, structured workplace exposure, and feedback loops that translate employer inputs into measurable training standards. The objective should be to convert the neutral segment into satisfied outcomes through targeted improvements in practical skills, soft skills, and job specific competencies.

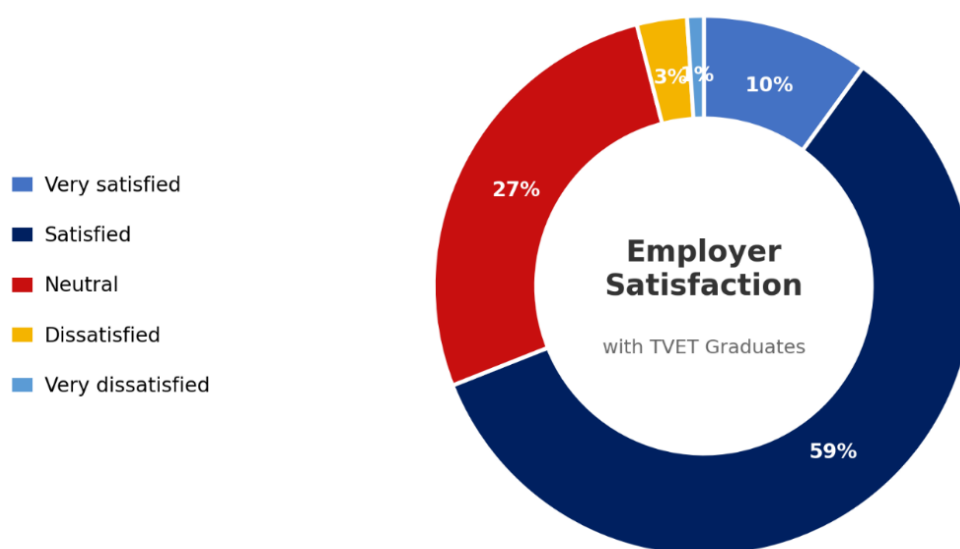


Figure 22: Employers' Satisfaction with Technical and Vocational Education and Training

4.15. Industry-Led Training Opportunities

Figure 23 presents the reported availability of organizational training programmes in Sindh including on the job training, apprenticeships, and online courses. The results indicate limited institutionalization of employer provided training. Only 20% of employers report that such training programmes are available, while 80% report that these are not available.

This distribution implies that most firms are relying on external hiring or informal learning rather than structured skills development within the workplace. For Sindh, this supports expanding employer linked training models that reduce the cost and risk for firms to train workers. Priority actions needed include formal apprenticeship and internship agreements, incentives for firm-based training delivery, and partnerships between training providers and employers to co-deliver short cycle modules that are aligned with the specific trades showing the highest demand in the province. Improving the availability of structured workplace learning is also necessary to convert neutral satisfaction responses into consistently positive employer outcomes as reported in Section 4.13.

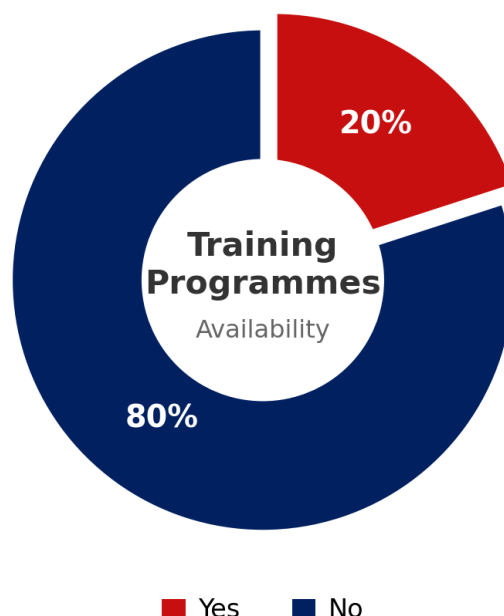


Figure 23: Industry-Led Training Opportunities

Figure 24 presents the types of training programmes offered by employers in Sindh. Inhouse practical training is the most commonly reported programme type at 54.2%. Machine Operating related training is reported by 14.3%. Preference of Apprenticeship is reported by 10.7%, of Computer AI course reported by 8.2%, of Solar installation training by 6.9%, and Crane operator training by 5.8%.

The distribution indicates that where employers do provide training, it is primarily focused on immediate operational needs and practical skill building rather than formal structured programmes. The dominance of inhouse practical training suggests that firms prefer short cycle, job specific learning delivered at the workplace. Apprenticeship preference reported by only 10.7% indicates that formal work-based learning models are still limited, which aligns with earlier findings on the low availability of organizational training programmes and limited presence of dedicated learning and development departments. For Sindh, this implies that skills policy should capitalise on what employers already do. Priority actions include formalizing inhouse practical training through standardized competency checklists and assessment, expanding apprenticeship arrangements through simplified agreements and incentives, and supporting targeted technical modules in machine operation, solar installation, and applied digital skills that can be co delivered with employers.

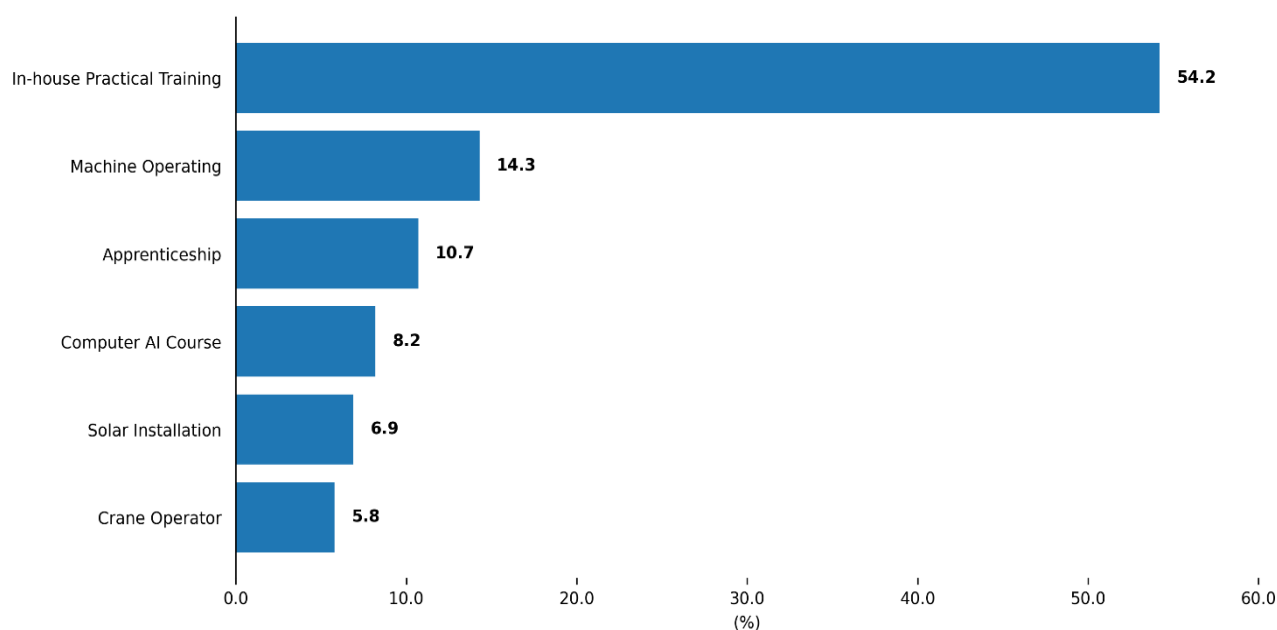


Figure 24: Industry-Led Training: Preferred Skills

4.16. Skills Assessment in Recruitment

Figure 25 presents the skills and competency assessment methods used by employers in Sindh during hiring. The most common method is personal interviews reported by 44.6% of employers. Practical skills assessments are the second most common at 36.9%. Written tests and exams are used by 13.3%. Group discussions and activities are used by 4.7%. Other methods are negligible at 0.5%.

The distribution indicates that recruitment decisions are driven primarily by subjective screening through interviews, with a sizeable but not universal use of practical testing. The limited use of written exams and group-based methods suggests that many firms may not be applying standardized competency verification across candidates. For Sindh, this supports strengthening employer adoption of structured competency-based hiring tools, particularly for high demand technical trades, by developing standard practical assessment templates, trade tests, and joint assessment protocols with training providers. Expanding the use and quality of practical testing can also improve employer confidence in TVET outputs and support better matching between training and workplace performance.

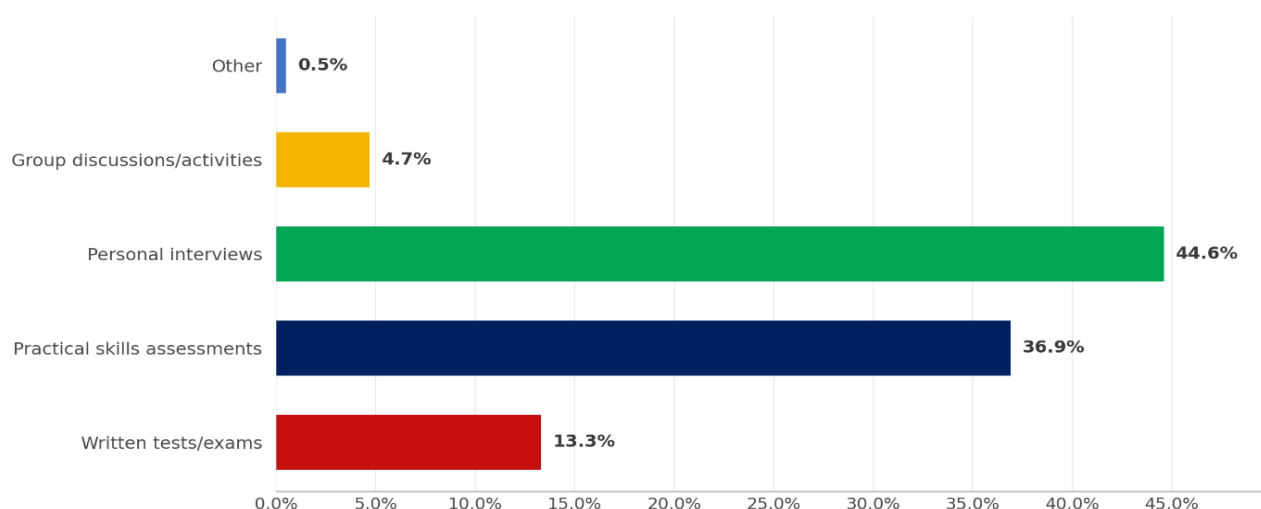


Figure 25: Methods of Skills and Competencies Assessment

4.17. Challenges in Hiring or Retaining Employees

4.17.1. Female Employees

Figure 26 presents the major challenges reported by employers in Sindh in hiring and retaining female employees. The most frequently cited challenge is workplace policies and culture not being conducive for women at 29.9%. Safety and security concerns are the second most cited reason at 25.2%. Occupational segregation and gender stereotypes are reported by 21.9%. Lack of flexible work arrangements is reported by 20.5%, while other challenges account for 2.6%.

The distribution indicates that barriers are primarily structural rather than related to technical capability. Nearly three quarters of responses are concentrated in workplace environment, safety, and gender norms, which collectively limit recruitment and retention even when demand exists. For Sindh, this implies that increasing female participation in high demand sectors requires complementary interventions beyond training supply. Priority actions include workplace policy reforms and codes of conduct, safe transport and secure worksite measures, structured anti-harassment mechanisms, and employer adoption of flexible work arrangements where feasible. Addressing occupational segregation also requires deliberate placement and mentorship pathways that support female entry into male dominated trades alongside targeted outreach and employer commitment.

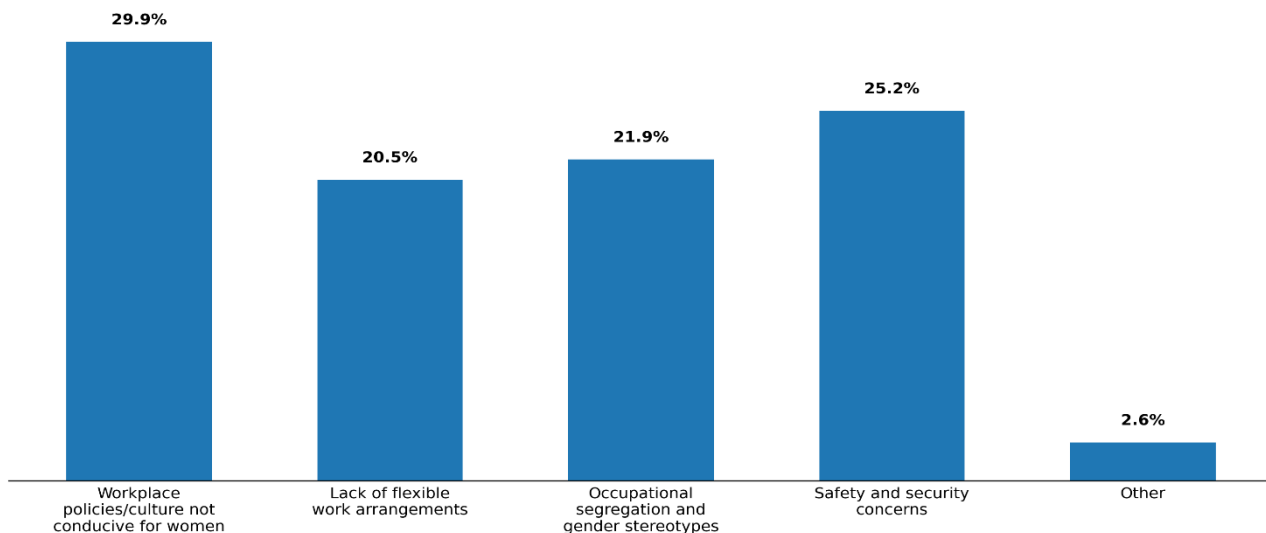


Figure 26: Challenges in Hiring or Retaining Female Employees

4.17.2. Disadvantaged Groups

Figure 27 presents the major challenges reported by employers in Sindh in hiring and retaining employees from disadvantaged groups. Accessibility issues related to physical infrastructure and transportation are the most frequently reported challenge at 32.4%. Inadequate skills training opportunities are the second most cited at 26.0%. Lack of supportive policies and reasonable accommodations is reported by 23.4%. Social stigma and discrimination are reported by 17.8%. Other challenges account for 0.4% of the challenges in integrating Disadvantaged groups in workforce.

The distribution indicates that barriers are primarily practical and institutional rather than purely attitudinal. Over half of responses are concentrated in accessibility constraints and lack of skills training opportunities, showing that potential candidates are being excluded before recruitment decisions are even possible. For Sindh, this implies that increasing inclusion requires coordinated action across training provision and workplace readiness. Priority actions include improving physical and transport accessibility in training centres and workplaces, embedding reasonable accommodations in employer HR practices, and expanding targeted training pathways for disadvantaged groups linked to placement. In parallel, stigma and discrimination must be addressed through employer sensitization and enforcement of non-discrimination practices so that inclusive recruitment is sustained after hiring.

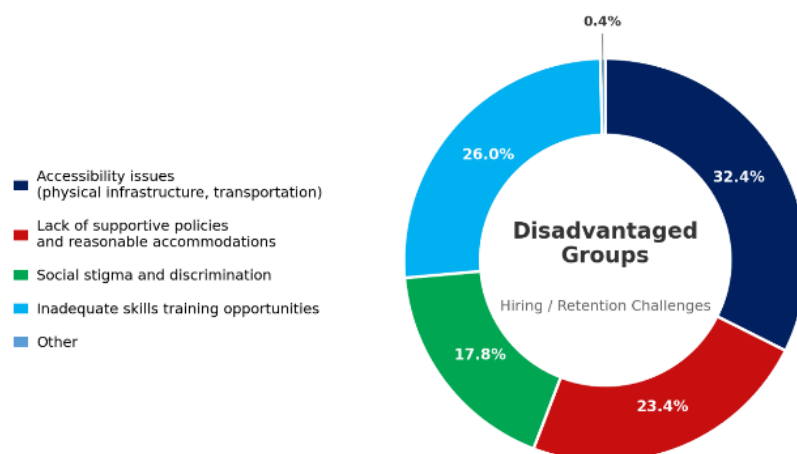


Figure 27: Challenges in Hiring/Retaining Employees from Disadvantaged Groups

4.17.3. Workforce Development and Skills Acquisition

Figure 28 presents the top challenges reported by employers in Sindh with regard to workforce development and skill acquisition. Skill misalignment remains the most pressing hurdle for the sector,

with 27.8% of organizations reporting a disconnect between academic training and practical job requirements. When combined with the technological shift (26.2%) and a tightening talent market (25.8%), the struggle to maintain a modern workforce is compounded by a lack of dedicated training resources (20.3%). This data underscores a dual challenge: a shortage of job-ready talent on the supply side and a lack of institutional capacity to upskill workers internally.

The high share for attracting and retaining talent suggests retention and wage competition pressures, while limited budgets signals constraints on in-house upgrading. For Sindh, this implies that skills policy must focus on stronger alignment and faster updating of training content. Priority actions include employer co-designed curricula, competency-based assessment with practical testing, regular review of trade standards to reflect technology shifts, and incentives for firm linked training such as apprenticeships and short cycle upskilling that reduce the cost burden on employers.

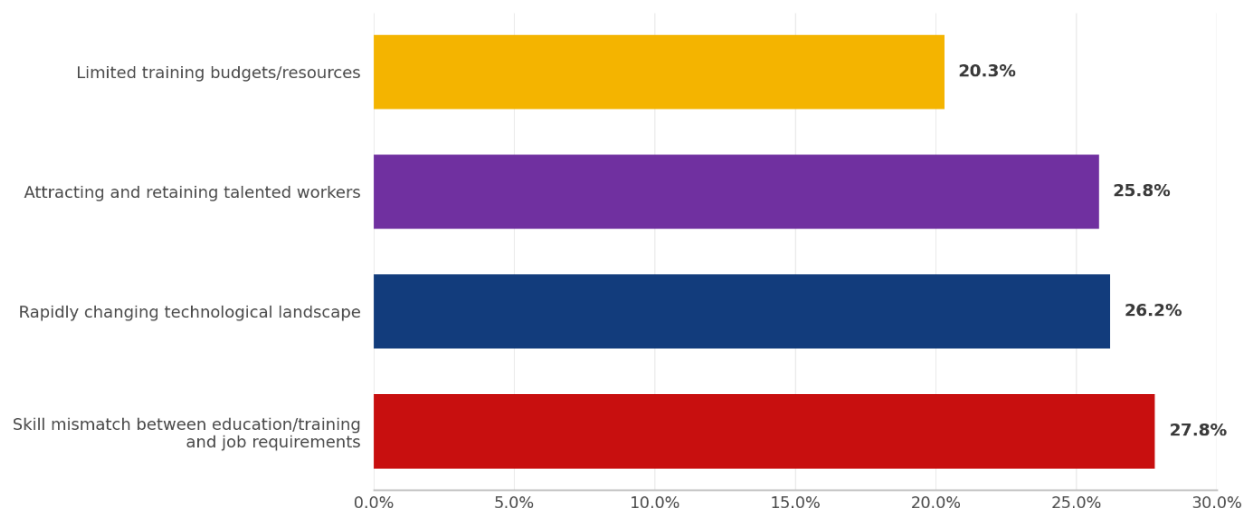


Figure 28: Key Issues Emerging in Workforce Development and Skill Acquisition

4.18. Suggestions for Improvement

4.18.1. Quality and Relevance of TVET Programmes

Figure 29 presents employer recommendations for improving the quality and labour market relevance of TVET programmes in Sindh. The most frequently identified priority is modernizing curricula and ensuring stronger alignment with industry needs, reported by 32.3% of employers. This is followed by the need to strengthen practical and hands-on skills development (24.6%), indicating employer concerns regarding the gap between classroom-based learning and workplace requirements. Improving the quality of instructors and trainers is identified by 22.0% of respondents, while enhancing industry exposure through internships and workplace learning opportunities is highlighted by 20.4%. Other suggestions account for a small proportion of responses (0.8%).

The findings demonstrate that employers view TVET improvement as a combination of curriculum relevance, practical skill development, trainer capacity, and stronger industry engagement. The emphasis on curriculum modernization highlights the need for training programmes to respond continuously to changing technologies, occupational requirements, and sector-specific skill demands. Similarly, the importance assigned to practical training reflects the need to move beyond theory-based instruction toward competency-based approaches supported by appropriate equipment and assessment systems.

For Sindh, these priorities require an integrated reform approach. Curricula should be regularly reviewed and co-designed with employers, particularly in high-demand sectors and occupations. Practical training hours, laboratory facilities, and trade-specific equipment should be strengthened, while instructors should receive continuous professional development and certification. In addition, structured internship and apprenticeship arrangements should be institutionalized to ensure that workplace exposure becomes a standard component of TVET delivery and improves graduate employability.

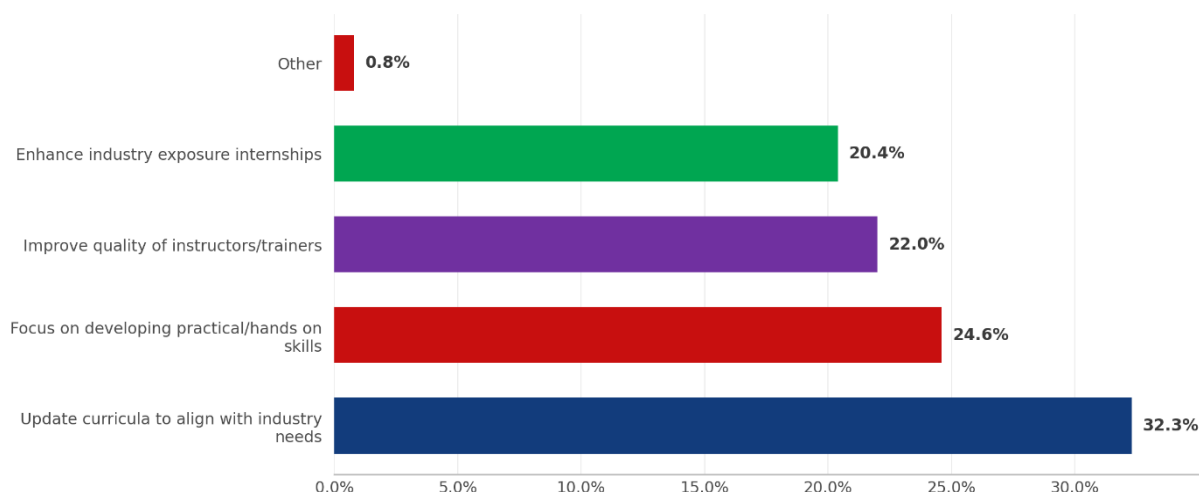


Figure 29: Suggestions for Improving the Quality and Relevance of TVET Programmes

4.18.2. Green Skills

Figure 30 presents the green skills that employers consider most relevant across industries in Sindh. Environmental compliance and regulations are the most frequently cited green skill area at 21.8%, followed closely by waste management and recycling at 21.2% and energy efficiency practices at 21.2%. Sustainable manufacturing processes are reported by 20.3% of employers, while digital skills linked to green and resource-efficient operations account for 14.7%. Other green skills represent 0.9% of responses.

The distribution indicates that employers are not treating green skills as a niche topic. The top four items are closely clustered around 20% to 22%, showing broad-based demand for practical competencies that support compliance, operational efficiency, and cleaner production. For Sindh, this supports integrating green modules across priority trades rather than creating stand-alone courses. Environmental compliance content should be embedded into sector-specific training standards. Waste management and recycling practices should be linked to manufacturing, construction, and municipal services occupations. Energy efficiency practices should be integrated into electrical, mechanical, and facility-related trades. Sustainable manufacturing processes should be aligned with the manufacturing and agribusiness value chains identified earlier in Chapter 4 so that training directly reflects the highest-demand sectors.

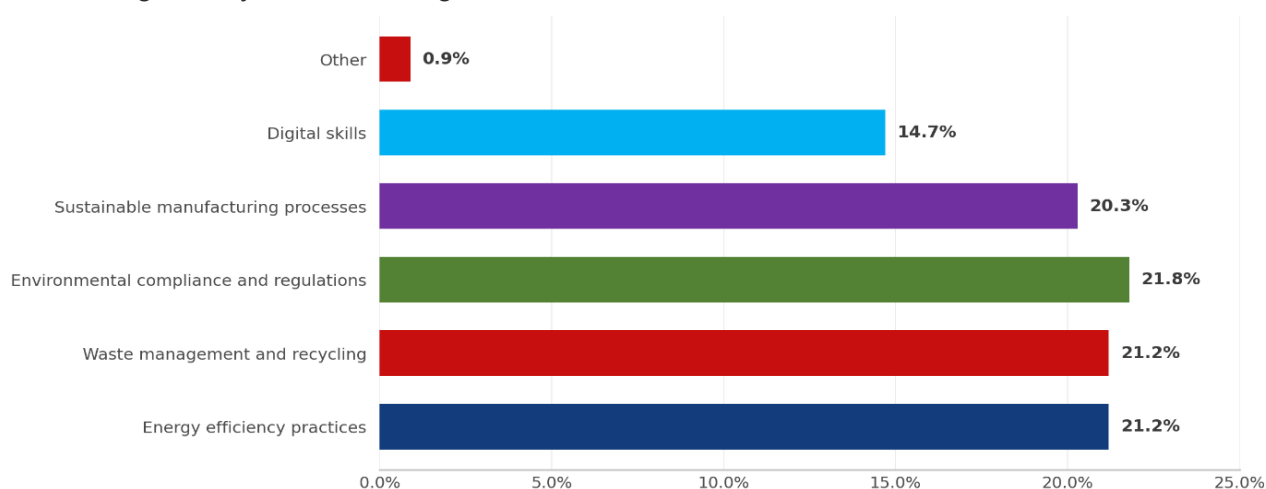


Figure 30: Green Skills across Industries

4.18.3. Introducing Green Skills in TVET Programmes

Figure 31 presents employer suggestions for integrating green and environmental skills within TVET programmes in Sindh. The most frequently reported suggestion for integrating green skills programmes in TVET is practical training in green technologies at 27.5%. Integrating concepts across relevant trades and disciplines is reported by 25.7%. Collaborations with environmental

organizations is reported by 23.3%. Incorporating dedicated courses or modules is reported by 22.6%. Other suggestions account for 1.0%.

The distribution indicates that employers prefer an applied integration model rather than purely classroom-based content. The strongest signal is for practical exposure to green technologies, supported by cross trade integration so that environmental competencies become part of core occupational standards. The sizeable share for collaboration with environmental organizations suggests a demand for external expertise, field exposure, and credible delivery partners. For Sindh, this implies that green integration should be implemented through three coordinated actions. First, develop trade specific practical modules and lab-based competencies in priority sectors such as manufacturing, construction, energy, and water. Second, embed environmental compliance, waste management, and energy efficiency competencies within existing curricula and assessments so that all graduates in relevant trades acquire these skills by default. Third, formalize partnerships with environmental agencies and organizations to support trainer upskilling, curriculum inputs, workplace exposure, and certification pathways.

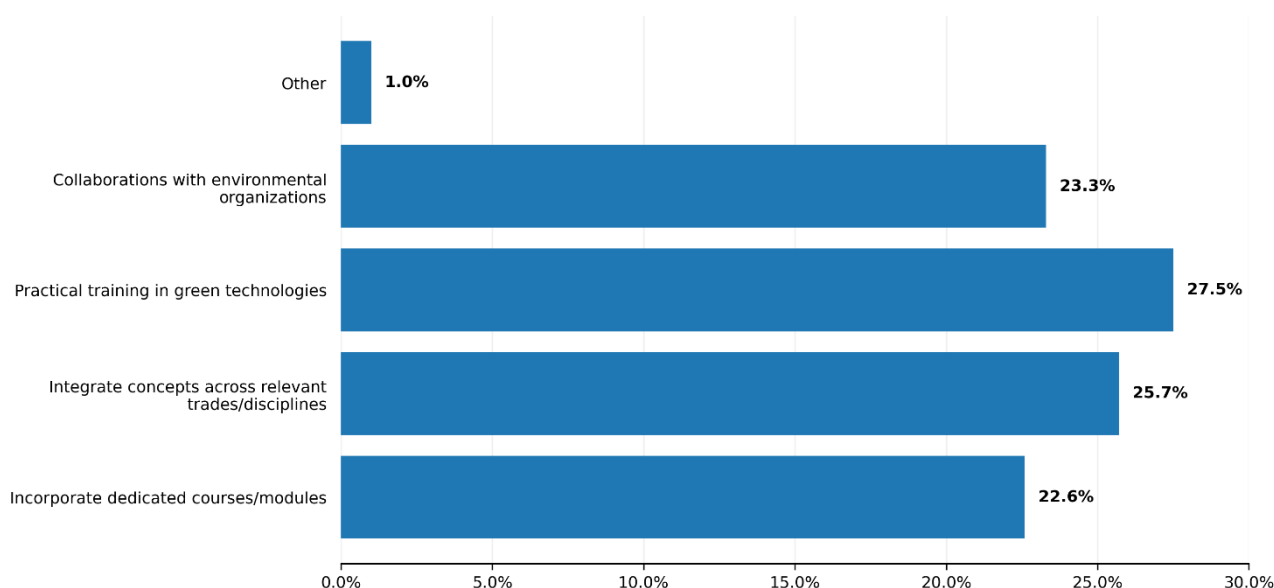


Figure 31: Suggestions for Introducing Green Skills in TVET Programmes

4.18.4. Improving the Overall TVET System

Figure 32 presents employer suggestions for improving the overall TVET system in Sindh. The most frequently reported suggestion is increasing funding and investment in TVET infrastructure at 31%. Better coordination between industry, government, and training providers is the second most cited at 26%. Stronger quality assurance and accreditation is reported by 23%. Promotion of TVET as a viable career pathway is reported by 20%. Other suggestions account for 1%.

The distribution indicates that employers see system performance as a governance and capacity issue rather than only a curriculum issue. The emphasis on infrastructure funding suggests gaps in equipment, labs, and delivery capacity needed for practical training. The strong call for coordination indicates weak linkages among key actors, which reduces alignment between training supply and labour market demand. Quality assurance and accreditation being cited by nearly a quarter of employers reflects concerns about inconsistent standards and outcomes across providers. For Sindh, these results support a system level reform package that increases investment in training infrastructure, formalizes industry government provider coordination mechanisms, strengthens accreditation and quality assurance processes, and improves public awareness and signalling so that TVET is perceived as a credible pathway into skilled employment.

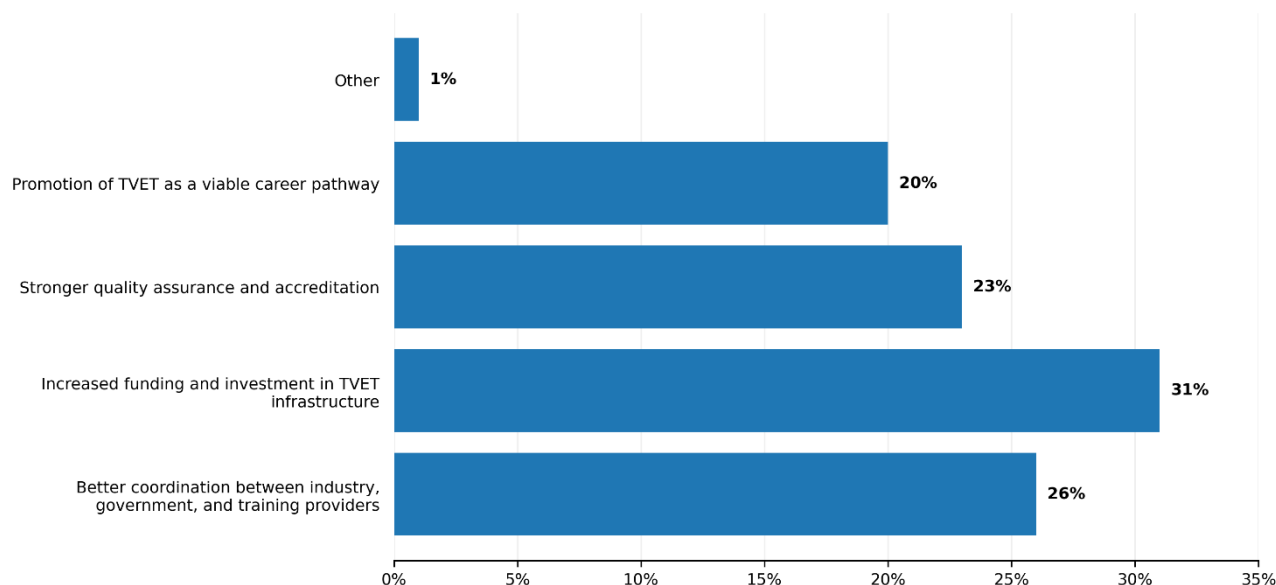


Figure 32: Suggestions for Improving Overall TVET System

4.18.5. Enterprises Having Dedicated Training Facilities

Figure 33 presents the availability of a dedicated training and learning and development department among surveyed employers in Sindh. Only 32.4% of employers report that they have a dedicated department, while 67.6% report that they do not.

The distribution indicates that structured in house capacity for workforce development is limited across firms. With over two thirds of employers lacking a dedicated function, training is likely being handled informally or on an ad hoc basis, which reduces consistency in skill upgrading and weakens the feedback loop between skill needs and internal training response. For Sindh, this implies that external support mechanisms and partnerships are essential to expand firm level training uptake. Priority actions include creating shared training platforms through employer associations, enabling pooled apprenticeship models, and providing standardized short course modules that firms can adopt without establishing full internal departments. This result is also consistent with the limited availability of organizational training programmes reported earlier in Section 4.14, reinforcing the need for system level incentives and coordination to increase workplace learning.

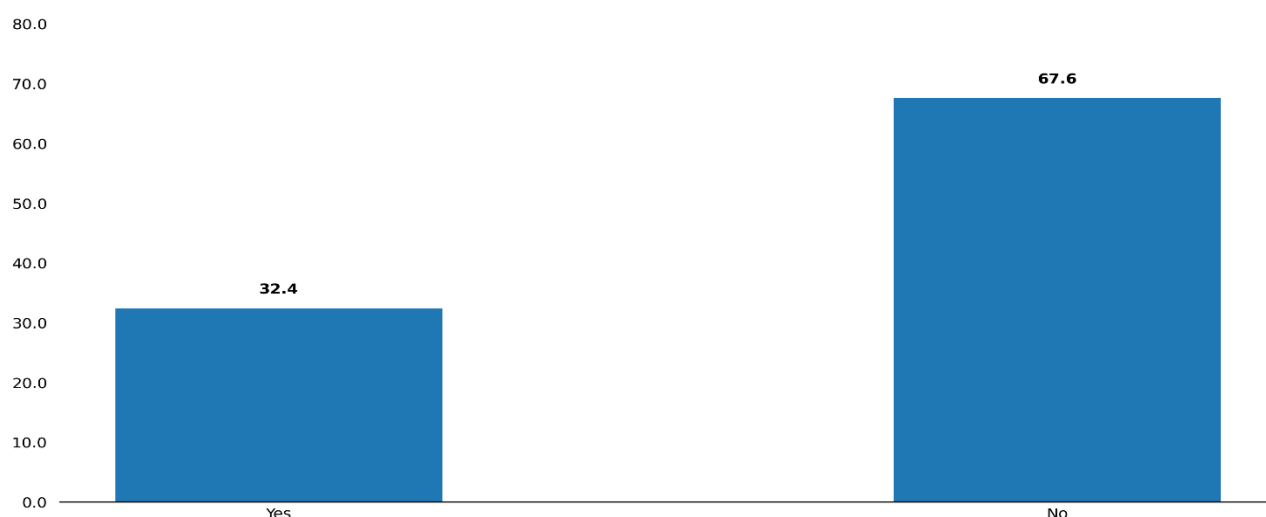


Figure 33: Availability of Dedicated Training Department

4.18.6. Identifying and Sourcing Potential Job Candidates

Figure 34 presents the preferred methods used by employers in Sindh for identifying and sourcing job candidates. Employee referrals are the most commonly reported method at 37.8%. Online job portals are the second most reported method at 30.5%. Print media advertisement is reported by

20.3%. Job fairs and career events are reported by 6.3% employers. Campus recruitment drives are reported by 4.0%. Other methods account for 1.0% of the methods used.

The distribution indicates that hiring is primarily driven by informal networks and digital platforms rather than institutional pipelines. Referrals and online portals together account for 68.3%, suggesting that many employers rely on low cost, rapid screening channels that may not systematically connect to TVET providers. The low use of campus recruitment and job fairs implies weak structured linkage between training institutions and employers at the point of hiring. For Sindh, this supports strengthening placement and job matching mechanisms, including formalizing employer engagement with training providers, establishing institutional job placement cells, and expanding work-based learning pathways that lead directly into recruitment. Improved employer provider coordination can also shift reliance away from referrals toward more transparent and competency verified hiring channels.

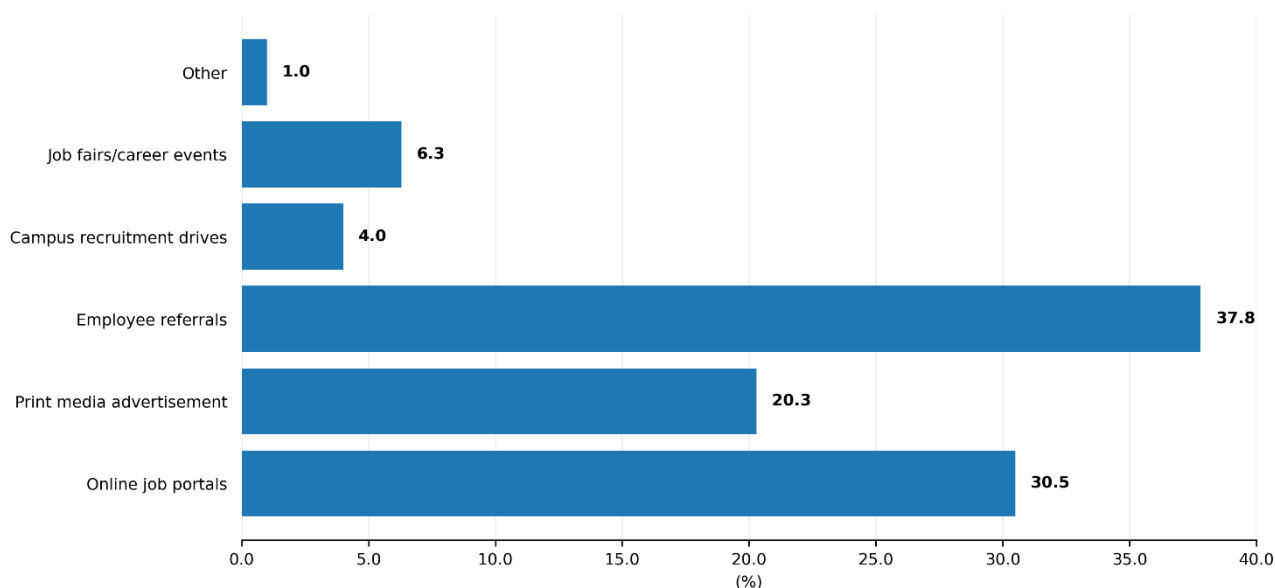


Figure 34: Preferred Methods for Identifying and Sourcing Potential Job Candidates

4.18.7. Competency Standards/Curricula

Figure 35 presents employers' willingness to participate in developing or reviewing competency standards and curricula in Sindh. A majority of employers (55%) expressed interest in contributing, while 45% were not willing to participate. This indicates a potential foundation for stronger industry engagement in improving TVET relevance, although participation requires structured mechanisms. Limited time, unclear benefits, weak coordination, or lack of formal engagement channels may explain lower participation. To strengthen collaboration, TVET institutions should establish sector-specific advisory groups, regular consultation mechanisms, and practical opportunities for employers to contribute through curriculum reviews, workplace assessments, and graduate performance feedback.

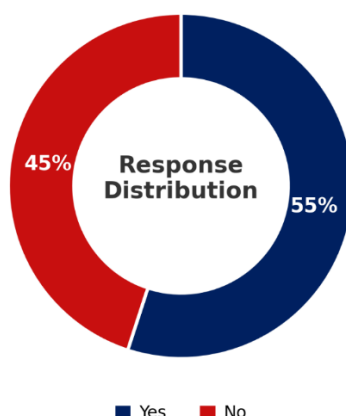


Figure 35: Interest in Participating in Development or Review of Competency Standards and Curricula

4.18.9. Employee Retention and Turnover

Figure 36 presents whether employers in Sindh face challenges related to employee retention and turnover. The responses are almost evenly split. 49% of employers report that they face retention and turnover challenges, while 51% report that they do not.

The near balance indicates that retention pressures are significant for a large share of employers, but not universal across the market. This suggests differences by sector, occupation, wage structures, and workplace conditions. For Sindh, this supports a targeted approach rather than a single province wide assumption. For employers reporting challenges, priority responses include strengthening career progression pathways, improving job matching and workplace readiness of recruits, and supporting firm linked upskilling that increases worker productivity and retention incentives. For employers not reporting challenges, the focus should be on sustaining good practices through continued employer engagement in training design and placement so that stable retention environments are reinforced.

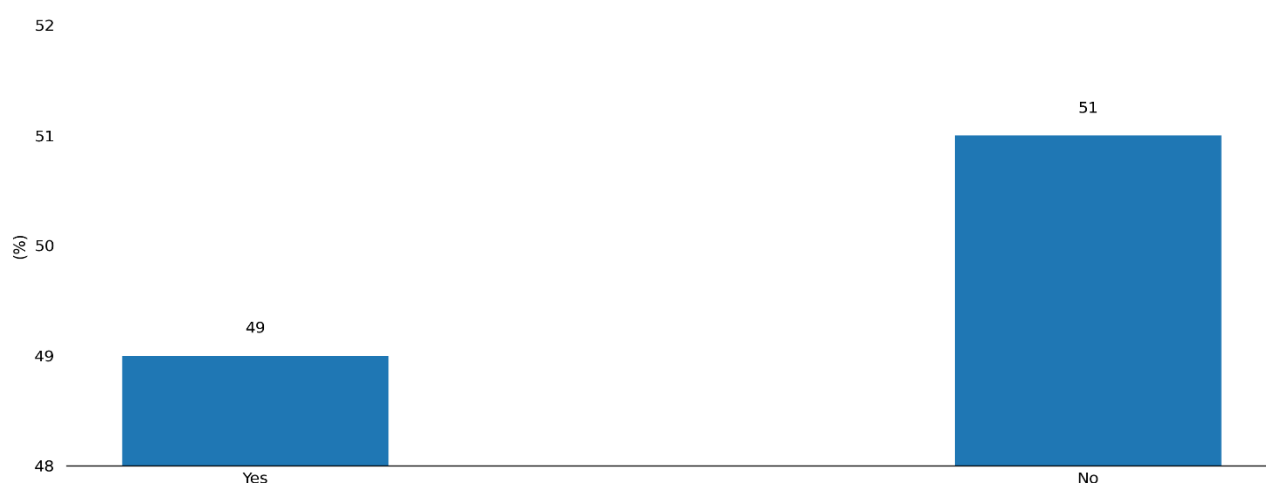


Figure 36: Challenges related to Employee Retention and Turnover

4.18.10. Digital/ICT Skills Workforce Demand

Figure 37 presents employer views on the importance of digital and ICT skills for current and future workforce needs in Sindh. The findings show that employers place strong importance on digital readiness. Overall, 54.7% of employers rated digital/ICT skills as very important or extremely important, comprising 44.2% who rated them very important and 10.5% who rated them extremely important. When the moderate category is also included, 82.4% of employers rated digital/ICT skills as at least moderately important. This means that the 82.4% figure includes the 54.7% who already rated these skills as very or extremely important, rather than representing an additional category.

The distribution indicates that digital capability is becoming a baseline requirement across sectors rather than a specialized skill limited to IT occupations. For Sindh, this implies that digital and ICT competencies should be embedded across priority trades and levels, including foundational digital literacy, workplace software use, data handling, inventory systems, communication tools, and technology-enabled processes in technical trades. The strong share of employers rating these skills as important also supports accelerating the integration of applied digital modules within TVET programmes, linked to practical workplace tasks and employer-validated tools, so that graduates can meet evolving labour market requirements.

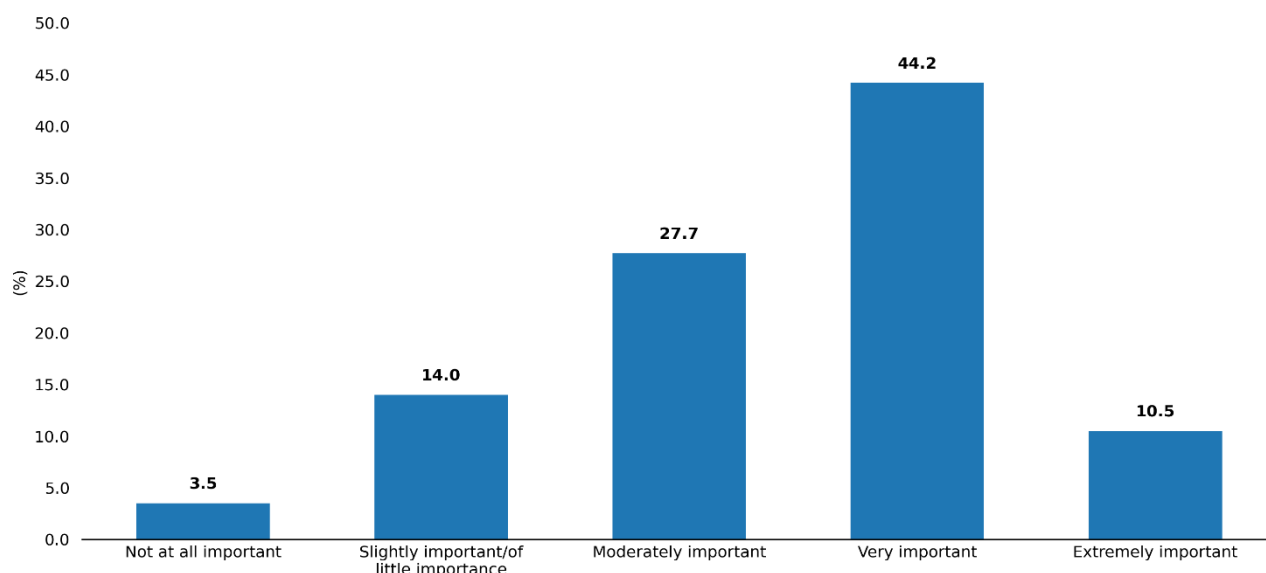


Figure 37: Importance of Digital and ICT Skills for Current and Future Workforce

4.18.11. Suggestions for Introducing New and Emerging Technologies in TVET Programmes

Figure 38 presents employer suggestions for introducing new and emerging technologies in TVET programmes in Sindh. The most frequently reported suggestion is to offer specialized courses and certifications at 31%. Collaboration with technology companies is the second most suggested option at 28%. Training instructors on new technologies is reported by 22%. Upgrading equipment and facilities is reported by 19%. Other suggestions account for 1%.

The distribution indicates that employers are prioritizing market relevant certification pathways and industry linked delivery mechanisms over standalone curriculum statements. The emphasis on specialized courses suggests demand for clear, competency-based credentials that signal employability. The strong call for collaboration with technology companies implies that providers need access to current tools, real use cases, and employer validated content. Trainer upskilling and facility upgrades are also central, indicating that delivery capacity and equipment readiness are binding constraints to technology adoption in training. For Sindh, this supports a sequenced approach that starts with targeted certifications in priority technologies, formal partnerships with technology firms for content and placements, structured trainer development, and phased upgrading of labs and workshops to meet competency requirements.

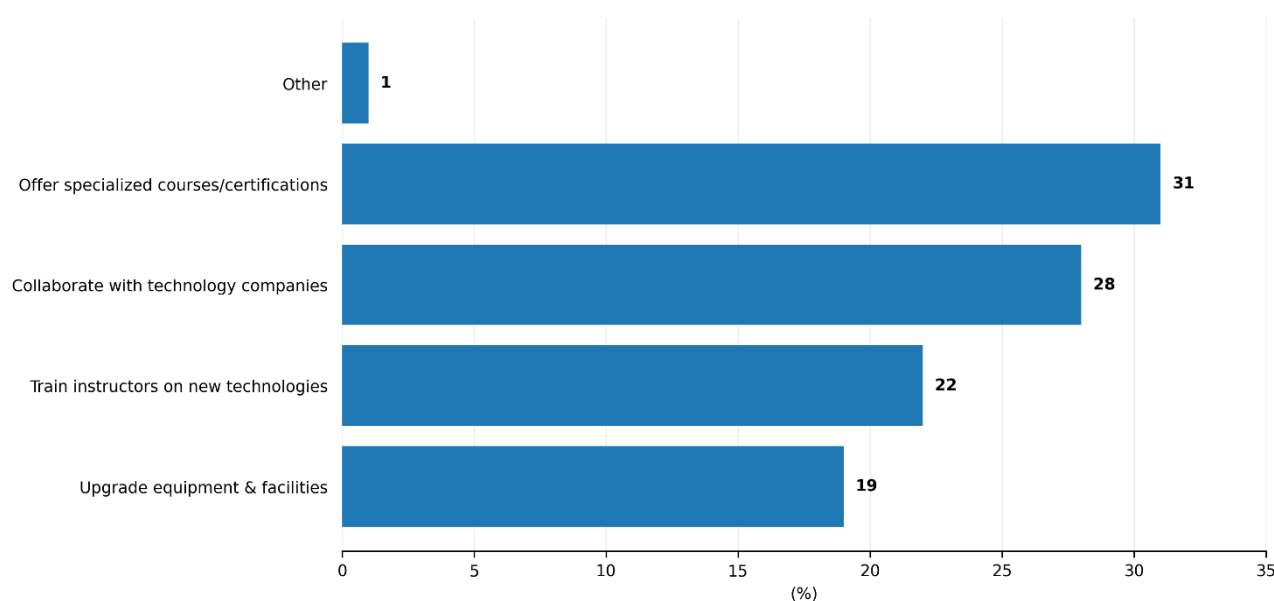


Figure 38: Suggestions for Introducing New and Emerging Technologies in TVET Programmes

4.18.12. Introducing New or Emerging Technologies

Figure 39 presents employer views on the key technologies that should be included in TVET programmes in Sindh. Artificial Intelligence as a demand has been the most frequently cited technology at 41%, 3D printing second most at 30%, Robotics at 22%, and other technologies at 6%.

The distribution indicates that employers are prioritizing advanced digital and automation related capabilities, with Artificial Intelligence receiving the strongest emphasis. The sizeable share for 3D printing suggests demand for design to production and prototyping related skills, while Robotics reflects interest in automation and mechatronics-oriented competencies. For Sindh, these results imply that technology integration should be applied and trade linked rather than offered as generic awareness modules. Artificial Intelligence content should be embedded through practical applications in relevant fields such as manufacturing quality, maintenance diagnostics, logistics, and service operations. Robotics and 3D printing should be developed through modular lab-based training aligned to technician and operator roles, with assessment based on hands on projects and industry validated equipment standards.

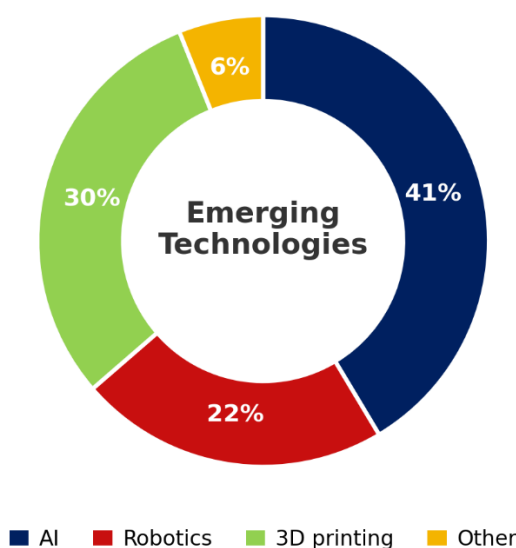


Figure 39: Demand in Niche and Emerging Areas

4.18.13. Key Digital Skills Expected to be in Demand in the Future Sindh

Figure 40 presents employer views on the top three digital skills expected to be in demand in the future in Sindh. Innovation and future skills are the most frequently reported area at 31.1%. Artificial intelligence and robotics are the second most cited area at 15.5%. Training and education enhancements is the third highest in demand at 11.9%. IT and digital skills is very close behind at 11.8% followed by specialized technical skills at 10.8%. Digital marketing and online work are reported by 9.9% and 3D printing and digital manufacturing by 9.0%.

The distribution indicates that employers are emphasizing future oriented capability building rather than narrow software use. The leading focus on innovation and future skills signals demand for adaptability problem solving and technology enabled work practices across occupations. The strong emphasis on artificial intelligence and robotics reinforces the need to embed applied automation and data driven workflows within relevant trades. For Sindh this supports designing modular pathways that combine foundational digital literacy with applied innovation projects and sector linked exposure to automation and emerging technology tools.

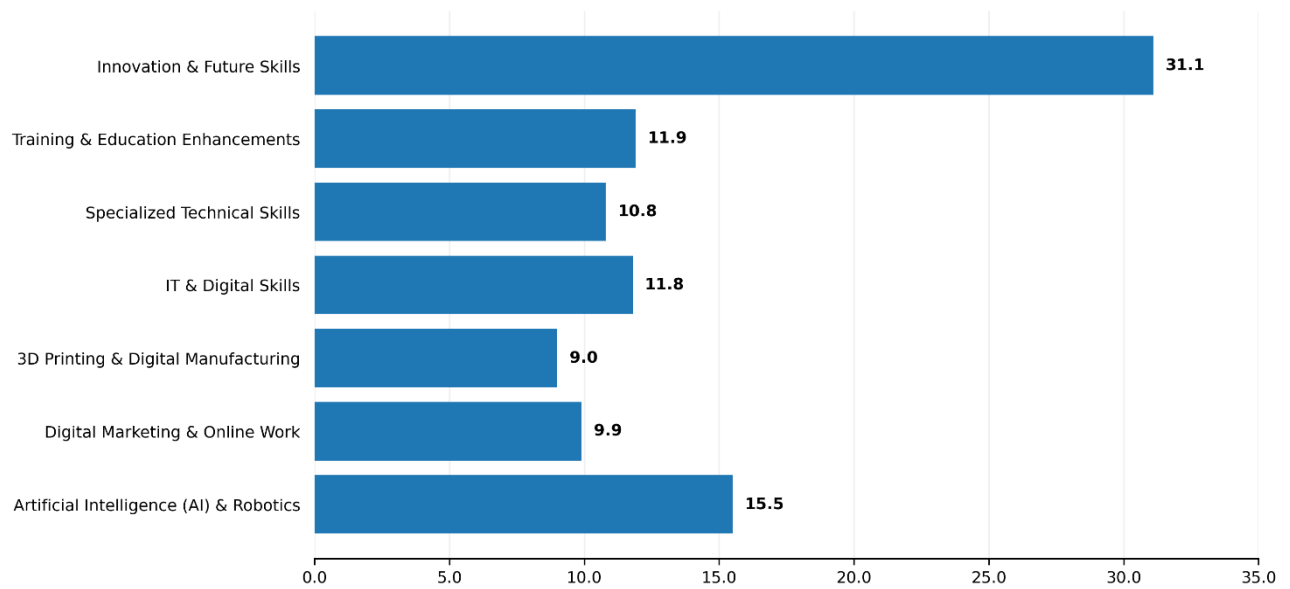


Figure 40: Key Digital Skills Expected to be in Demand in the Future

5. CONCLUSION

This assessment provides a robust baseline of Sindh's labour market and skills ecosystem and translates the analysis into findings that can be used for practical decision-making. The evidence shows that the province is characterized by concentrated industrial and service clusters, significant differences across districts, and a skills development system that needs to be aligned with actual job tasks rather than broad or generic sector classifications.

The labour market in Sindh is organized around a dominant metropolitan core and a set of secondary corridors and belts with distinct employment structures. Karachi functions as the primary node for maritime logistics, wholesale and retail trade, chemicals and pharmaceuticals, packaging, health services, and a growing ICT and business services layer. Outside Karachi, the Hyderabad–Jamshoro–Nooriabad corridor emerges as a second axis with light manufacturing, packaging, and services, while North Sindh is structured around wholesale intermediation, warehousing, and transport functions linked to inter-provincial trade routes. Coastal districts show a different profile where fisheries, landing sites, ice plants, cold storage, and connected logistics shape demand, while agrarian districts sustain agriculture, livestock, first-stage processing, and extensive retail and services around market towns. Sindh should not be interpreted as a single integrated labour market. It behaves as a set of connected but differentiated cluster economies. This finding justifies district-based stratification and prevents Karachi's scale from masking labour market realities in coastal, inland, and agrarian belts.

The study's two-tier sector taxonomy is a necessary design response to Sindh's industrial realities. Tier-A core sectors captured the province's major demand lanes for skills planning, while Tier-B sectors retained Sindh-specific distinctive functions, especially ports and logistics, chemicals and pharmaceuticals, ICT and business services, and wholesale and retail. As such, manufacturing could not be treated as one aggregated block if the goal is actionable skills policy. Textiles and garments, chemicals and pharmaceuticals, light manufacturing, and sports goods have been found to differ in compliance regimes, production technologies, and training pipelines. Treating them separately has not been a technical preference but a requirement to avoid generic findings that cannot be operationalized by TVET providers or employer partners.

Across priority sectors, demand is best described in occupation families and task bundles rather than in broad job titles. Ports and logistics require equipment operation, warehousing, inventory systems, customs documentation, and safety-critical handling. Chemicals and pharmaceuticals require documentation discipline, quality and process support roles, utilities operations, and safety awareness. Packaging and printing require pre-press workflows, press operations, finishing, and quality checking, often under supplier compliance expectations. Construction demand combines core trades with safety and efficiency-oriented work practices. Health services demand includes technician-level roles and facility-linked support functions. ICT and business services demand centres on service desk operations, productivity tools, customer support workflows, and basic cyber hygiene.

A key conclusion is that “digital” and “green” capabilities do not function as standalone sectors within Sindh's labour market. Instead, they emerge as cross-cutting competency layers integrated into traditional occupations, such as energy-efficiency tasks within maintenance roles, inverter diagnostics and photovoltaic (PV) operations and maintenance within electrical career pathways, and technology-enabled workflows in logistics and retail. This finding is central for curriculum design because it shifts the emphasis from isolated new trades to modular add-ons that can be embedded in existing certification pathways.

Participation patterns and workplace conditions operate as binding constraints on employability and retention. Women's feasible entry points are primarily found in occupations with more predictable and safer working conditions, including health services, education, packaging and quality inspection, retail and customer support, and selected functions within the hospitality sector.

Youth pathways are visible in warehousing, retail operations, supervised construction trades, PV installation assistance, and call-centre roles with structured progression. Persons with disabilities and transgender persons have feasible access points in roles where accommodation and anti-discrimination practices are present, particularly in customer support, documentation, and packaging environments. Inclusion cannot be treated as an outreach narrative. This constitutes both a survey

design requirement and a policy interpretation requirement. In the absence of explicit quotas, screening of workplace conditions, and sector-specific inclusion lenses, aggregate findings would systematically understate barriers faced by women and other disadvantaged groups, while overstating the availability of jobs that are not realistically accessible.

The findings imply that Sindh's training system must be oriented around cluster realities, employer-defined task profiles, and competency-based modular pathways that map cleanly to occupation families. Skills planning must follow district concentrations and industrial estate catchments so that recommendations are anchored where firms actually recruit and where occupational demand is sustained. Sector coding rules and the PSIC crosswalk are therefore not an annex exercise; they are a governance mechanism that protects comparability, prevents misclassification, and enables implementable training recommendations.

6. RECOMMENDATIONS

Some policy recommendations are made here to help make Sindh's TVET system more responsive, inclusive, and aligned with labour market needs.

1. Adopt a province-wide “demand-led planning rule” that ties all public TVET funding, institute intake, and short-course approvals to the Employer Skills Survey and the PSIC-to-study sector crosswalk: This makes the skills system accountable to employer task demand rather than legacy course portfolios. Require every provider to submit an annual intake plan showing how seats map to the study sectors and priority occupation families. Any programme that cannot demonstrate employer relevance should be paused and restructured.

2. Create sector-specific Centres of Excellence linked to real industrial clusters rather than administrative convenience: Establish cluster-anchored CoEs for maritime, ports and logistics (Karachi South and Keamari); chemicals and pharmaceuticals with GMP and utilities capability (Korangi, Landhi, Bin Qasim); printing and packaging with pharma supplier focus (Korangi, SITE, Nooriabad); industrial maintenance and PV service skills (Hyderabad, Jamshoro, Nooriabad); renewables including wind corridor competencies (Thatta and Jamshoro); and cold-chain and fisheries handling (Thatta, Sujawal, Badin). Each CoE must have industry-grade equipment, standardized assessment, and guaranteed work placements.

3. Build mandatory employer governance into institutes through functional Institute Management Committees and sector employer councils with decision authority: IMCs should be required for public institutes, with named employer representation from the relevant district clusters. Sector councils must formally approve competency bundles, practical assessment methods, and minimum equipment standards before programmes receive public funding. This shifts governance from administrative compliance to labour market relevance.

4. Make apprenticeship and structured workplace learning a financing condition for priority technical programmes: Mandate every priority programme to demonstrate access to real workplaces, equipment, and supervised task exposure before it can enrol students. Employers receiving facilitation through SEZ and EPZ regimes should be obligated to host apprentices and participate in competency assessment. This operationalizes the industry-institute linkage instead of treating it as a recommendation.

5. Scale Recognition of Prior Learning as the fastest route to productivity gains in Sindh's high-informality segments: Launch RPL pipelines for construction, retail and repair, logistics and warehousing, agri-processing, and small manufacturing so informal workers can become certifiable and mobile. Pair RPL with short top-up modules for safety, quality, and digital tools. This reduces retraining time and raises employability without forcing workers into long-duration programmes.

6. Rationalize the training portfolio annually by phasing out oversupplied and low-absorption courses and reallocating seats to priority sectors: Institute a formal mechanism to filter programmes with persistently weak placement or wage outcomes to be paused and replaced. Seats should shift toward the sectors already embedded in the demand signal, including agriculture and livestock (230), wholesale, retail and repair (190), construction (153), allied health (141), and the industrial lanes such as textiles, packaging, chemicals and pharmaceuticals, and hospitality. This prevents supply drifting into politically convenient but economically weak course mixes.

7. Treat women's participation as a binding constraint and adopt a targeted policy package anchored to the female refined LFPR of 15.2 percent: Design training and placement around safe transport, childcare options, harassment safeguards, and facility standards, including sanitation and secure shift policies. Focus early scale-up on sectors with feasible entry roles identified in the inclusion pathways, then deliberately expand into higher-wage lanes such as PV after-sales and diagnostics, warehousing documentation, and quality inspection. Success should be measured through verified placement, retention, and wage progression, not enrolments alone.

8. Institutionalize inclusion quotas and accessibility screening for youth, persons with disabilities, and transgender persons across both survey operations and programme delivery: Require training providers to meet basic accessibility standards and employers to demonstrate minimal accommodation capacity for partnership eligibility. Build role-specific pathways where inclusion is feasible, such as customer support, documentation, packaging QA, EMR support, and selected service desk functions. This turns inclusion into an operational requirement rather than a narrative section.

9. Mainstream digital competence as a baseline requirement across all trades, not only ICT programmes: Introduce a standardized “digital essentials” module stack for inventory systems, service ticketing, basic spreadsheets, data hygiene, and workplace communication, then embed it into certification pathways for logistics, retail, allied health administration, and manufacturing support. This aligns with the reality that modern workflows depend on digital tools even in traditional sectors. Certification should require practical digital demonstration, not only theory.

10. Mainstream green skills through modular add-ons to conventional occupations and prioritize energy-transition roles where Sindh already has clusters: Build green competency modules into electrical, industrial maintenance, construction, cold-chain, utilities, and water-linked tasks, so green transition becomes system-wide rather than siloed. Prioritize PV installation and commissioning, inverter diagnostics, preventive maintenance, energy efficiency practices, and safe working procedures, especially in the Hyderabad–Jamshoro corridor and the Thatta–Jhimpir wind corridor. Make green competence measurable through task-based assessments.

11. Strengthen occupational health and safety as a core labour market competency and integrate it into both training and employer partnership rules: Require OHS modules and practical checks for construction, ports and logistics, chemicals and pharmaceuticals, and fisheries cold-chain environments. Providers without enforceable OHS capacity should not run high-risk trades. Employers should be screened for basic safety compliance before being used as apprenticeship sites.

12. Convert SEZ and EPZ facilitation into enforceable workforce development commitments for tenant firms: Tenants to be mandated to host apprentices, participate in competency assessment, and provide structured workplace exposure for priority occupations relevant to their sector. Tie compliance to facilitation services and renewals so the policy instrument has teeth. This anchors training pipelines where actual jobs exist.

13. Establish a Sindh Labour Market Information System that integrates employer demand, TVET supply, placement outcomes, and district-level labour market indicators into one decision dashboard: The system should consolidate Sindh-specific evidence, including refined labour force participation rate, crude labour force participation rate, unemployment rate, sector-wise skilled workforce demand, district-wise demand, trade-wise shortage and surplus, training supply by institute and trade, graduate placement rates, employer satisfaction, retention outcomes, and inclusion indicators for women and disadvantaged groups. The dashboard should be updated periodically using administrative data from TVET providers, employer feedback, placement records, and refreshed labour market surveys. This will create a single Sindh-specific evidence base for annual intake planning, course rationalisation, district prioritisation, apprenticeship targeting, and monitoring of whether training investments are translating into employment outcomes.

14. Adopt a Sindh youth transition package linked to verified district and sector demand, with structured entry roles and progression ladders: Youth-focused TVET pathways should be designed around the occupations and sectors where employer demand is already visible in Sindh, including retail operations, warehouse and logistics support, supervised construction trades, PV installation assistance, allied health support roles, computer operations, and customer service functions. The package should combine career counselling, short-cycle competency-based training, apprenticeships, internships, workplace readiness modules, and placement support. Employers participating in these pathways should define progression milestones so that young workers do not remain trapped in casual or low-skill roles. Outcomes should be tracked through Sindh-specific indicators such as placement rates, three- and six-month retention, wage progression, employer satisfaction, and transition from entry-level roles into certified skilled occupations.

15. Implement district-specific value chain training packages for coastal and agrarian belts to protect non-Karachi economic identities: In coastal districts, prioritize fisheries handling, cold-chain operations, equipment maintenance, quality and safety routines, and logistics linkage to Karachi wholesale and export channels. In agrarian districts, prioritize post-harvest handling, packhouse operations, basic QA, and small equipment maintenance with productivity focus. This prevents one-size-fits-all training that fails outside Karachi.

16. Create a province-wide placement and employer feedback mechanism that is continuous, standardized, and used to trigger programme redesign: Require employers engaged through the survey and partnerships to provide structured feedback on graduate task readiness, practical competence, and soft skills. Providers should be encouraged to use this feedback in bi-annual programme reviews that lead to curriculum adjustment or programme discontinuation. This is the operational backbone of responsiveness.

17. Adopt performance-based funding for public providers tied to verified placement, retention, and employer satisfaction rather than enrolments: A simple provider scorecard should be introduced, covering placement rates, three- and six-month retention, employer satisfaction, and wage progression against a baseline reference. Providers that demonstrate stronger outcomes should receive priority access to additional funding, new course approvals, equipment support, and partnership opportunities. Providers with weak outcomes should be required to improve through corrective action plans, stronger employer linkages, curriculum revision, and placement support before receiving expanded allocations. This will shift incentives away from enrolment-based delivery and toward measurable labour market outcomes.

18. Implement an accountability mechanism that converts the assessment into timed actions with named owners, measurable targets, and annual public reporting: Assign each priority sector lane a lead agency and partner agencies, set annual targets for training seats, apprenticeships, RPL certifications, and inclusion outcomes, and publish progress updates. Require each district to produce a short annual “district skills action note” that lists its clusters, priority occupations, inclusion plan, and seat allocation. This ensures the assessment functions as an implementation tool rather than a one-off report.

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ANNEXURES

Annex A: Skilled Workforce Demand Side Questionnaire

Confidentiality Statement: The information collected through this questionnaire will be kept strictly confidential and will be used only for research, planning, and strengthening of the National Skills Information System, NAVTTC, Government of Pakistan.

Introduction and Consent

Assalam-o-Alaikum,

My name is **[Enumerator Name]**, and I am representing the **TVET Sector Support Programme**, implemented by GIZ in close coordination with NAVTTC and TEVTAs in Sindh. This survey aims to support the improvement of Technical and Vocational Education and Training (TVET) programmes by identifying labour market needs, skills gaps across key economic sectors, weaknesses in current training provision, and emerging growth areas.

As part of this study, we are collecting information from employers to better understand the current state of labour market demand, skilled workforce requirements, and the relevance and effectiveness of existing TVET initiatives in both public and private sectors. Your participation is important for generating evidence that can guide policy planning, programme design, and future skills development interventions.

Your participation in this survey is completely voluntary. There is no payment for participation. Your responses will remain confidential and anonymous. The information collected will be analysed and reported only in aggregated form, without mentioning the names or identities of respondents or establishments.

Please feel free to ask for clarification about the study at any time. For any questions or concerns, you may also contact the Team Leader of the survey:

Mr. Himat Ullah

Phone: 0334-5259088

Do you agree to participate in this survey?

1. Yes
2. No — End interview

Section A: Basic Information

Item	Response
Name of Organization / Establishment	_____
Date	_____
Province	_____
District	_____
Email	_____
Name of Industrial Zone, if applicable	_____
Address of Establishment	_____

A1.1 Size of Enterprise / Establishment

Select only one answer.

Code	Response
1	Large, 250 persons and more
2	Medium, 100–249 persons
3	Small, 10–99 persons
4	Micro, less than 10 persons

A1.2 Enterprise Ownership

Select only one answer.

Code	Response
1	Public
2	Private

A1.3 Location of Enterprise

Select only one answer.

Code	Response
1	Urban
2	Rural

A1.4 Legal Form of Enterprise

Select only one answer.

Code	Response
1	Limited liability company
2	Joint-stock company
3	Cooperative
4	Private enterprise
5	Public enterprise
6	Other form, please specify: _____

A1.5 Economic Activity of Enterprise

In which economic activity does the enterprise mainly operate? Select only one answer.

Code	Economic Activity
1	Agriculture, Livestock and Forestry
2	Fishing
3	Textile & Garments
4	Chemicals & Pharmaceuticals
5	Leather and Footwear
6	Lighting and Other Manufacturing
7	Sports Goods & Equipment
8	Mining and Quarrying
9	Water Supply, Sewerage, Waste Management and Remediation Activities
10	Hospitality & Tourism
11	ICT & Business Services
12	Construction
13	Printing and Packaging
14	Allied Health Services
15	Wholesale, Retail & Repair
16	Energy
17	Automobile Repairing
18	Maritime, Ports & Logistics
19	Other Services

A1.6 Current Employment in the Establishment

How many people are currently employed in your establishment, including full-time and part-time workers?

S. No.	Period	Full-Time Employees	Part-Time Employees
1	Currently, as of _____ 2025	_____	_____
2	Last year, as of _____ 2024	_____	_____

Section B: Skilled Workforce Demand

B2.1 Future Skilled Workforce Demand

Please list the occupations for which your establishment expects to require skilled workers in the future. Examples may include Machine Operator, Software Developer, Building Electrician, Nurse, Technician, Plumber, Welder, etc.

S. No.	Occupation / Trade	Expected Number Required per Annum — Male	Expected Number Required per Annum — Female	Total Expected Demand per Annum	Remarks, if any
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					

B2.2 Quotas for Disadvantaged Groups

Does your organization reserve quotas or specific seats for any of the following groups: minorities, persons with disabilities, transgender individuals, or elderly persons?

Code	Response
1	Yes
2	No

B2.3 Details of Reserved Seats for Disadvantaged Groups

If the answer to B2.2 is “Yes”, please provide details in the table below.

Codes for disadvantaged groups:

- 1 = Persons with Disabilities
- 2 = Minorities
- 3 = Transgender Individuals
- 4 = Elderly Persons

S. No.	Occupation / Trade	Disadvantaged Group Code	Number of Reserved Seats
1			
2			
3			
4			
5			

Section-C

Employer Questionnaire on TVET Programmes, Skills Needs, and Industry Collaboration

Instruction: Please select the most appropriate response option. For open-ended questions, kindly provide brief details where required.

S. No.	Question	Response Options
1	How satisfied are you with TVET graduates?	1. Very satisfied 2. Satisfied 3. Neutral 4. Dissatisfied 5. Very dissatisfied
2	What specific skills do you find lacking in TVET graduates?	Open-ended response
3	Does your organization offer training programmes such as on-the-job training, apprenticeships, or online	1. Yes 2. No

S. No.	Question	Response Options
	courses?	
4	If yes, what types of training programmes does your organization offer?	Open-ended response
5	How do you currently assess the skills and competencies of job candidates during the hiring process?	1. Written tests/exams 2. Practical skills assessments 3. Personal interviews 4. Group discussions/activities 5. Other, please specify
6	Do you have any specific suggestions for improving the quality and relevance of TVET programmes?	1. Update curricula to align with industry needs 2. Focus on developing practical/hands-on skills 3. Improve quality of instructors/trainers 4. Enhance industry exposure and internships 5. Other, please specify
7	Which green skills are relevant for your organization or sector?	1. Energy efficiency practices 2. Waste management and recycling 3. Environmental compliance and regulations 4. Sustainable manufacturing processes 5. Digital skills 6. Other, please specify: _____
8	Do you have any specific suggestions for introducing green skills or environmental skills in TVET programmes?	1. Incorporate dedicated courses/modules 2. Integrate concepts across relevant trades/disciplines 3. Provide practical training in green technologies 4. Establish collaborations with environmental organizations 5. Other, please specify
9	What are the major challenges in hiring or retaining female employees?	1. Workplace policies/culture not conducive for women 2. Lack of flexible work arrangements 3. Occupational segregation and gender stereotypes 4. Safety and security concerns 5. Other, please specify
10	What are the major challenges in hiring or retaining employees from disadvantaged groups?	1. Accessibility issues, including physical infrastructure and transportation 2. Lack of supportive policies and reasonable accommodations 3. Social stigma and discrimination 4. Inadequate skills training opportunities 5. Other, please specify
11	What are the top three challenges your organization faces in workforce development and skill acquisition?	1. Skills mismatch between education/training and job requirements 2. Rapidly changing technological landscape 3. Attracting and retaining talented workers 4. Limited training budgets and resources 5. Other, please specify
12	Do you have any suggestions for improving the overall TVET system in your region?	1. Better coordination between industry, government, and training providers 2. Increased funding and investment in TVET infrastructure 3. Stronger quality assurance and accreditation mechanisms 4. Promotion of TVET as a viable career pathway 5. Other, please specify
13	Do you have any specific suggestions for promoting greater industry-TVET collaboration?	1. Establish formal industry advisory boards/committees 2. Involve industry in curriculum design and review 3. Facilitate more internship and apprenticeship opportunities 4. Encourage industry contributions, including equipment, instructors, or funding 5. Other, please specify: _____
14	Which technologies should TVET programmes include?	1. Artificial Intelligence 2. Robotics 3. 3D printing 4. Other, please specify
15	Do you have any specific suggestions for introducing new or emerging technologies in TVET programmes?	1. Upgrade equipment and facilities 2. Train instructors on new technologies 3. Collaborate with technology companies 4. Offer specialized courses/certifications 5. Other, please specify
16	Is there anything else you would like to	1. Yes, please specify

S. No.	Question	Response Options
	add or suggest regarding TVET programmes and industry collaboration?	2. No
17	Does your organization/enterprise have a dedicated training, learning, or development department/team?	1. Yes 2. No
18	What is your preferred method for identifying and sourcing potential job candidates?	1. Online job portals 2. Print media advertisement 3. Employee referrals 4. Campus recruitment drives 5. Job fairs/career events 6. Other, please specify
19	Would you be interested in participating in the development or review of competency standards/curricula for relevant trades or occupations?	1. Yes 2. No
20	Does your organization/enterprise face any challenges related to employee retention and turnover?	1. Yes, please specify the major reasons: a. Low wages b. Limited career progression c. Work-life balance issues d. Other, please specify 2. No
21	How important are digital/ICT skills for the current and future workforce needs of your organization?	1. Not at all important 2. Slightly important/of little importance 3. Moderately important 4. Very important 5. Extremely important
22	Do you have any suggestions for enhancing the development of digital/ICT skills in TVET programmes?	1. Yes, please specify 2. No
23	What are the three digital skills you think will be in demand in the future?	1. _____ 2. _____ 3. _____
24	Do you have any additional comments or suggestions for improving TVET programmes?	1. Yes, please specify 2. No

Annex B: Demand-Supply Gap in Sindh

Table 11: Trade-wise Demand-Supply Gap

Trade	Supply	Demand	Gap
Lathe Machine Operator	0	5,961	(5,961)
CNC Machine Operator	0	8,468	(8,468)
Textile Machine Operator	0	2,708	(2,708)
Java Script	100	3,360	(3,260)
Web Designing	245	2,916	(2,671)
Computer Operator	300	1,590	(1,290)
Cyber Security	0	1,392	(1,392)
PHP	0	984	(984)
Artificial Intelligence	0	762	(762)
Software Technician	545	27,410	(26,865)
Computer Operator	3,352	17,642	(14,290)
Accountant	0	8,356	(8,356)
Sales & Marketing	234	7,639	(7,405)
Construction Worker	0	7,255	(7,255)
Nurse	0	7,230	(7,230)
Quality Assurance	0	5,889	(5,889)
Sewing Machine Operator	117	5,130	(5,013)
Electrician	2,692	5,076	(2,384)
Civil Technician	3,220	5,049	(1,829)

Plumbing & Pipe Fitter	721	4,660	(3,939)
Packaging Machine Operation	0	4,457	(4,457)
Industrial Stitching Machine Operator	451	4,442	(3,991)
Cane Inspector	0	4,242	(4,242)
Helper	0	3,941	(3,941)
Chef	100	3,697	(3,597)
Manager	0	3,287	(3,287)
Lab Technician	0	3,158	(3,158)
Auto Mechanic	1,288	3,049	(1,761)
Fitter	0	3,009	(3,009)
CRM (Customer Relationship Management)	0	2,859	(2,859)
Solar PV System Technician	591	2,850	(2,259)
Electrical Technician	3,409	2,835	574
Trained Labour	0	2,744	(2,744)
Driver	20	2,743	(2,723)
Solar Technician	0	2,718	(2,718)
Dyeing Machine Operator	0	2,584	(2,584)
Production Helper	0	2,549	(2,549)
Micro Paper Cutting Machine Operator	0	2,513	(2,513)
Graphic Designer	3,640	2,364	1,276
Welder	1,264	2,245	(981)
Cutting Machine Operator	0	2,093	(2,093)
Digital Marketing & Search Engine Optimization (SEO)	1,518	2,070	(552)
Mechanical Technician	2,638	1,966	672
Cook	699	1,908	(1,209)
D.I.T.	5,235	1,815	3,420
Crane Operator	0	1,666	(1,666)
Medical Technician	0	1,494	(1,494)
Pharmacy Technician	0	1,479	(1,479)
Printing Machine Operator	117	1,438	(1,321)
Production Technician	0	1,409	(1,409)
Mason	0	1,395	(1,395)
Information Technology (C.I.T.)	3,823	1,391	2,432
Building Electrician	0	1,336	(1,336)
Waiter	0	1,312	(1,312)
Supply Chain & Logistics	0	1,244	(1,244)
Packing Machine Operator	0	1,161	(1,161)
Architectural Technician	0	1,100	(1,100)
Maintenance Technician	0	1,059	(1,059)
Sweeper	0	1,017	(1,017)
Artificial Intelligence	117	1,014	(897)
Medicine Packing Machine Operator	0	984	(984)
Chemical Technician	332	968	(636)
Fabric Cutting Machine Operator	0	968	(968)
Auto Mechanic Technician	0	963	(963)
Product Manager	0	946	(946)
Fish Drying Machine Operator	0	924	(924)
Chef BBQ	0	920	(920)
Call Centre Operator	0	910	(910)
Storekeeper	0	894	(894)
Concrete Machine Operator	0	873	(873)
Water Treatment Plant Operator	0	852	(852)
Carpentry	482	844	(362)
Doctor (MBBs)	0	840	(840)
Safety Officer	0	826	(826)
Auto Electrician	0	805	(805)
Veterinary Technician	0	782	(782)
Textile Designing with CAD	275	756	(481)

Business Management	0	755	(755)
Cleaner	0	750	(750)
Finishing Supervisor	0	748	(748)
Logistic & Supply Technician	0	748	(748)
Livestock Technician	0	730	(730)
Sanitation Technician	0	725	(725)
Auxiliary Staff	0	724	(724)
Boiler Plant Operator	0	720	(720)
Architecture Technician	0	717	(717)
Medical Credential	0	696	(696)
Heavy Machinery Operator (including Bulldozer, Loader, Crane, Dozer, Grader, Excavator, etc.)	111	678	(567)
Data Analyst	0	676	(676)
Agricultural Technician	0	659	(659)
Fire & Safety Management	0	658	(658)
Childcare Attendant/Aya	0	642	(642)
Environmental Engineer (Water & Wastewater)	0	642	(642)
Peco Machine Expert	0	640	(640)
Seeds Expert	0	632	(632)
Knitting Machine Operator	0	624	(624)
Fashion Design & Dress Making	2,281	621	1,660
Web Designing & Development	1,575	619	956
Lineman	0	617	(617)
Industrial Electrician	561	596	(35)
Distributor	0	585	(585)
3D Animation (Realistic Rendering)	160	584	(424)
X-Ray Technician	0	584	(584)
Tiles Fixer	0	552	(552)
Networking Installation Technician	0	546	(546)
Paramedic	0	540	(540)
Livestock Helper	0	532	(532)
Machinist	1,090	524	566

Note: Figures in brackets indicate demand-supply shortages, while positive values indicate surplus supply.

